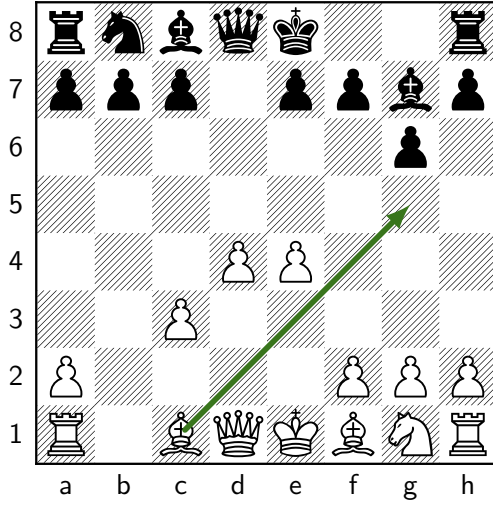


Melanie-MASTERS (human masters) — Grunfeld

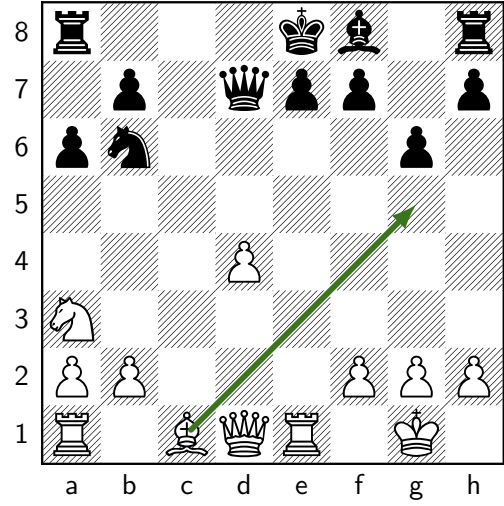
Melanie (Deep_Vision)

Abstract

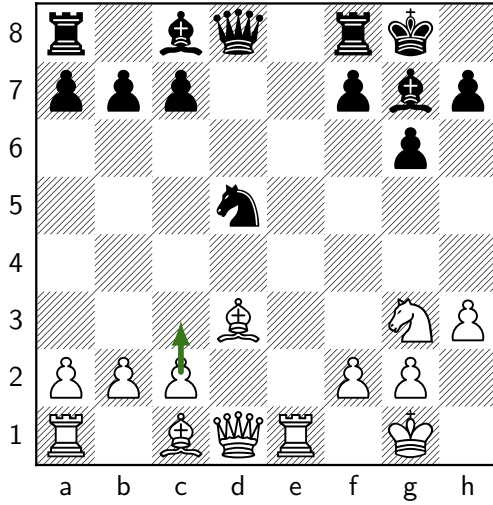
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



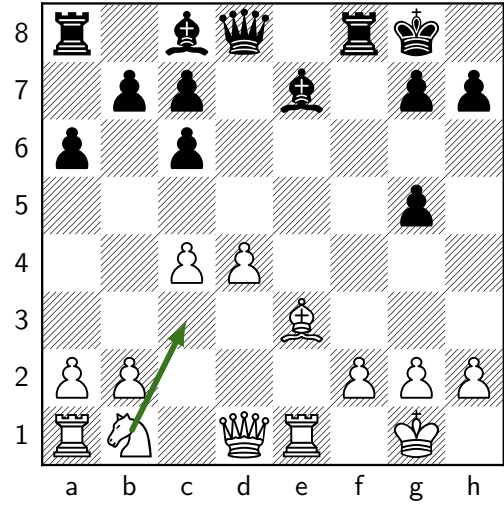
(a)



(b)

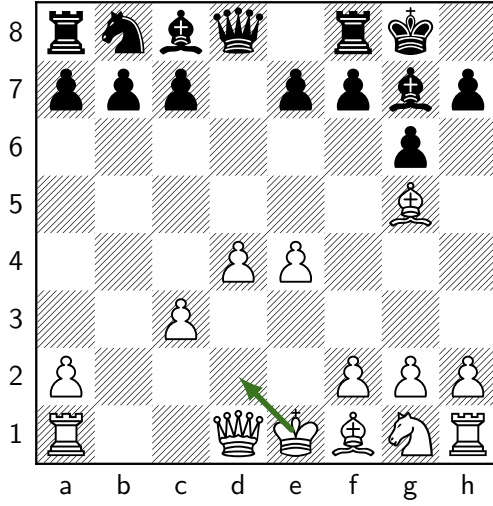


(c)

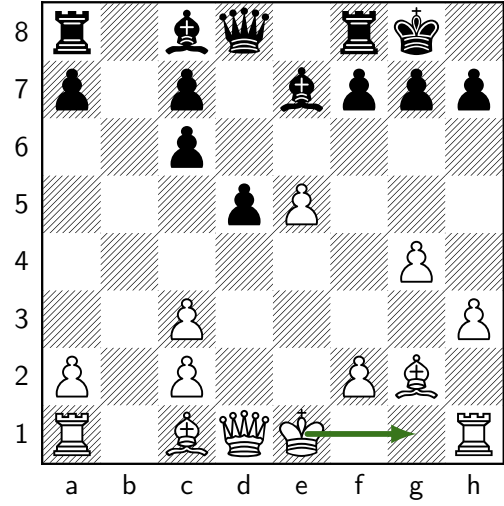


(d)

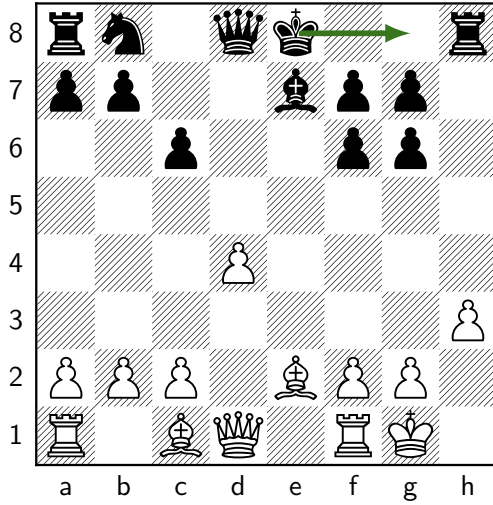
Figure 1: **Move 7, White: Bg5.** (a) Our position; Melanie plays Bg5 (c1g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0012#2995` (ply 28) [structural distance $d=75$, QAP obj=57]: the master played Bg5; under μ our *Bc1* plays the role of that game’s *Bc1*; cross-type roles: our *qd8* as their *Re1*, our *Ng1* as their *Pb2*; transferred move c1g5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0019#1765` (ply 26) [structural distance $d=65$, QAP obj=51]: the master played c3; under μ our *Pc3* plays the role of that game’s *Pc2*; cross-type roles: our *Bf1* as their *Ra1*, our *Qd1* as their *Bc1*; transferred move c3c4. (d) Retrieved from `lichess_elite_2024-09_p0003#3753` (ply 26) [structural distance $d=76$, QAP obj=48]: the master played Nc3; under μ our *Bc1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Ng1* as their *Re1*, our *Rh1* as their *Pa2*; transferred move c1e3. Synthesis: 1 retrieved analog transferred to Bg5 (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



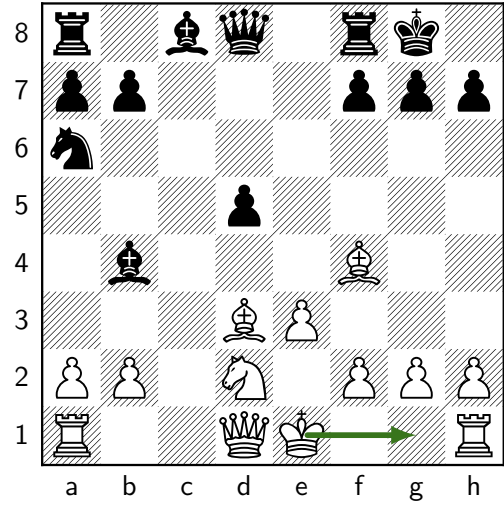
(a)



(b)

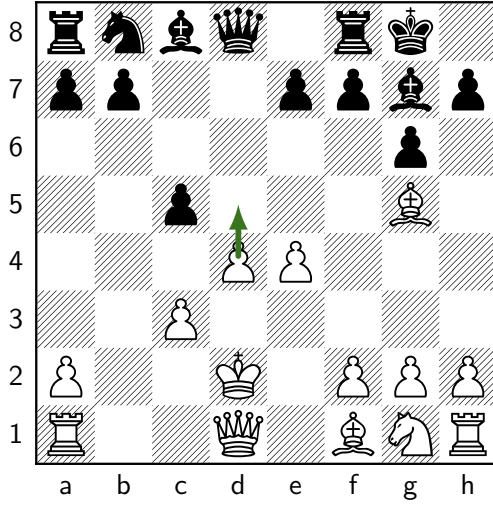


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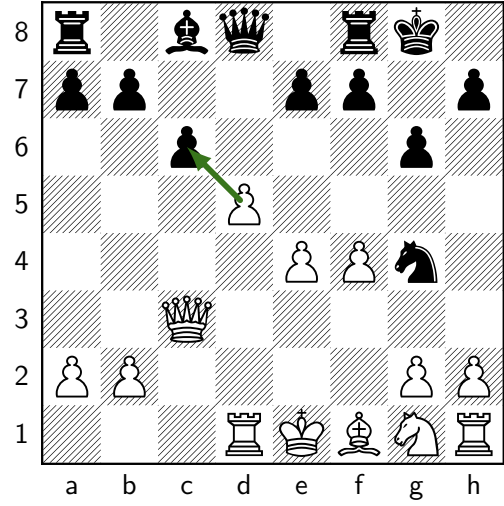


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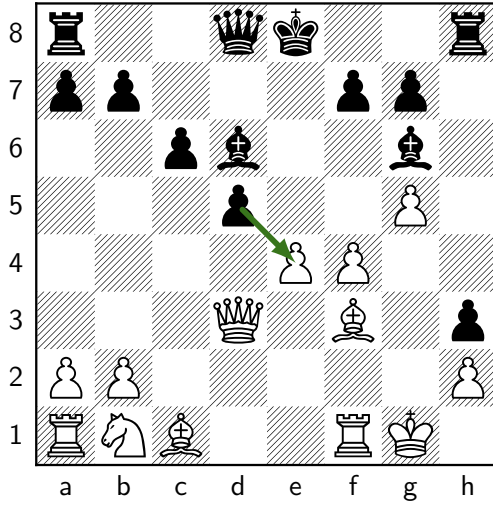
Figure 2: **Move 8, White: Kd2.** (a) Our position; Melanie plays Kd2 (e1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0016#776` (ply 22) [structural distance $d=59$, QAP obj=58]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *pc7* as their *Bc1*, our *Qd1* as their *Rh1*; transferred move e1d2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#2697` (ply 21) [structural distance $d=72$, QAP obj=38]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *pc7* as their *Nb8*, our *Bg5* as their *Rh8*; transferred move e1d2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0003#3389` (ply 22) [structural distance $d=51$, QAP obj=20]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *bc8* as their *Ra1*, our *Pd4* as their *Nd2*; transferred move e1d2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kd2 (summed QAP objective 116, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



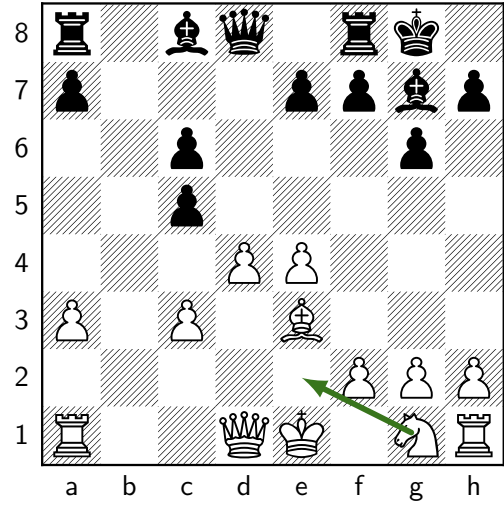
(a)



(b)

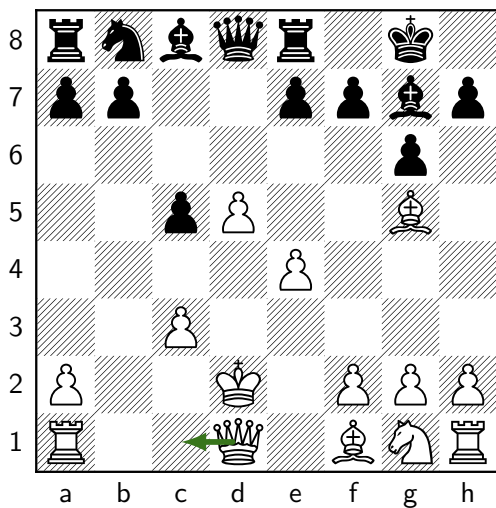


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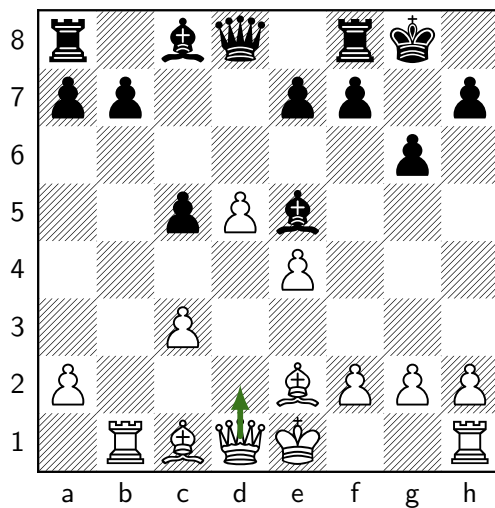


(d)

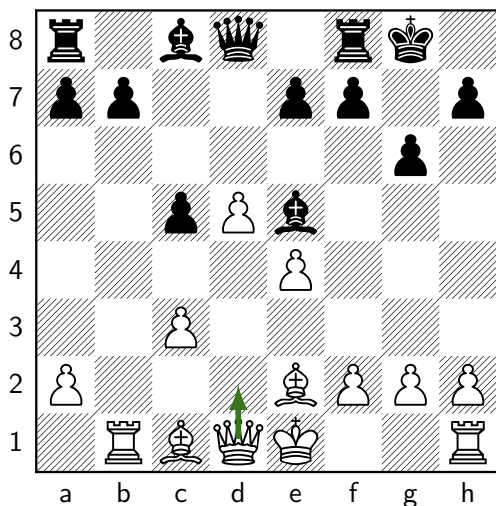
Figure 3: **Move 9, White: d5.** (a) Our position; Melanie plays d5 (d4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0025#2186` (ply 26) [structural distance $d=52$, QAP obj=98]: the master played dxc6; under μ our *Pd4* plays the role of that game’s *Pd5*; cross-type roles: our *Qd1* as their *Rd1*, our *ph7* as their *Rh1*; transferred move d4d5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0020#2744` (ply 33) [structural distance $d=74$, QAP obj=87]: the master played dxe4; under μ our *Pd4* plays the role of that game’s *Pd4*; cross-type roles: our *Ph2* as their *Rh8*, our *Bf1* as their *Pa7*; transferred move d4d5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0005#2753` (ply 20) [structural distance $d=42$, QAP obj=119]: the master played Ne2; under μ our *Ng1* plays the role of that game’s *Ng1*; cross-type roles: our *Bf1* as their *Ra1*, our *Ra1* as their *Pf2*; transferred move g1e2. Synthesis: 2 retrieved analogs transferred to d5 (summed QAP objective 186, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



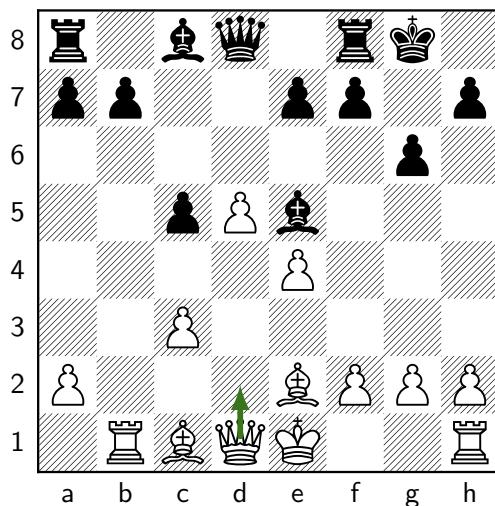
(a)



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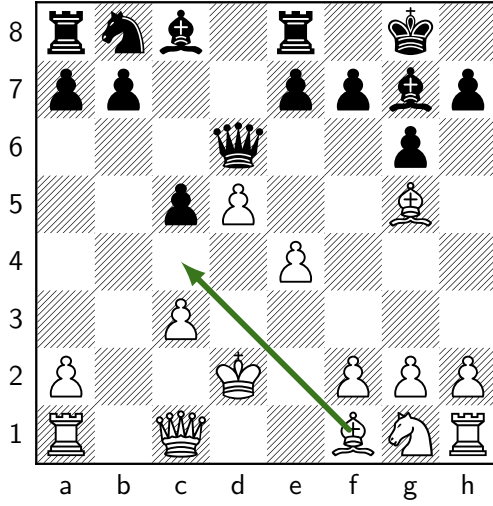


(c)

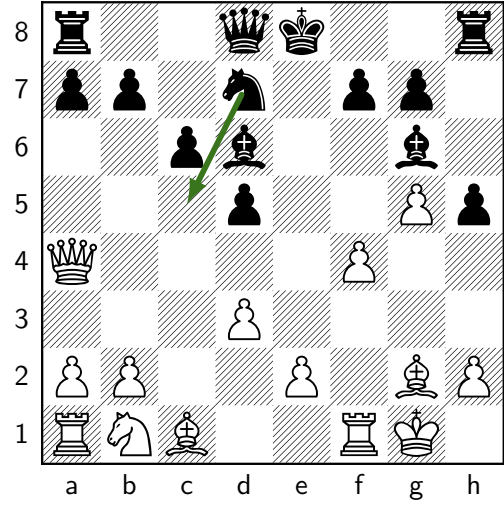


(d)

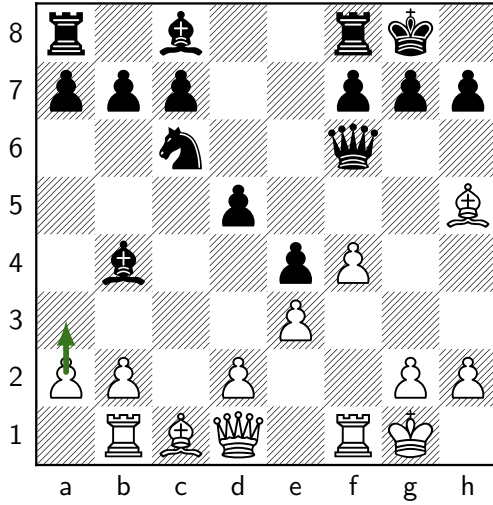
Figure 4: **Move 10, White: Qc1**. (a) Our position; Melanie plays Qc1 (d1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#1262` (ply 22) [structural distance $d=63$, QAP obj=71]: the master played Qd2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our bg7 as their Rb1, our ph7 as their Rh1; transferred move d1c1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0012#2244` (ply 22) [structural distance $d=63$, QAP obj=71]: the master played Qd2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our bg7 as their Rb1, our ph7 as their Rh1; transferred move d1c1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0021#3485` (ply 22) [structural distance $d=63$, QAP obj=71]: the master played Qd2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our bg7 as their Rb1, our ph7 as their Rh1; transferred move d1c1 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qc1 (summed QAP objective 356, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



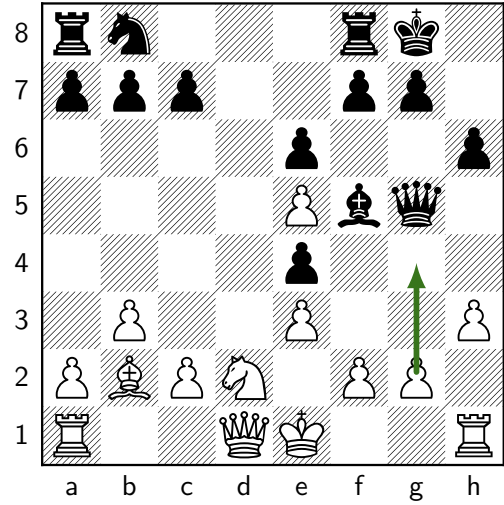
(a)



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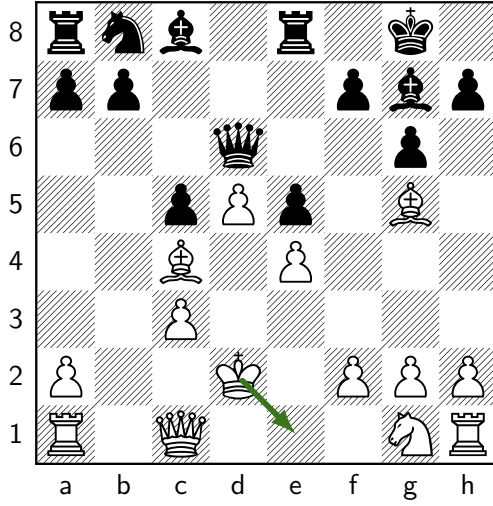


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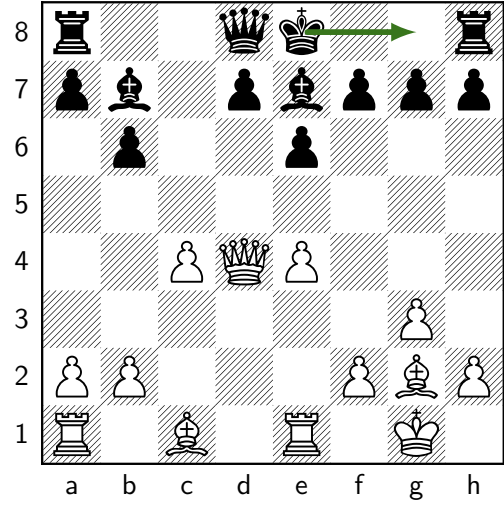


(d)

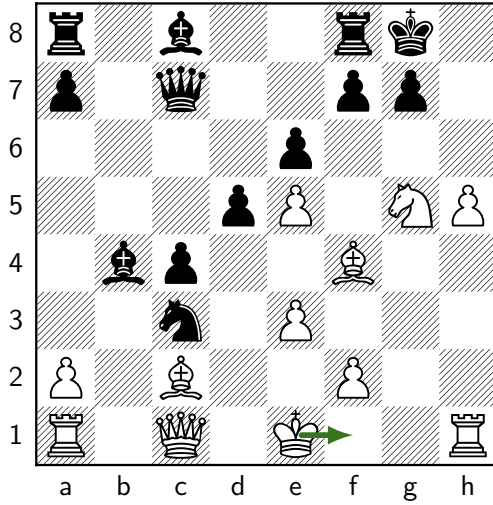
Figure 5: **Move 11, White: Bc4.** (a) Our position; Melanie plays Bc4 (f1c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0020#2744` (ply 25) [structural distance $d=72$, QAP obj=55]: the master played Nc5; under μ our Bf1 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Pf2 as their Ra8, our pb7 as their Rh8; transferred move f1c4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0016#145` (ply 30) [structural distance $d=82$, QAP obj=54]: the master played a3; under μ our Bf1 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our Pf2 as their Bc1, our Pc3 as their Qd1; transferred move f1c4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0030#2804` (ply 22) [structural distance $d=85$, QAP obj=50]: the master played g4; under μ our Ph2 plays the role of that game’s Pg2; cross-type roles: our Bf1 as their Ra1, our Rh1 as their Pa2; transferred move h2h3. Synthesis: 2 retrieved analogs transferred to Bc4 (summed QAP objective 109, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



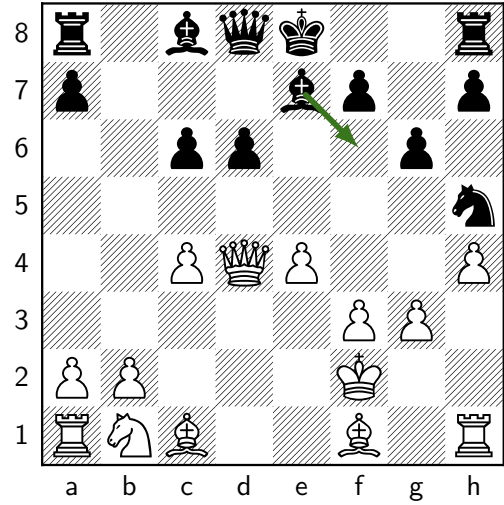
(a)



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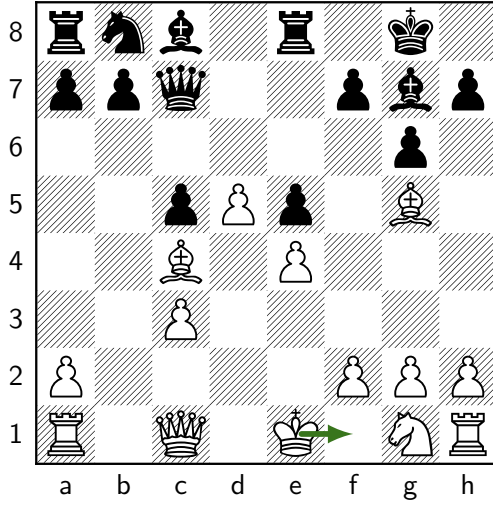


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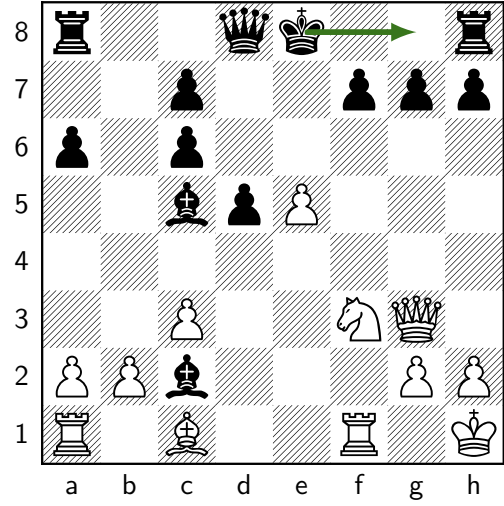


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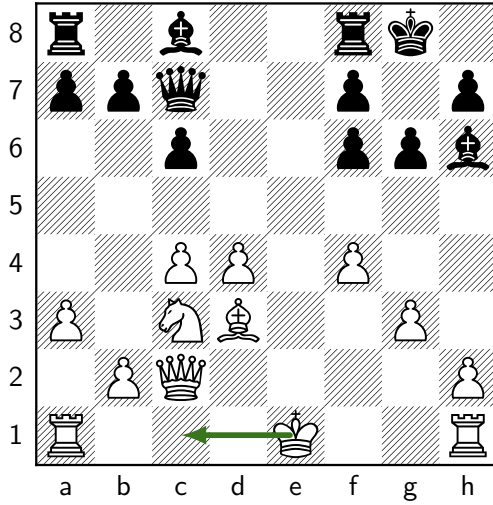
Figure 6: **Move 12, White: Ke1.** (a) Our position; Melanie plays Ke1 (d2e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#2395` (ply 25) [structural distance $d=89$, QAP obj=41]: the master played 0-0; under μ our $Kd2$ plays the role of that game’s $Ke1$; cross-type roles: our $pe5$ as their $Ra8$, our $Bg5$ as their $Rh8$; transferred move d2e1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0016#3412` (ply 36) [structural distance $d=86$, QAP obj=34]: the master played Kf1; under μ our $Kd2$ plays the role of that game’s $Ke1$; cross-type roles: our $Bg5$ as their $Rh1$, our $Ng1$ as their $Pf2$; transferred move d2e1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0017#2042` (ply 23) [structural distance $d=88$, QAP obj=38]: the master played Bf6; under μ our $Bc4$ plays the role of that game’s $Be2$; cross-type roles: our $Pe4$ as their $Ra8$, our $Pd5$ as their $Rh8$; transferred move c4f1. Synthesis: 2 retrieved analogs transferred to Ke1 (summed QAP objective 75, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



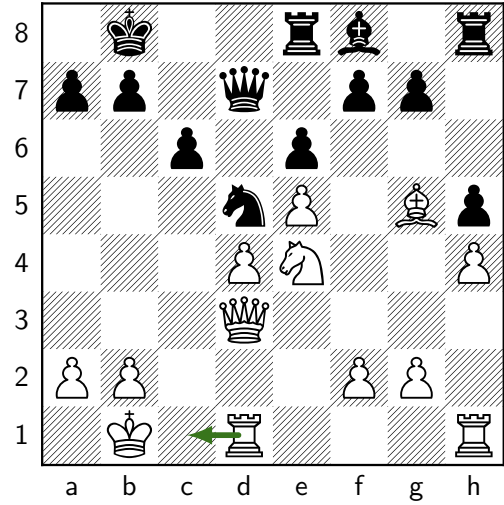
(a)



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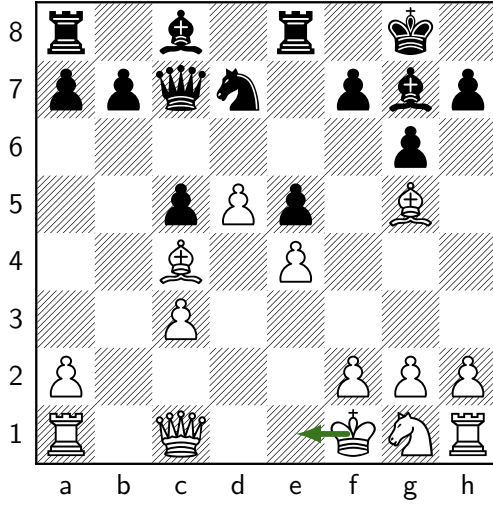


(c)

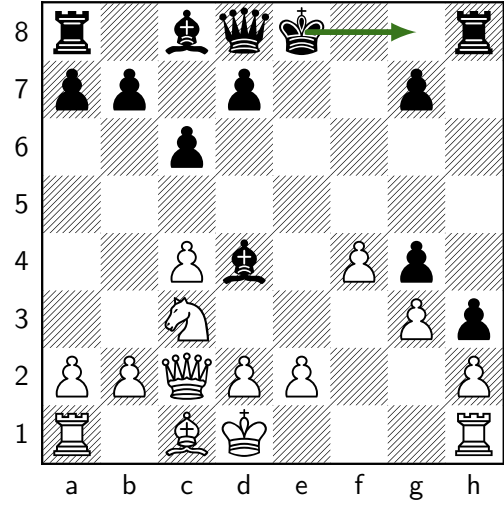


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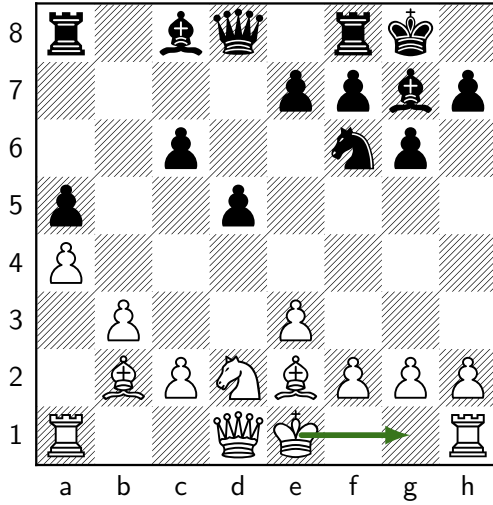
Figure 7: **Move 13, White: Kf1**. (a) Our position; Melanie plays Kf1 (e1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0018#1078` (ply 23) [structural distance $d=82$, QAP obj=8]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *nb8* as their *Pg7*, our *bc8* as their *Pc6*; transferred move e1f1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0016#2049` (ply 30) [structural distance $d=92$, QAP obj=6]: the master played 0-0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc4* as their *Ra1*, our *Qc1* as their *Rh1*; transferred move e1f1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0019#1202` (ply 36) [structural distance $d=72$, QAP obj=9]: the master played Rc1; under μ our *Qc1* plays the role of that game’s *Rd1* (a **cross-type** correspondence); cross-type roles: our *Ng1* as their *Pb2*, our *Rh1* as their *Pg2*; transferred move c1d1. Synthesis: 2 retrieved analogs transferred to Kf1 (summed QAP objective 14, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



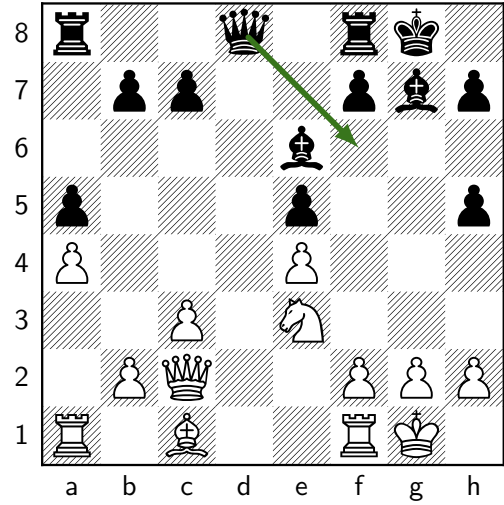
(a)



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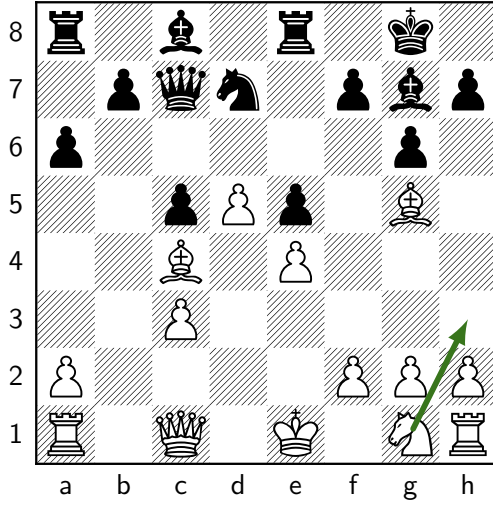


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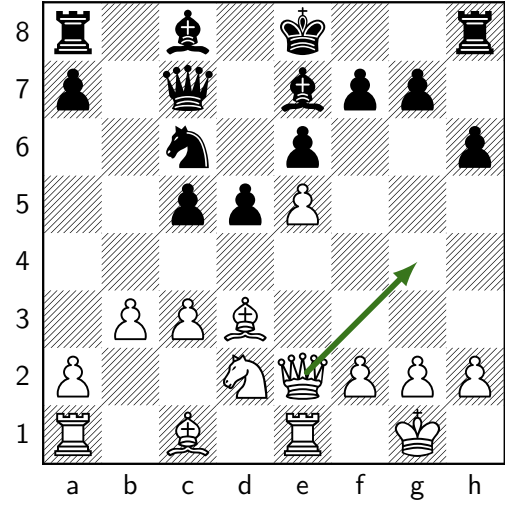


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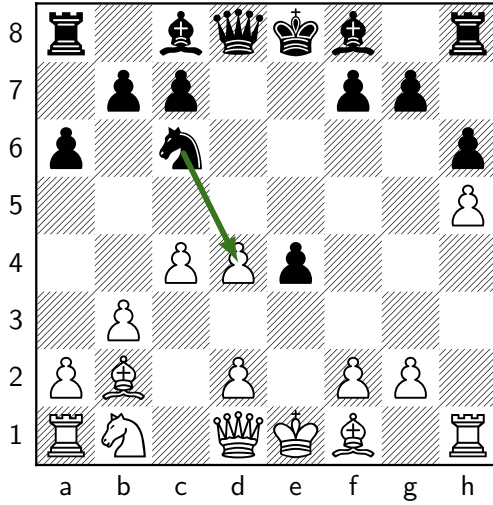
Figure 8: **Move 14, White: Ke1**. (a) Our position; Melanie plays Ke1 (f1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0024#2213 (ply 35) [structural distance $d=77$, QAP obj=6]: the master played 0-0; under μ our $Kf1$ plays the role of that game’s $Ke1$; cross-type roles: our $nd7$ as their $Ra8$, our $pg6$ as their $Bc8$; transferred move f1e1 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0023#1815 (ply 20) [structural distance $d=77$, QAP obj=5]: the master played 0-0; under μ our $Kf1$ plays the role of that game’s $Ke1$; cross-type roles: our $Bg5$ as their $Ra1$, our $Bc4$ as their $Rh1$; transferred move f1e1 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0023#1745 (ply 27) [structural distance $d=79$, QAP obj=6]: the master played Qf6; under μ our $Bg5$ plays the role of that game’s $Qd1$ (a **cross-type** correspondence); cross-type roles: our $Bc4$ as their $Rf8$, our $Ng1$ as their $Pb7$; transferred move g5e3. Synthesis: 2 retrieved analogs transferred to Ke1 (summed QAP objective 11, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



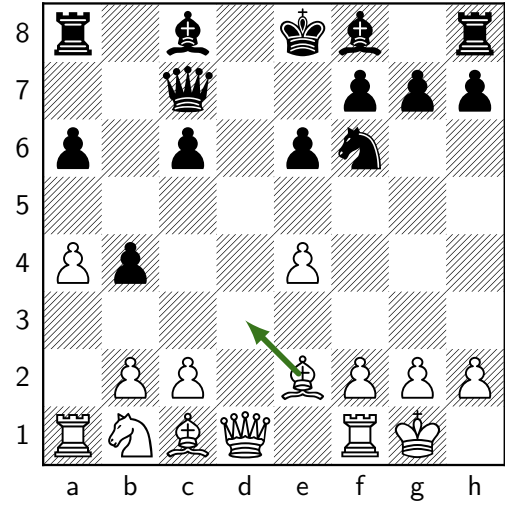
(a)



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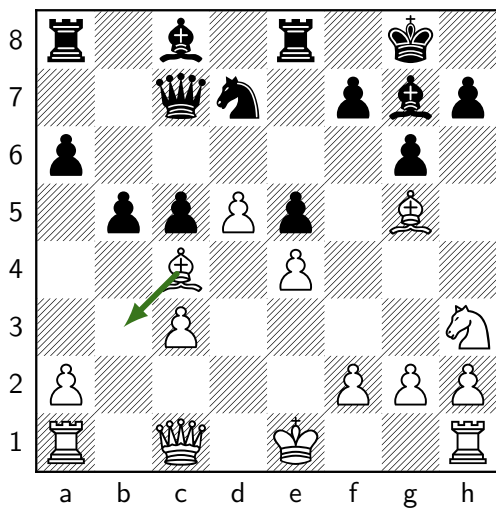


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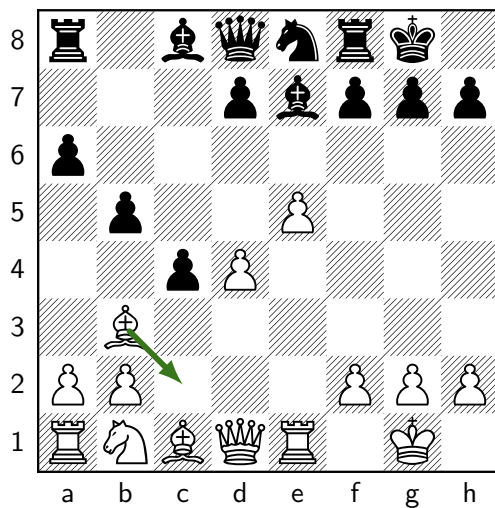


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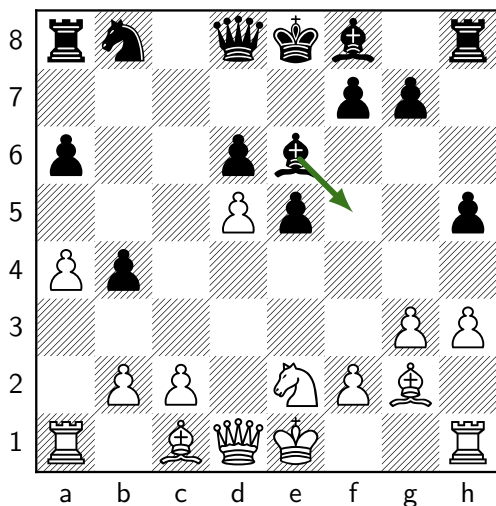
Figure 9: **Move 15, White: Nh3**. (a) Our position; Melanie plays Nh3 (g1h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#3944` (ply 26) [structural distance $d=89$, QAP obj=39]: the master played Qg4; under μ our Ng1 plays the role of that game’s Qe2 (a **cross-type** correspondence); cross-type roles: our Pc3 as their Bc1, our Qc1 as their Re1; transferred move g1h3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#3992` (ply 21) [structural distance $d=72$, QAP obj=29]: the master played Nxd4; under μ our Bc4 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our pe5 as their Ra8, our pg6 as their Bc8; transferred move c4d3. (d) Retrieved from `lichess_elite_2024-09_p0020#537` (ply 20) [structural distance $d=74$, QAP obj=26]: the master played Bd3; under μ our Bc4 plays the role of that game’s Be2; cross-type roles: our Bg5 as their Ra1, our pg6 as their Bc1; transferred move c4b3. Synthesis: 1 retrieved analog transferred to Nh3 (summed QAP objective 39, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



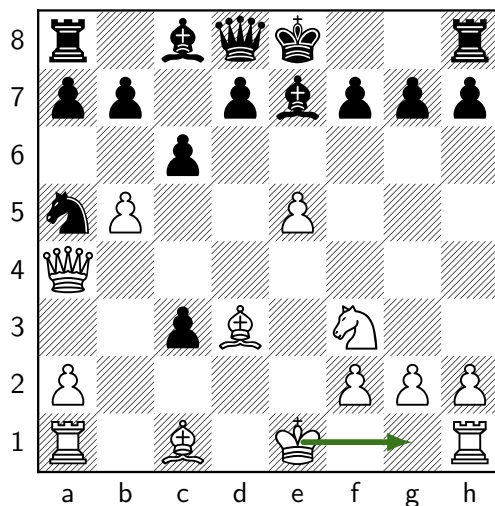
(a)



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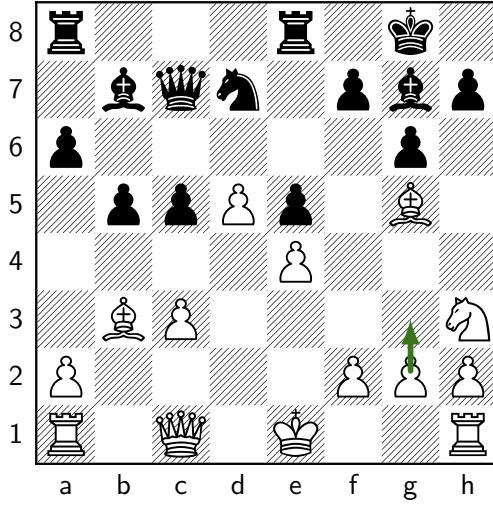


(c)

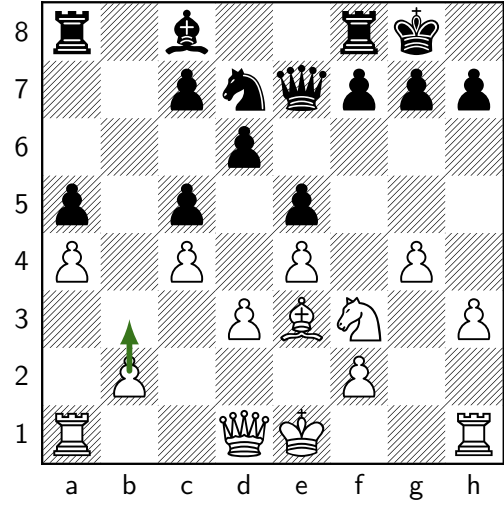


(d)

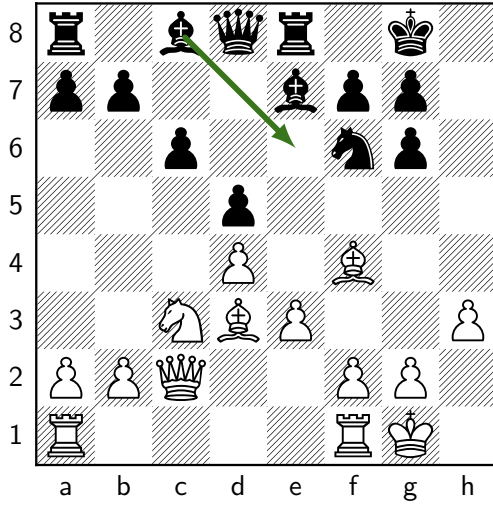
Figure 10: **Move 16, White: Bb3**. (a) Our position; Melanie plays Bb3 (c4b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0020#2011` (ply 24) [structural distance $d=62$, QAP obj=33]: the master played Bc2; under μ our Bc4 plays the role of that game’s Bb3; cross-type roles: our Ra1 as their Bc1, our pf7 as their Qd1; transferred move c4b3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0023#705` (ply 23) [structural distance $d=70$, QAP obj=32]: the master played Bf5; under μ our Bc4 plays the role of that game’s Be3; cross-type roles: our pe5 as their Ra8, our pg6 as their Nb8; transferred move c4d3. (d) Retrieved from `lichess_elite_2024-09_p0023#897` (ply 22) [structural distance $d=74$, QAP obj=25]: the master played 0-0; under μ our Ke1 plays the role of that game’s Ke1; cross-type roles: our Nh3 as their Ra1, our pc5 as their Bc1; transferred move e1f1. Synthesis: 2 retrieved analogs transferred to Bb3 (summed QAP objective 56, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



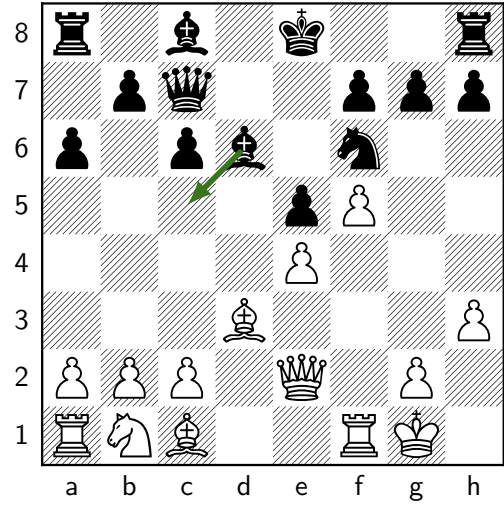
(a)



(b)

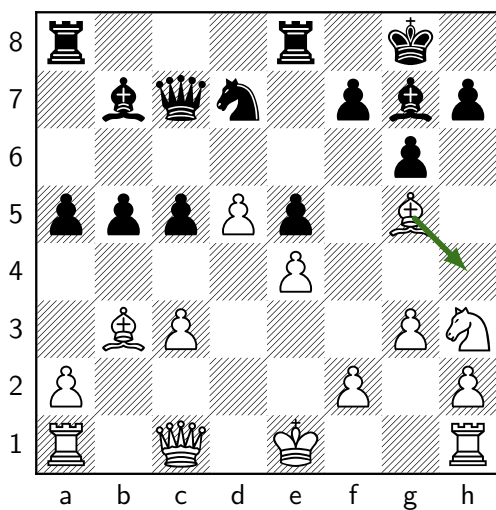


(c)

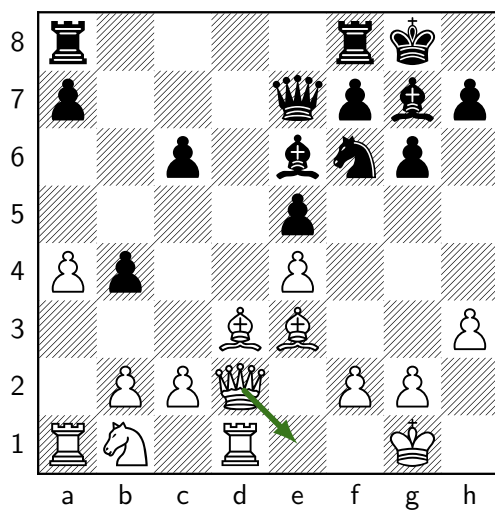


(d)

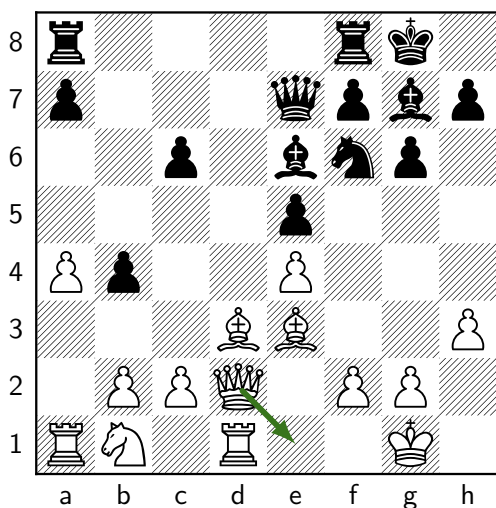
Figure 11: **Move 17, White: g3.** (a) Our position; Melanie plays g3 (g2g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#230` (ply 28) [structural distance $d=70$, QAP obj=36]: the master played b3; under μ our $Pg2$ plays the role of that game’s $Pb2$; cross-type roles: our $Pe4$ as their $Ra1$, our $qc7$ as their $Pf2$; transferred move g2g3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#3951` (ply 25) [structural distance $d=72$, QAP obj=29]: the master played Be6; under μ our $Bg5$ plays the role of that game’s $Bc1$; cross-type roles: our $Bb3$ as their $Qd8$, our $Qc1$ as their $Re8$; transferred move g5e3. (d) Retrieved from `lichess_elite_2024-09_p0025#523` (ply 21) [structural distance $d=66$, QAP obj=28]: the master played Bc5+; under μ our $Bb3$ plays the role of that game’s $Bd3$; cross-type roles: our $qc7$ as their $Ra8$, our $Pa2$ as their $Qc7$; transferred move b3c4. Synthesis: 1 retrieved analog transferred to g3 (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



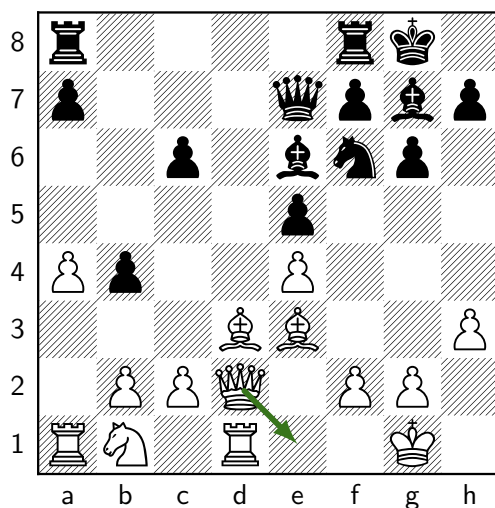
(a)



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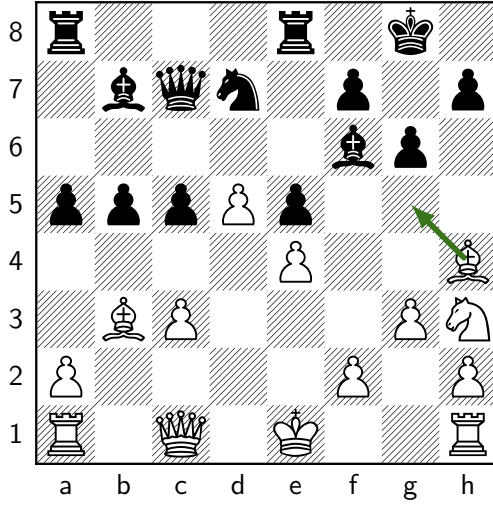


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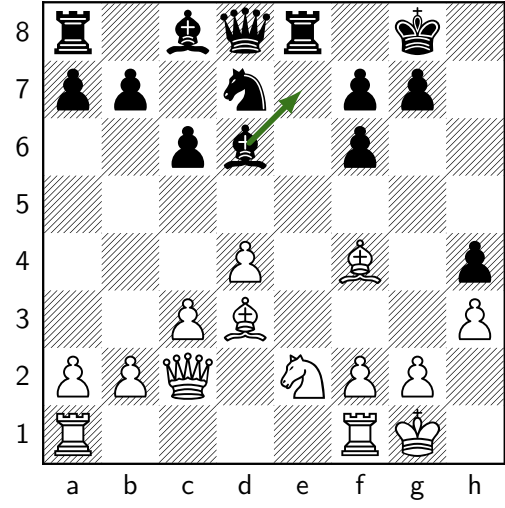


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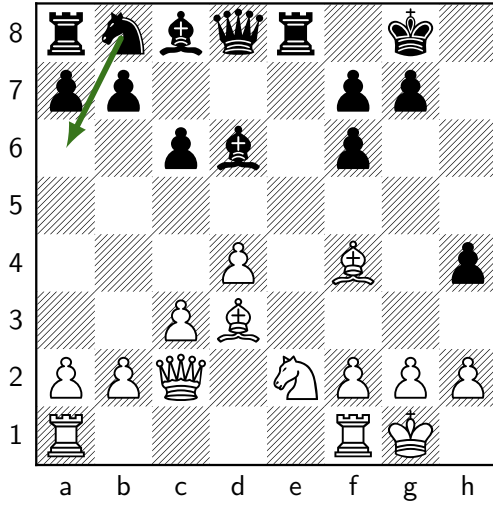
Figure 12: **Move 18, White: Bh4**. (a) Our position; Melanie plays Bh4 (g5h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0025#3095 (ply 28) [structural distance $d=77$, QAP obj=31]: the master played Qe1; under μ our Bg5 plays the role of that game’s Qd2 (a **cross-type** correspondence); cross-type roles: our Nh3 as their Ra1, our pg6 as their Nb1; transferred move g5h4 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0026#1175 (ply 28) [structural distance $d=77$, QAP obj=31]: the master played Qe1; under μ our Bg5 plays the role of that game’s Qd2 (a **cross-type** correspondence); cross-type roles: our Nh3 as their Ra1, our pg6 as their Nb1; transferred move g5h4 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0030#3040 (ply 28) [structural distance $d=77$, QAP obj=31]: the master played Qe1; under μ our Bg5 plays the role of that game’s Qd2 (a **cross-type** correspondence); cross-type roles: our Nh3 as their Ra1, our pg6 as their Nb1; transferred move g5h4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bh4 (summed QAP objective 155, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



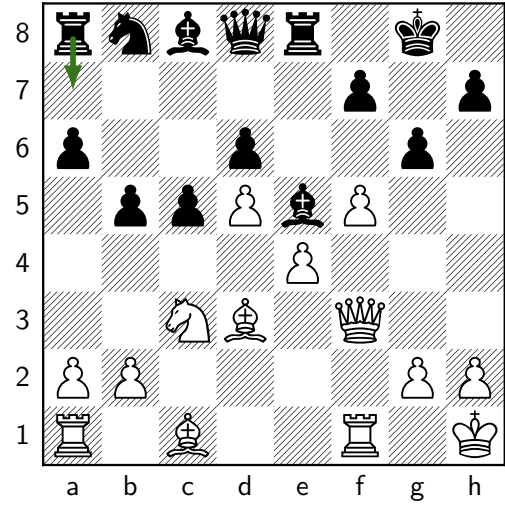
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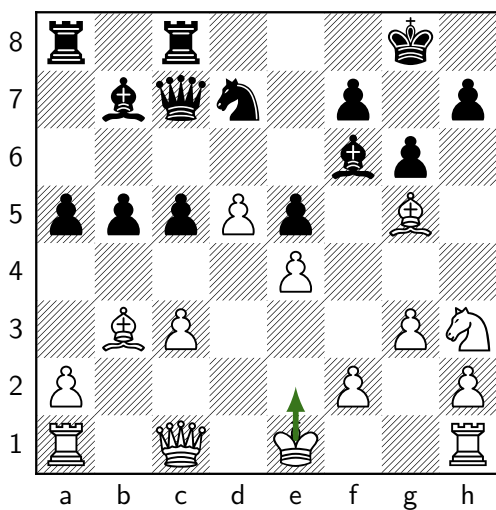


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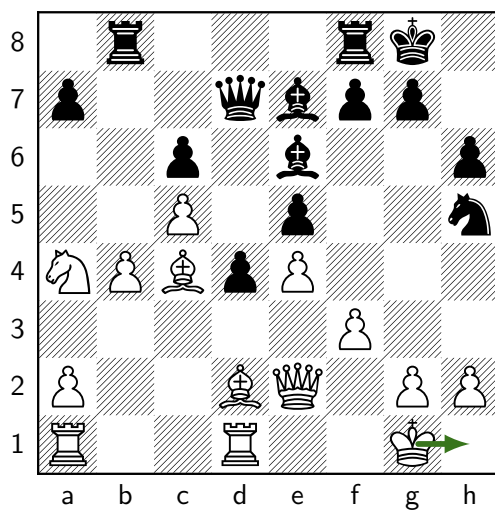


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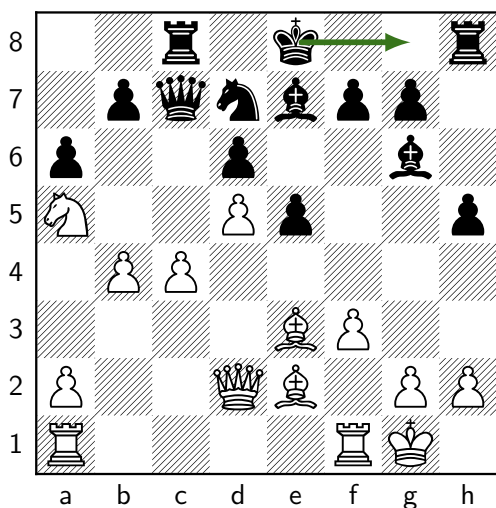
Figure 13: **Move 19, White: Bg5.** (a) Our position; Melanie plays Bg5 (h4g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0025#2700` (ply 23) [structural distance $d=82$, QAP obj=48]: the master played Bc7; under μ our Bh4 plays the role of that game’s Bd3; cross-type roles: our Qc1 as their Ra8, our pa5 as their Bc8; transferred move h4g5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#3576` (ply 21) [structural distance $d=86$, QAP obj=47]: the master played Na6; under μ our Qc1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our pa5 as their Ra8, our Pg3 as their Bc8; transferred move c1b1. (d) Retrieved from `lichess_elite_2024-09_p0025#2053` (ply 29) [structural distance $d=85$, QAP obj=47]: the master played Ra7; under μ our Qc1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Bb3 as their Nb8, our Pc3 as their Bc8; transferred move c1a3. Synthesis: 1 retrieved analog transferred to Bg5 (summed QAP objective 48, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



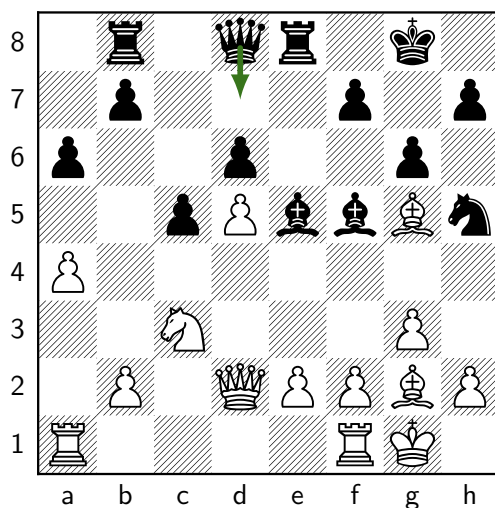
(a)



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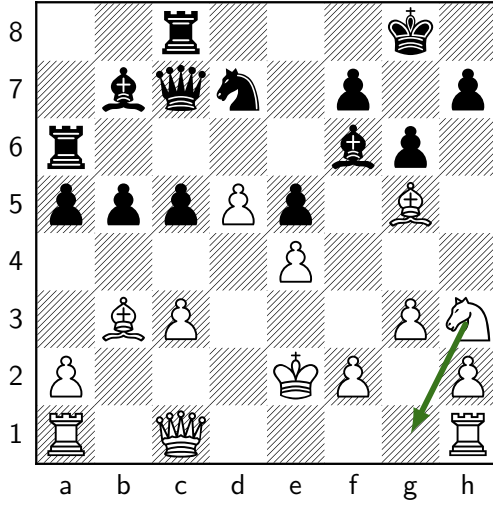


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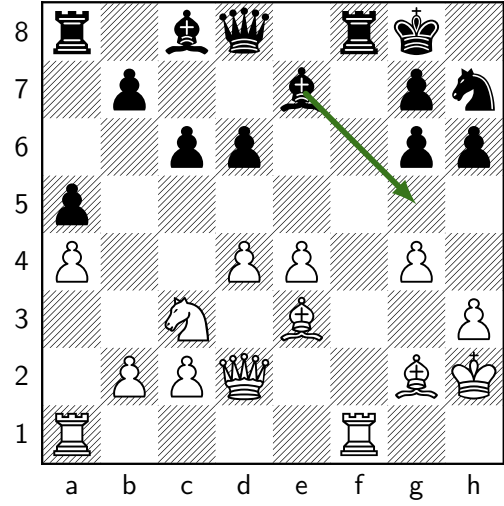


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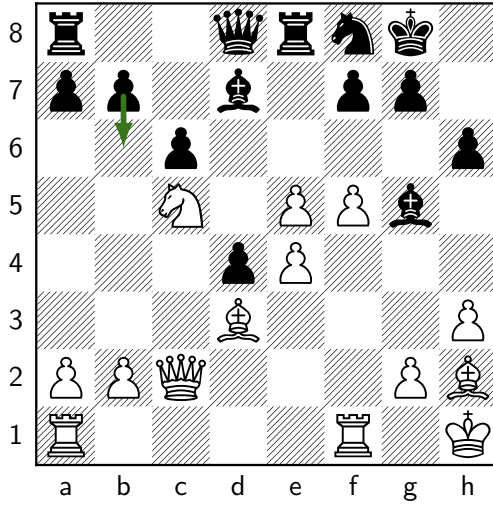
Figure 14: **Move 20, White: Ke2.** (a) Our position; Melanie plays Ke2 (e1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#3602` (ply 36) [structural distance $d=73$, QAP obj=29]: the master played Kh1; under μ our Ke1 plays the role of that game’s Kg1; cross-type roles: our Ph2 as their Rd1, our pf7 as their Qe2; transferred move e1e2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0012#2806` (ply 31) [structural distance $d=77$, QAP obj=17]: the master played 0-0; under μ our Ke1 plays the role of that game’s Ke1; cross-type roles: our Qc1 as their Rc8, our Bg5 as their Qc7; transferred move e1e2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0028#1927` (ply 33) [structural distance $d=82$, QAP obj=37]: the master played Qd7; under μ our Bg5 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our pb5 as their Rb8, our Qc1 as their Re8; transferred move g5h4. Synthesis: 2 retrieved analogs transferred to Ke2 (summed QAP objective 46, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



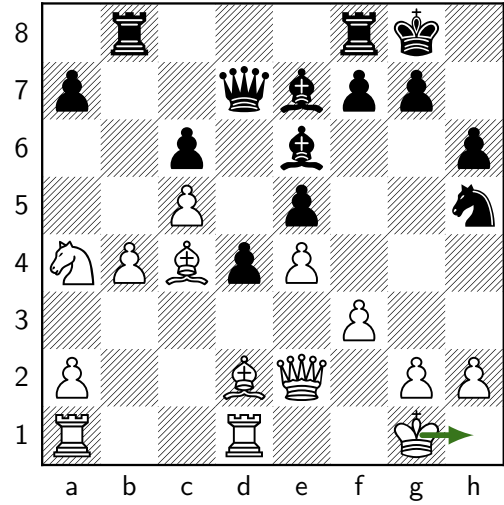
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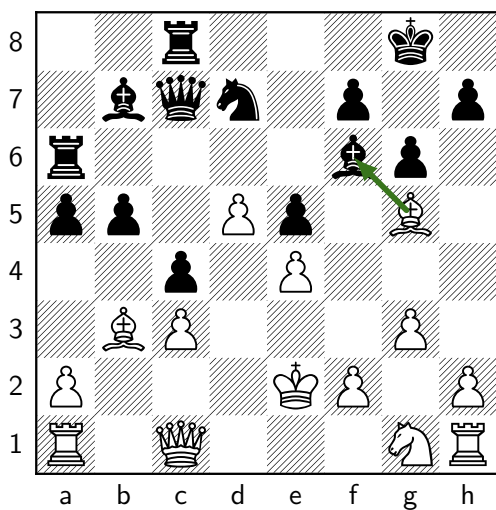


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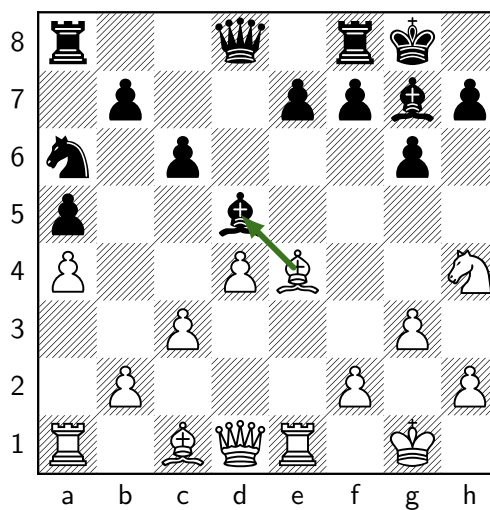


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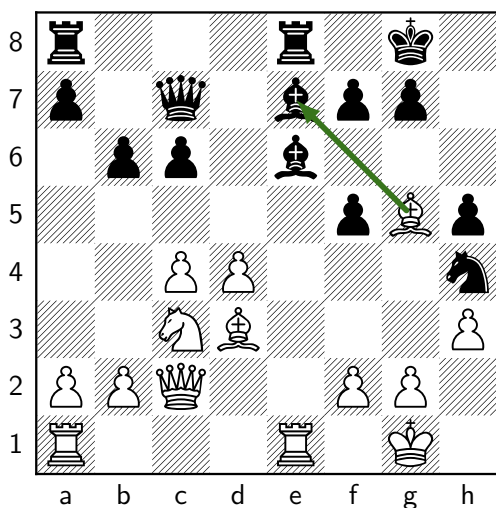
Figure 15: **Move 21, White: Ng1**. (a) Our position; Melanie plays Ng1 (h3g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0020#1905` (ply 29) [structural distance $d=84$, QAP obj=16]: the master played Bg5; under μ our *Nh3* plays the role of that game’s *Be2* (a **cross-type** correspondence); cross-type roles: our *Bb3* as their *Ra8*, our *pf7* as their *Bc8*; transferred move h3g1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0021#406` (ply 39) [structural distance $d=84$, QAP obj=16]: the master played b6; under μ our *Nh3* plays the role of that game’s *Pb2* (a **cross-type** correspondence); cross-type roles: our *Qc1* as their *Ra8*, our *bb7* as their *Qd8*; transferred move h3g1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0003#3602` (ply 36) [structural distance $d=72$, QAP obj=27]: the master played Kh1; under μ our *Ke2* plays the role of that game’s *Kg1*; cross-type roles: our *Pd5* as their *Rd1*, our *pe5* as their *Bd2*; transferred move e2f3. Synthesis: 2 retrieved analogs transferred to Ng1 (summed QAP objective 32, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



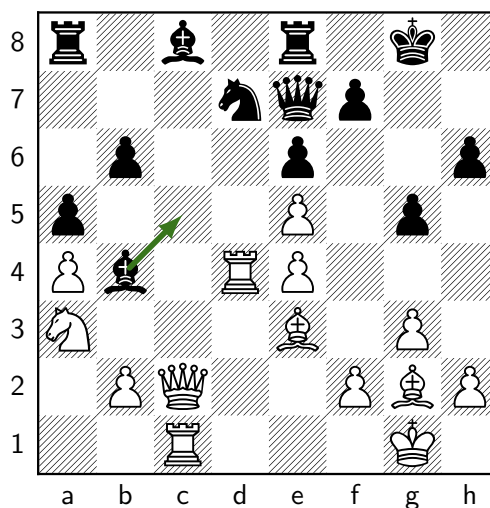
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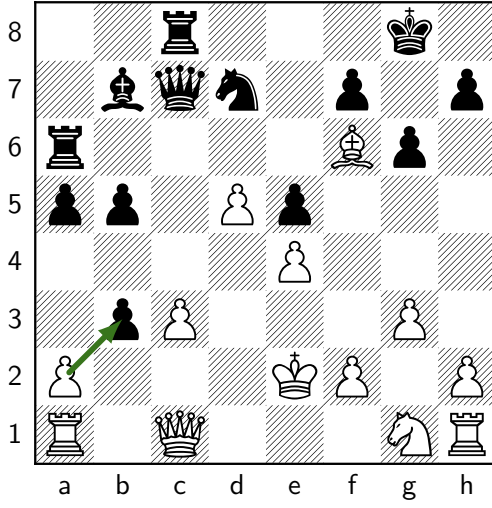


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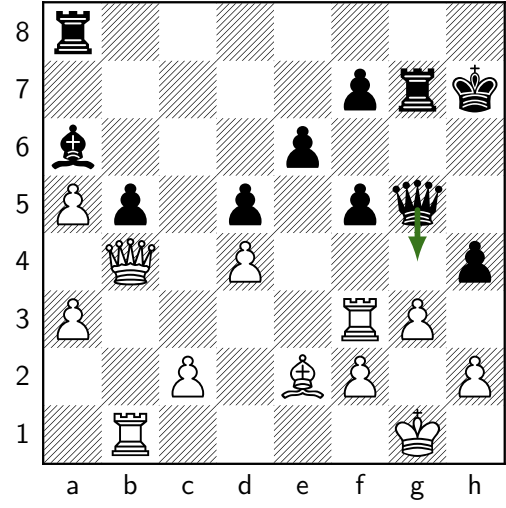


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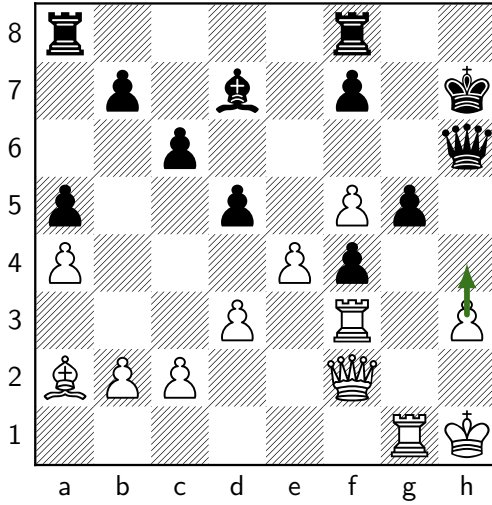
Figure 16: **Move 22, White: Bxf6**. (a) Our position; Melanie plays Bxf6 (g5f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#29` (ply 26) [structural distance $d=78$, QAP obj=42]: the master played Bxd5; under μ our Bg5 plays the role of that game’s Be4; cross-type roles: our Pe4 as their Ra1, our pb5 as their Re1; transferred move g5f6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0010#392` (ply 36) [structural distance $d=74$, QAP obj=41]: the master played Bxe7; under μ our Bg5 plays the role of that game’s Bg5; cross-type roles: our Ng1 as their Re1, our Pc3 as their Qc2; transferred move g5f6 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0017#2100` (ply 51) [structural distance $d=77$, QAP obj=37]: the master played Bc5; under μ our Bg5 plays the role of that game’s Bb5; cross-type roles: our pc4 as their Bc8, our Ng1 as their Re8; transferred move g5f4. Synthesis: 2 retrieved analogs transferred to Bxf6 (summed QAP objective 83, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



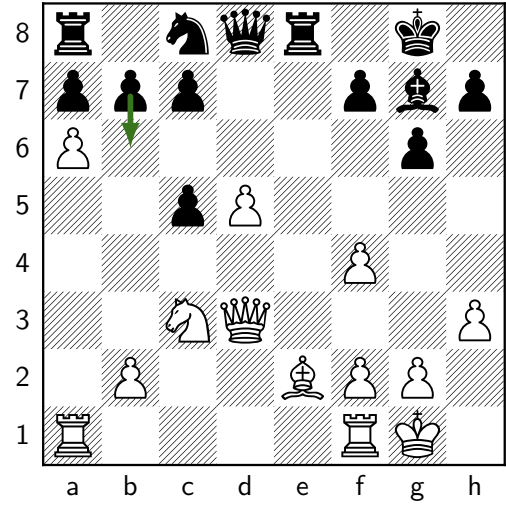
(a)



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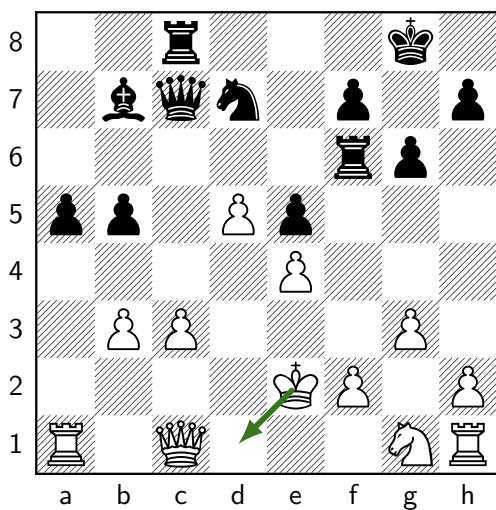


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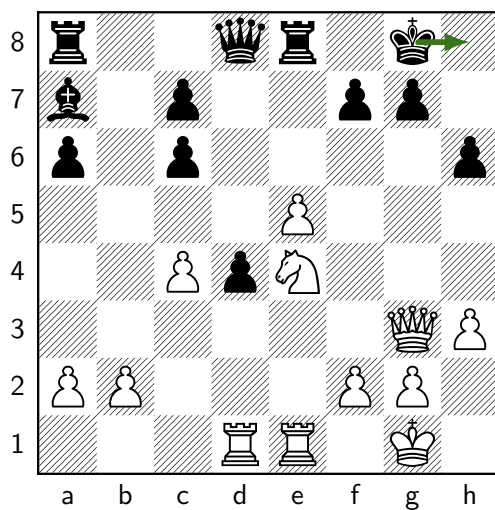


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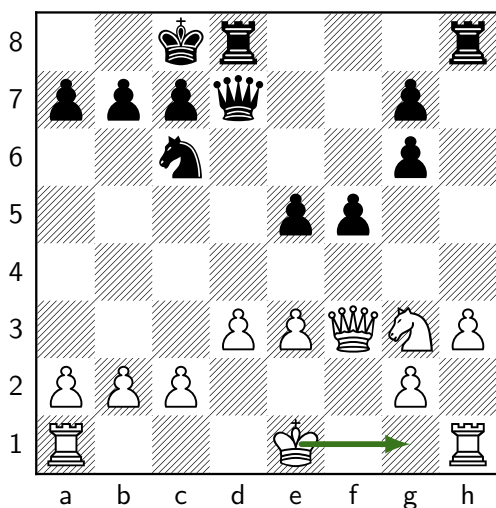
Figure 17: **Move 23, White: axb3**. (a) Our position; Melanie plays axb3 (a2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0020#2079` (ply 57) [structural distance $d=79$, QAP obj=48]: the master played Qg4; under μ our *Pa2* plays the role of that game’s *Qg4* (a **cross-type** correspondence); cross-type roles: our *Pf2* as their *Ra8*, our *Qc1* as their *Pf7*; transferred move a2b3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#2658` (ply 54) [structural distance $d=73$, QAP obj=63]: the master played h4; under μ our *Pc3* plays the role of that game’s *Ph3*; cross-type roles: our *Ng1* as their *Rg1*, our *Qc1* as their *Ba2*; transferred move c3c4. (d) Retrieved from `lichess_elite_2025-11_p0000#3728` (ply 33) [structural distance $d=71$, QAP obj=57]: the master played b6; under μ our *Pa2* plays the role of that game’s *Pb2*; cross-type roles: our *Ng1* as their *Ra8*, our *ra6* as their *Nc8*; transferred move a2a3. Synthesis: 1 retrieved analog transferred to axb3 (summed QAP objective 48, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



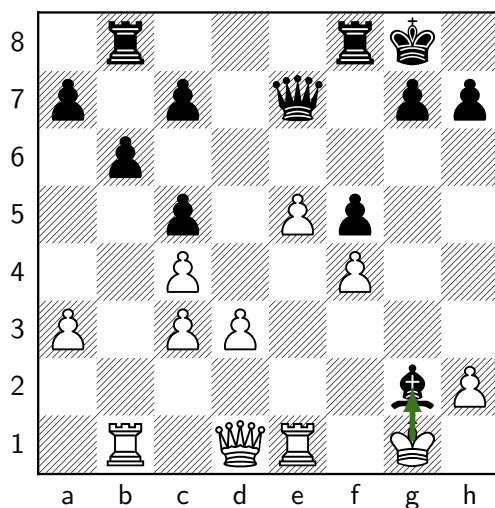
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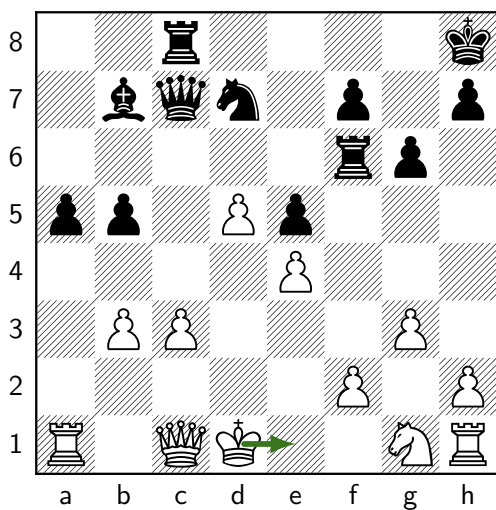


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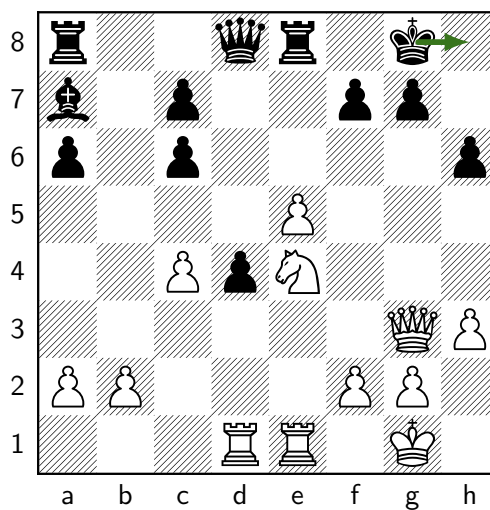


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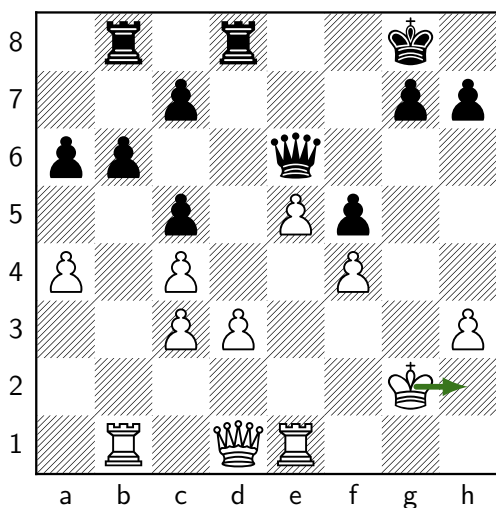
Figure 18: **Move 24, White: Kd1.** (a) Our position; Melanie plays Kd1 (e2d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#983` (ply 39) [structural distance $d=85$, QAP obj=59]: the master played Kh8; under μ our Ke2 plays the role of that game’s Kg1; cross-type roles: our qc7 as their Re8, our Pd5 as their Ba7; transferred move e2d1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0023#3276` (ply 32) [structural distance $d=87$, QAP obj=46]: the master played 0-0; under μ our Ke2 plays the role of that game’s Ke1; cross-type roles: our Qc1 as their Rh1, our bb7 as their Qf3; transferred move e2d1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0023#3721` (ply 40) [structural distance $d=78$, QAP obj=20]: the master played Kxg2; under μ our Ke2 plays the role of that game’s Kg1; cross-type roles: our Pb3 as their Qd1, our Qc1 as their Re1; transferred move e2d1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kd1 (summed QAP objective 125, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



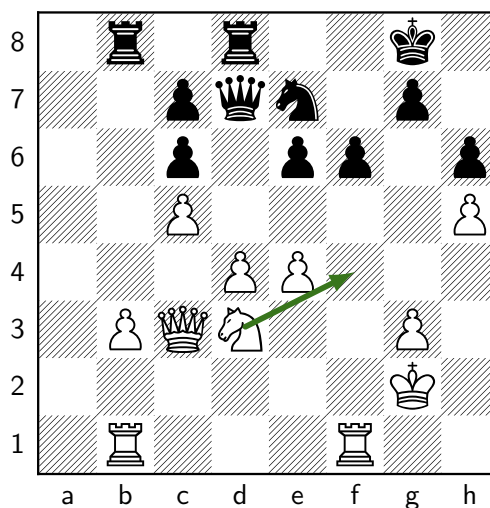
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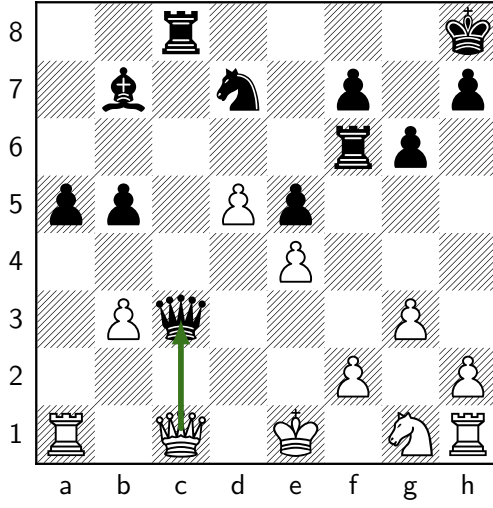


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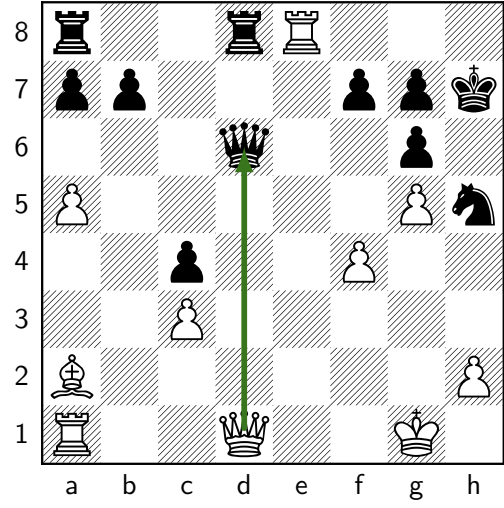


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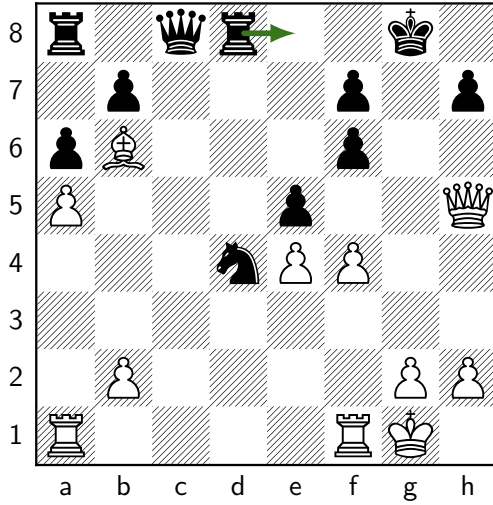
Figure 19: **Move 25, White: Ke1.** (a) Our position; Melanie plays Ke1 (d1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#983` (ply 39) [structural distance $d=76$, QAP obj=57]: the master played Kh8; under μ our Kd1 plays the role of that game’s Kg1; cross-type roles: our qc7 as their Re8, our ph7 as their Ba7; transferred move d1e1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0023#3721` (ply 46) [structural distance $d=79$, QAP obj=22]: the master played Kh2; under μ our Kd1 plays the role of that game’s Kg2; cross-type roles: our Ng1 as their Ph3, our bb7 as their qe6; transferred move d1e1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0003#423` (ply 64) [structural distance $d=80$, QAP obj=67]: the master played Nf4; under μ our Pg3 plays the role of that game’s Nd3 (a **cross-type** correspondence); cross-type roles: our pf7 as their Rb1, our Ra1 as their Pg3; transferred move g3g4. Synthesis: 3 retrieved analogs transferred to Ke1 (summed QAP objective 101, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



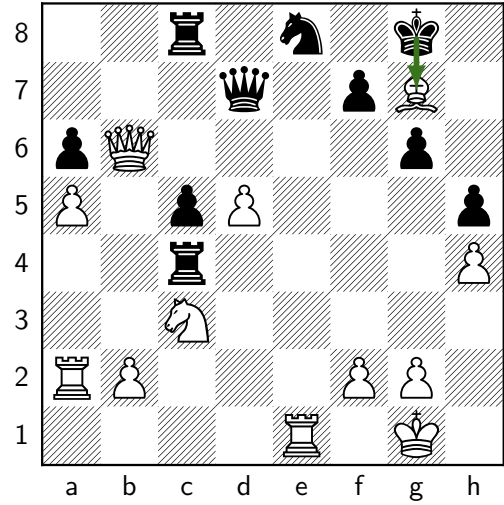
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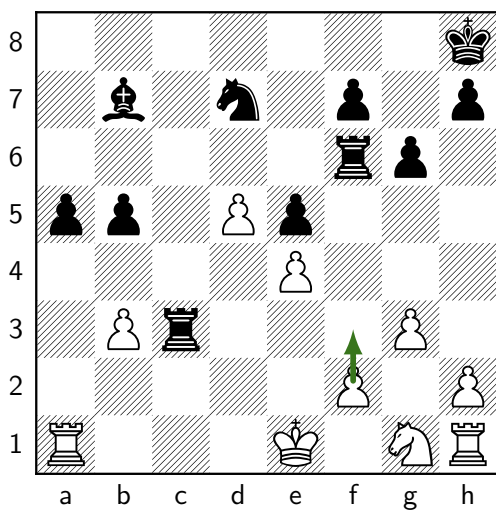


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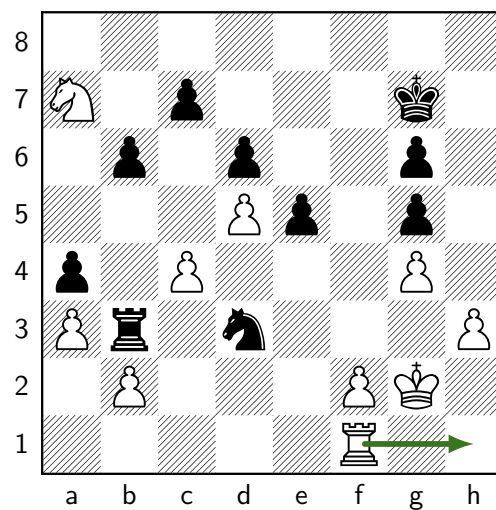


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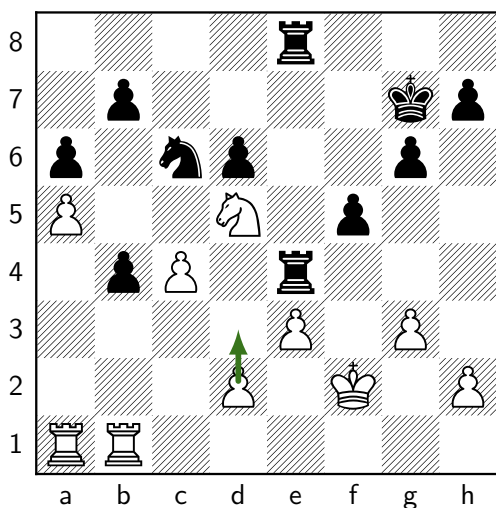
Figure 20: **Move 26, White: Qxc3**. (a) Our position; Melanie plays Qxc3 (c1c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#1777` (ply 50) [structural distance $d=90$, QAP obj=6]: the master played Qxd6; under μ our Qc1 plays the role of that game’s Qd1; cross-type roles: our bb7 as their Ra1, our Pb3 as their Ba2; transferred move c1c3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#3375` (ply 45) [structural distance $d=98$, QAP obj=4]: the master played Re8; under μ our Qc1 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Ra1 as their Qc8, our Ng1 as their Pb7; transferred move c1c3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0019#2263` (ply 53) [structural distance $d=99$, QAP obj=7]: the master played Kxg7; under μ our Ke1 plays the role of that game’s Kg1; cross-type roles: our Qc1 as their Rc8, our Pf2 as their Ne8; transferred move e1e2. Synthesis: 2 retrieved analogs transferred to Qxc3 (summed QAP objective 10, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



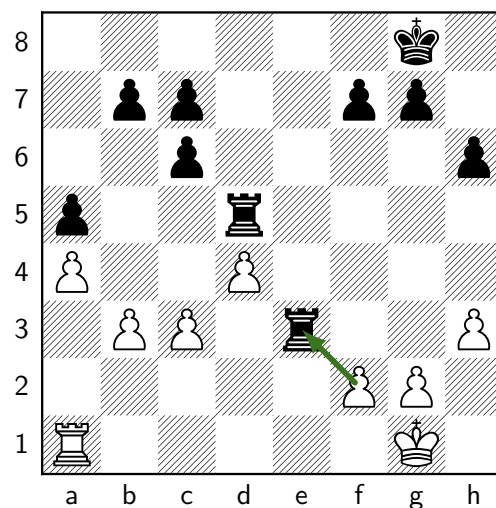
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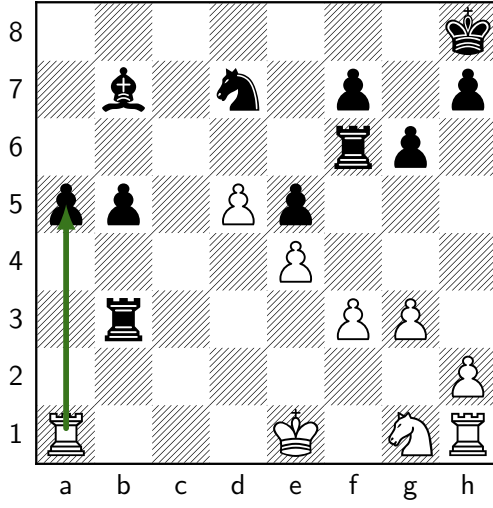


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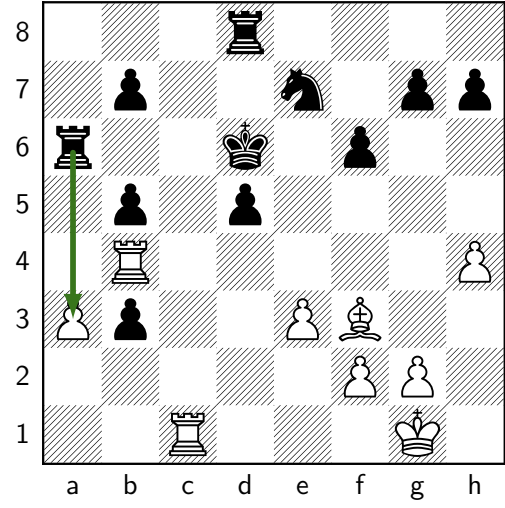


(d)

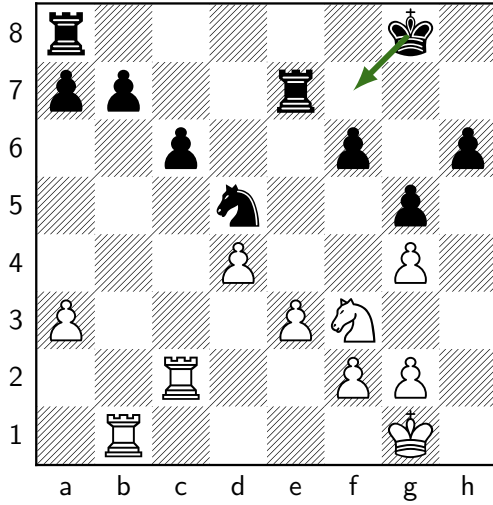
Figure 21: **Move 27, White: f3 (f2f3).** (a) Our position; Melanie plays f3 (f2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0025#791` (ply 64) [structural distance $d=71$, QAP obj=44]: the master played Rh1; under μ our Pf2 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our rc3 as their nd3, our Ra1 as their Ph3; transferred move f2f3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0016#2466` (ply 64) [structural distance $d=69$, QAP obj=40]: the master played d3; under μ our Pf2 plays the role of that game’s Pd2; cross-type roles: our bb7 as their Rb1, our Rh1 as their Ph2; transferred move f2f3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0021#236` (ply 44) [structural distance $d=55$, QAP obj=4]: the master played fxe3; under μ our Pf2 plays the role of that game’s Pf2; cross-type roles: our rf6 as their pa5, our pe5 as their rd5; transferred move f2f3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to f3 (summed QAP objective 88, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



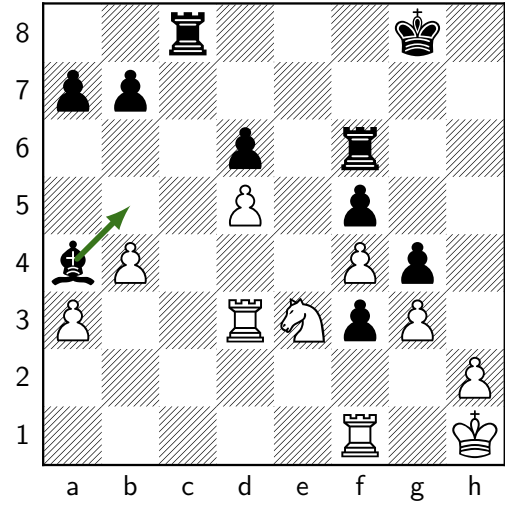
(a)



(b)



(c)



(d)

Figure 22: **Move 28, White: Rxa5**. (a) Our position; Melanie plays Rxa5 (a1a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#3264` (ply 55) [structural distance $d=58$, QAP obj=31]: the master played Rxa3; under μ our *Ra1* plays the role of that game’s *Ra3*; cross-type roles: our *Rh1* as their *Ph7*, our *bb7* as their *rb4*; transferred move **a1a5 (endorses this move)**. (c) Retrieved from `lichess_elite_2024-09_p0019#2694` (ply 47) [structural distance $d=53$, QAP obj=53]: the master played Kf7; under μ our *Ke1* plays the role of that game’s *Kg1*; cross-type roles: our *pg6* as their *Ra8*, our *rf6* as their *Nd5*; transferred move **e1e2**. (d) Retrieved from `lichess_elite_2024-09_p0023#1714` (ply 61) [structural distance $d=58$, QAP obj=47]: the master played Bb5; under μ our *Ph2* plays the role of that game’s *Ba5* (a **cross-type** correspondence); cross-type roles: our *Rh1* as their *Pa7*, our *Pg3* as their *Rf6*; transferred move **h2h4**. Synthesis: 1 retrieved analog transferred to Rxa5 (summed QAP objective 31, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

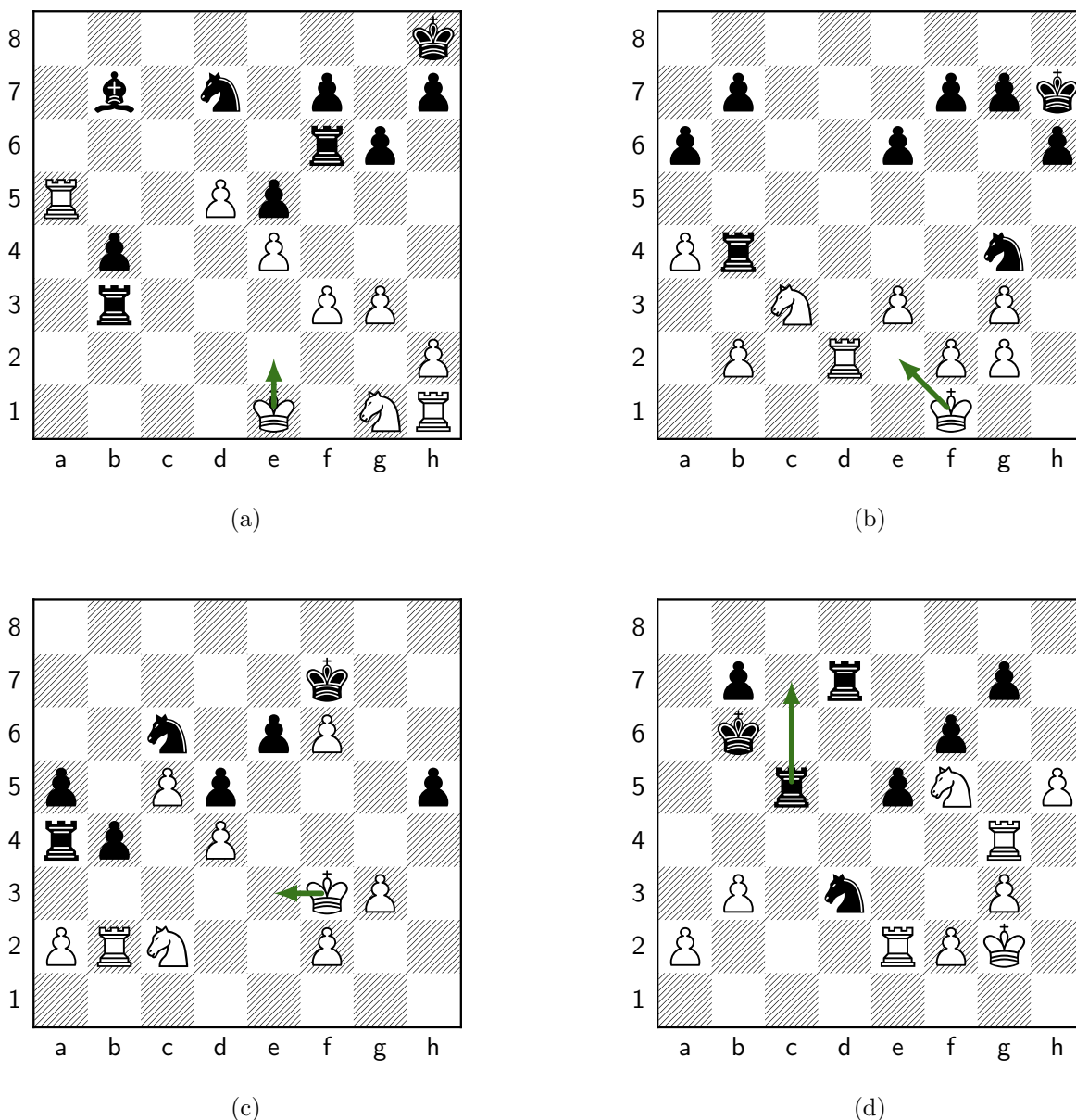
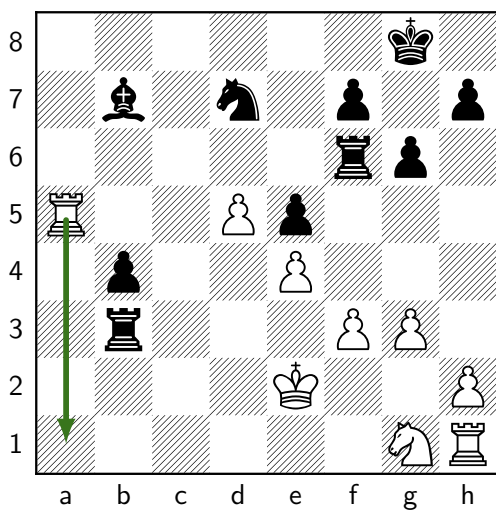
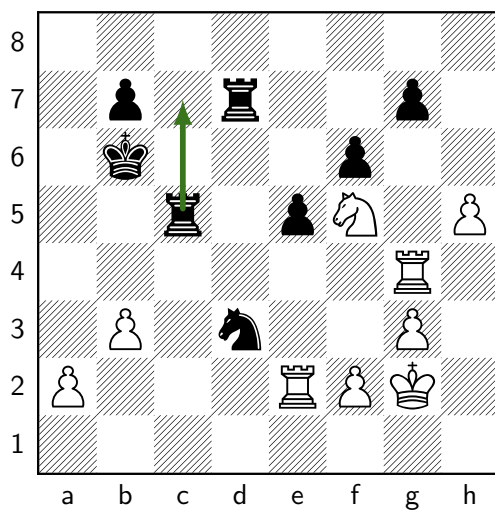


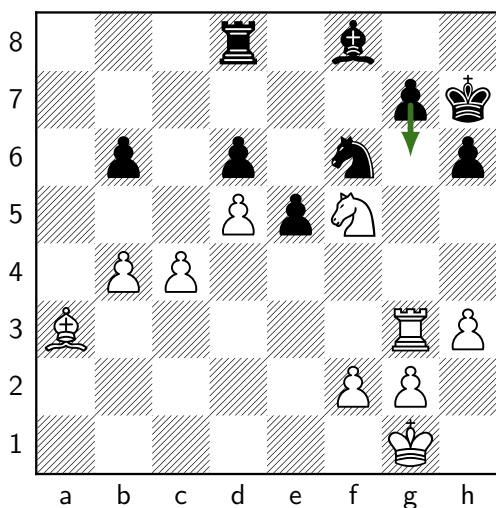
Figure 23: **Move 29, White: Ke2.** (a) Our position; Melanie plays Ke2 (e1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0010#248 (ply 54) [structural distance $d=62$, QAP obj=28]: the master played Ke2; under μ our Ke1 plays the role of that game's Kf1; cross-type roles: our Pe4 as their Nc3, our bb7 as their ng4; transferred move e1e2 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0010#65 (ply 68) [structural distance $d=64$, QAP obj=26]: the master played Ke3; under μ our Ke1 plays the role of that game's Kf3; cross-type roles: our bb7 as their Rb2, our bb7 as their Nc2; transferred move e1e2 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0031#3337 (ply 73) [structural distance $d=59$, QAP obj=52]: the master played Rcc7; under μ our Ra5 plays the role of that game's Rc4; cross-type roles: our pb4 as their Rd7, our rb3 as their nf5; transferred move a5a2. Synthesis: 2 retrieved analogs transferred to Ke2 (summed QAP objective 54, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



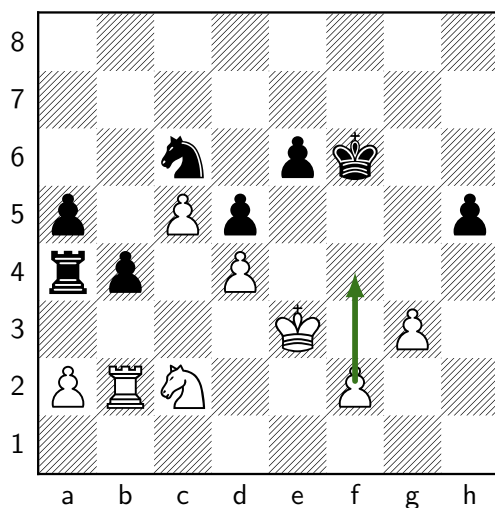
(a)



(b)

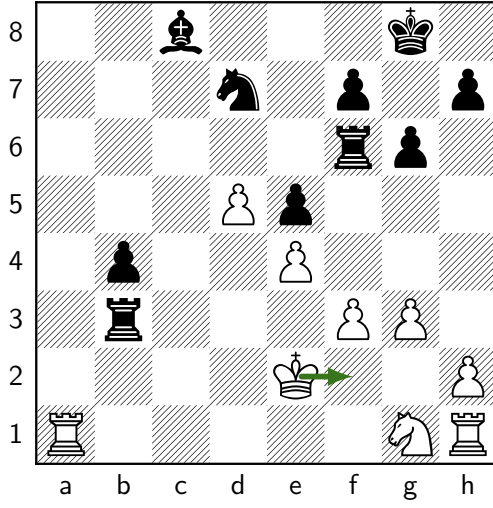


(c)

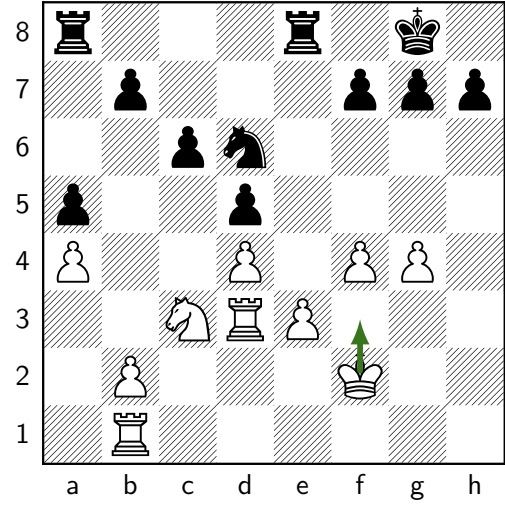


(d)

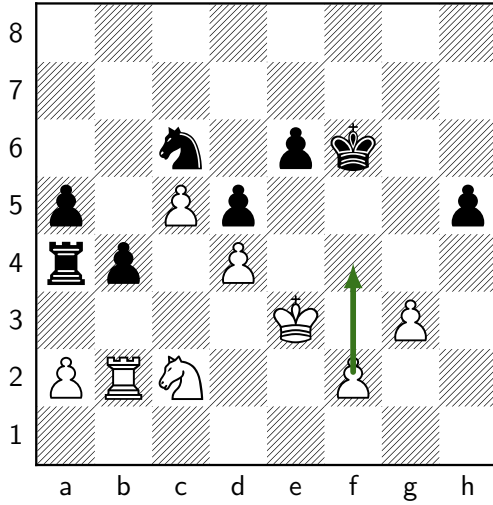
Figure 24: **Move 30, White: Ra1**. (a) Our position; Melanie plays Ra1 (a5a1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#3337` (ply 73) [structural distance $d=54$, QAP obj=53]: the master played Rcc7; under μ our Ra5 plays the role of that game’s Rc4; cross-type roles: our Ng1 as their Pb7, our pb4 as their Rd7; transferred move a5a1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0031#455` (ply 63) [structural distance $d=69$, QAP obj=66]: the master played g6; under μ our Pd5 plays the role of that game’s Pg2; cross-type roles: our Pe4 as their Bf8, our rf6 as their Nf6; transferred move d5d6. (d) Retrieved from `lichess_elite_2024-09_p0010#65` (ply 70) [structural distance $d=65$, QAP obj=54]: the master played f4; under μ our Ng1 plays the role of that game’s Pf2 (a **cross-type** correspondence); cross-type roles: our rf6 as their Nc2, our bb7 as their ra4; transferred move g1h3. Synthesis: 1 retrieved analog transferred to Ra1 (summed QAP objective 53, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



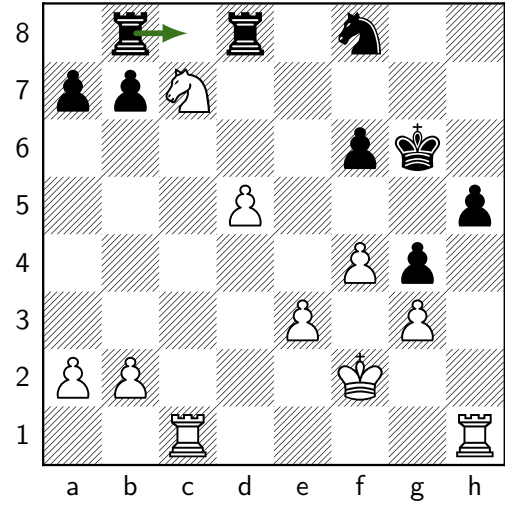
(a)



(b)

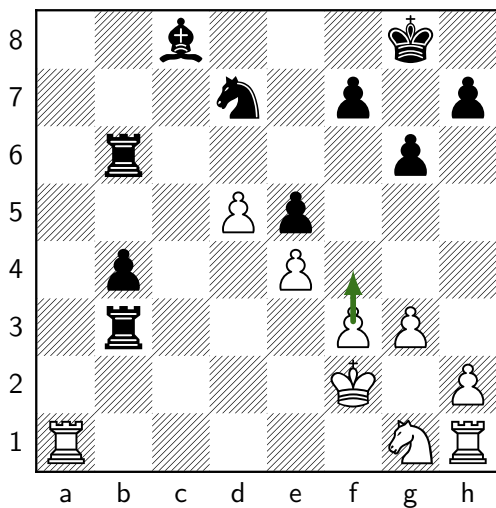


(c)

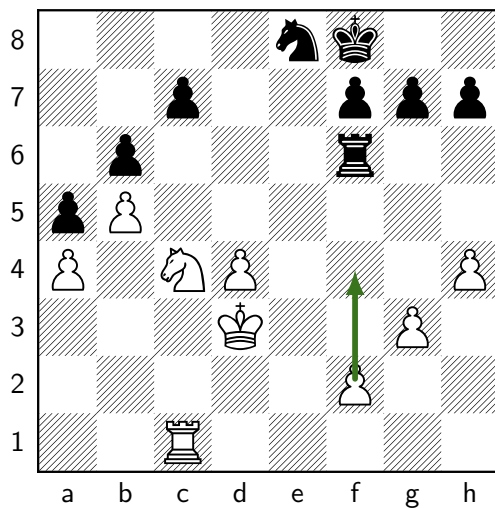


(d)

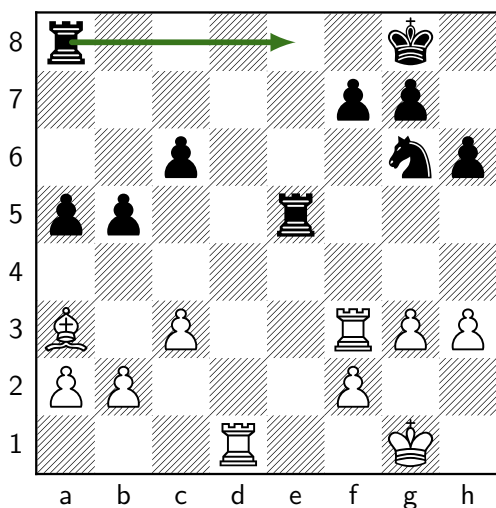
Figure 25: **Move 31, White: Kf2.** (a) Our position; Melanie plays Kf2 (e2f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#3307` (ply 46) [structural distance $d=60$, QAP obj=20]: the master played Kf3; under μ our Ke2 plays the role of that game’s Kf2; cross-type roles: our rf6 as their Nc3, our Ng1 as their Rd3; transferred move e2f2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0010#65` (ply 70) [structural distance $d=67$, QAP obj=36]: the master played f4; under μ our Ng1 plays the role of that game’s Pf2 (a **cross-type** correspondence); cross-type roles: our rb3 as their Nc2, our Ra1 as their ph5; transferred move g1h3. (d) Retrieved from `lichess_elite_2024-09_p0028#3188` (ply 55) [structural distance $d=71$, QAP obj=31]: the master played Rbc8; under μ our Pd5 plays the role of that game’s Rb1 (a **cross-type** correspondence); cross-type roles: our pf7 as their nc7, our Ra1 as their Pg4; transferred move d5d6. Synthesis: 1 retrieved analog transferred to Kf2 (summed QAP objective 20, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



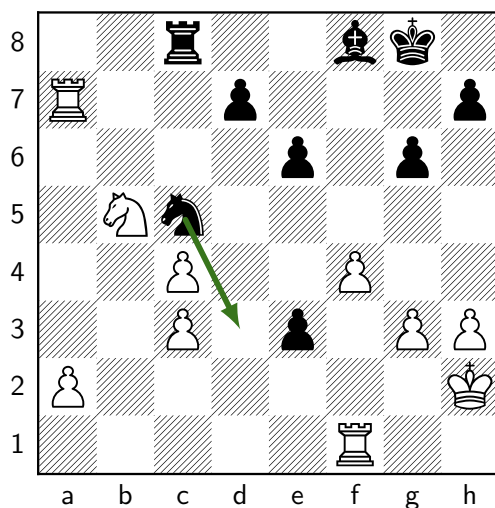
(a)



(b)

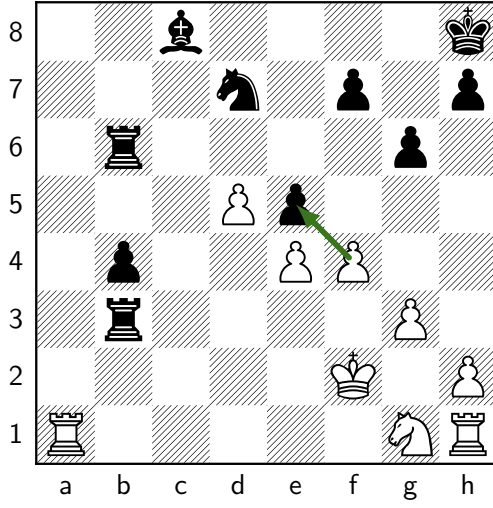


(c)

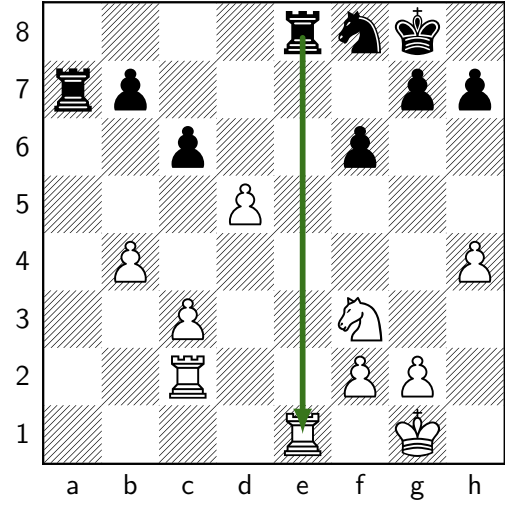


(d)

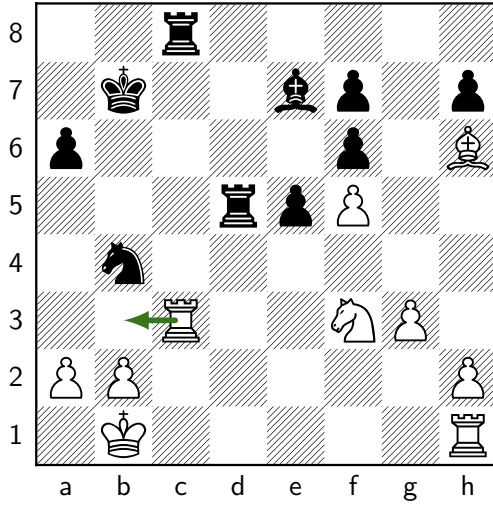
Figure 26: **Move 32, White: f4.** (a) Our position; Melanie plays f4 (f3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#2017` (ply 52) [structural distance $d=61$, QAP obj=18]: the master played f4; under μ our *Pf3* plays the role of that game’s *Pf2*; cross-type roles: our *rb3* as their *Nc4*, our *Ng1* as their *Pd4*; transferred move **f3f4 (endorses this move)**. (c) Retrieved from `lichess_elite_2024-09_p0020#2779` (ply 49) [structural distance $d=61$, QAP obj=23]: the master played *Rae8*; under μ our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *Ng1* as their *Pg7*, our *pg6* as their *Ng6*; transferred move **a1a2**. (d) Retrieved from `lichess_elite_2024-09_p0003#865` (ply 65) [structural distance $d=47$, QAP obj=19]: the master played *Nd3*; under μ our *Pd5* plays the role of that game’s *Nc4* (a **cross-type** correspondence); cross-type roles: our *bc8* as their *Rc8*, our *Ng1* as their *Bf8*; transferred move **d5d6**. Synthesis: 2 retrieved analogs transferred to **f4** (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



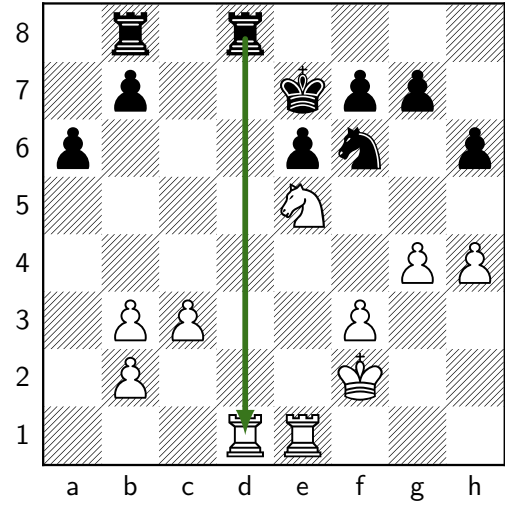
(a)



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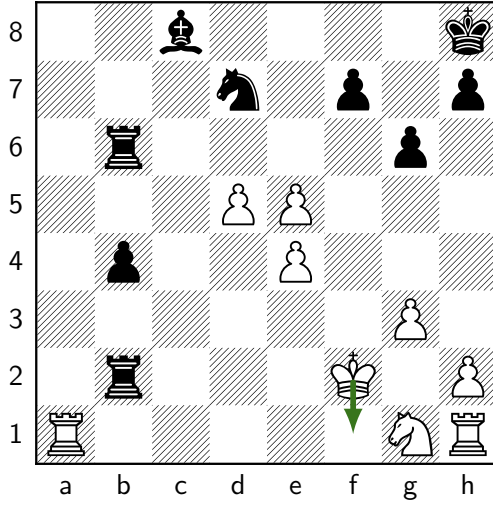


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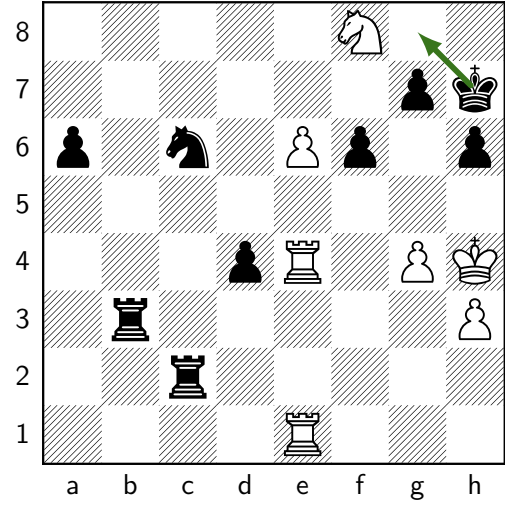


(d)

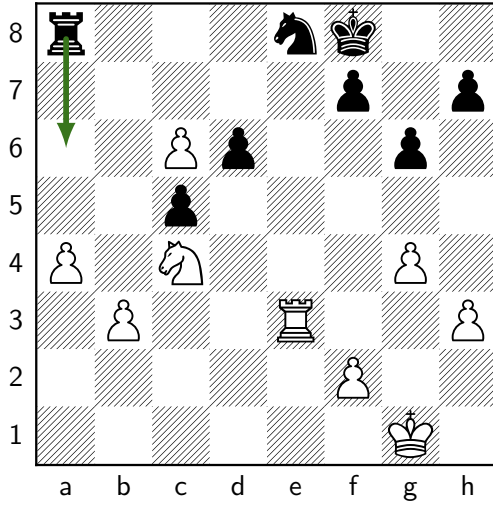
Figure 27: **Move 33, White: fxe5**. (a) Our position; Melanie plays fxe5 (f4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0031#3571 (ply 61) [structural distance $d=57$, QAP obj=56]: the master played Rxe1+; under μ our Pf4 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Pg3 as their Nf8, our rb3 as their Pg7; transferred move f4e5 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0017#3850 (ply 50) [structural distance $d=63$, QAP obj=38]: the master played Rb3; under μ our Pf4 plays the role of that game’s Rc3 (a **cross-type** correspondence); cross-type roles: our Ng1 as their Rh1, our Pd5 as their Nf3; transferred move f4e5 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0019#3571 (ply 55) [structural distance $d=63$, QAP obj=29]: the master played Rxd1; under μ our Pf4 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our rb3 as their Pa6, our rb6 as their Ph6; transferred move f4e5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to fxe5 (summed QAP objective 153, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



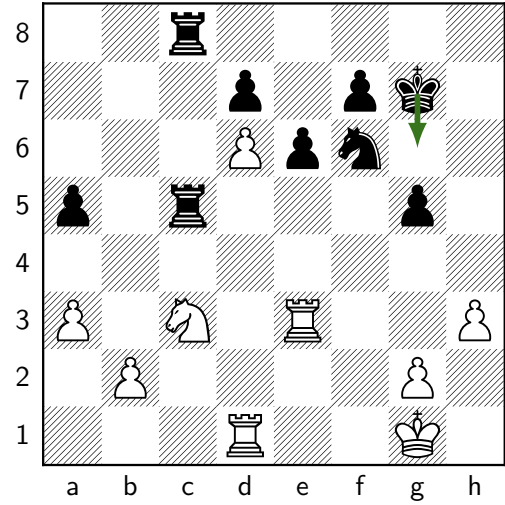
(a)



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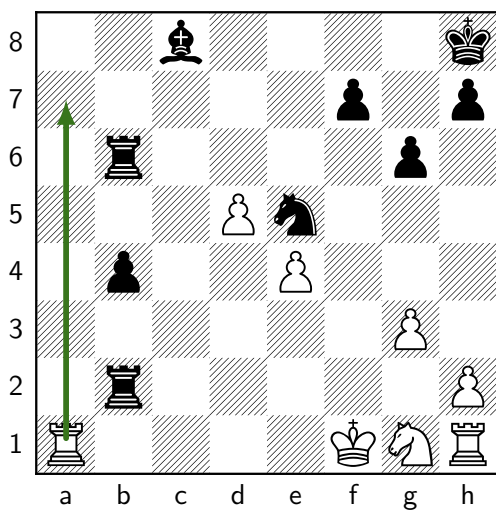


(c)

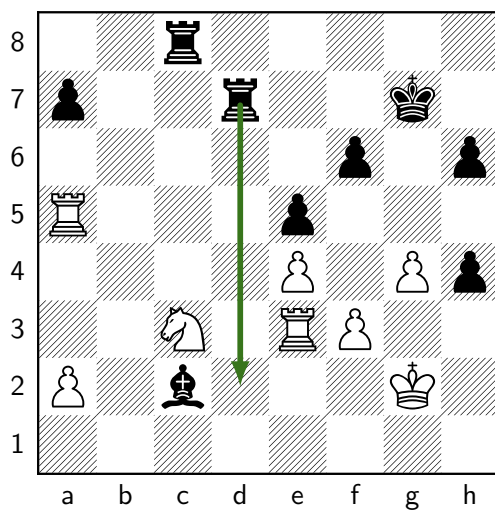


(d)

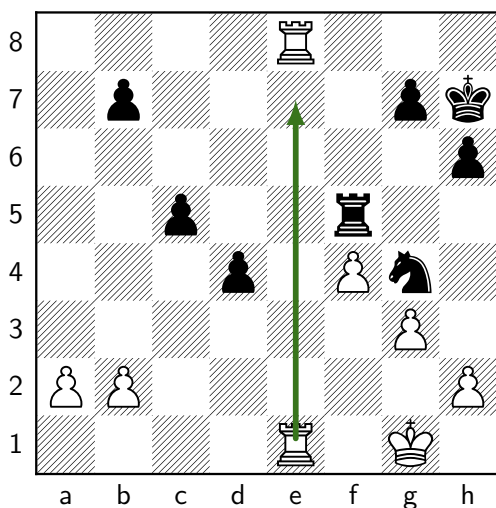
Figure 28: **Move 34, White: Kf1**. (a) Our position; Melanie plays Kf1 (f2f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#3178` (ply 83) [structural distance $d=49$, QAP obj=38]: the master played Kg8; under μ our Kf2 plays the role of that game’s Kh2; cross-type roles: our rb2 as their nf8, our Ng1 as their Pg7; transferred move f2f1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0010#2276` (ply 79) [structural distance $d=54$, QAP obj=5]: the master played Ra6; under μ our Ng1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Pe4 as their Ne8, our Rh1 as their Ph7; transferred move g1e2. (d) Retrieved from `lichess_elite_2024-09_p0013#1516` (ply 57) [structural distance $d=47$, QAP obj=2]: the master played Kg6; under μ our Kf2 plays the role of that game’s Kg2; cross-type roles: our bc8 as their Rc8, our rb6 as their Pe6; transferred move f2f3. Synthesis: 1 retrieved analog transferred to Kf1 (summed QAP objective 38, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



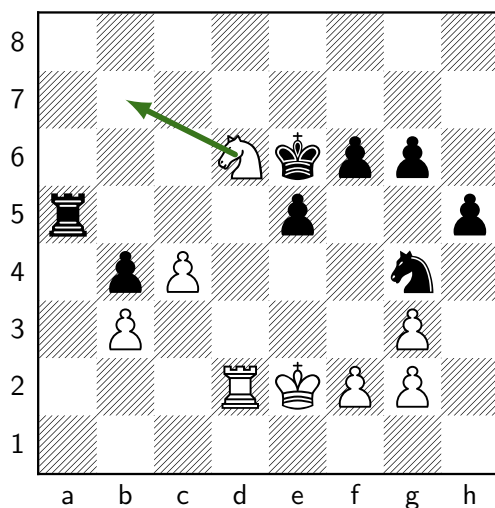
(a)



(b)

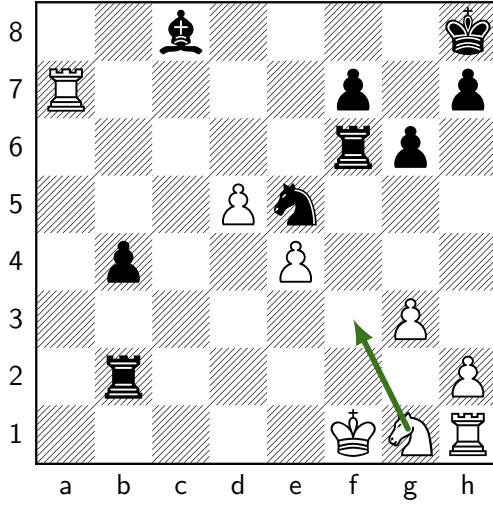


(c)

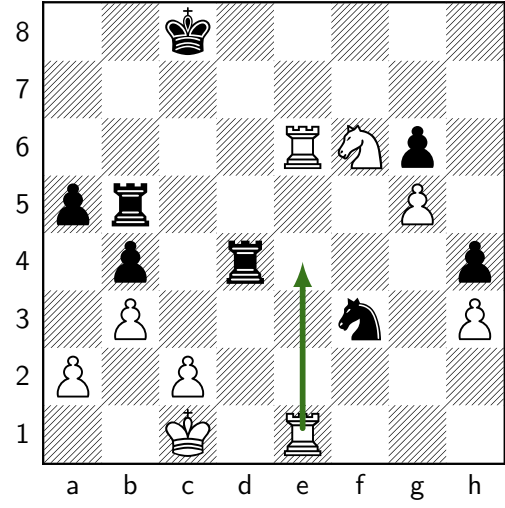


(d)

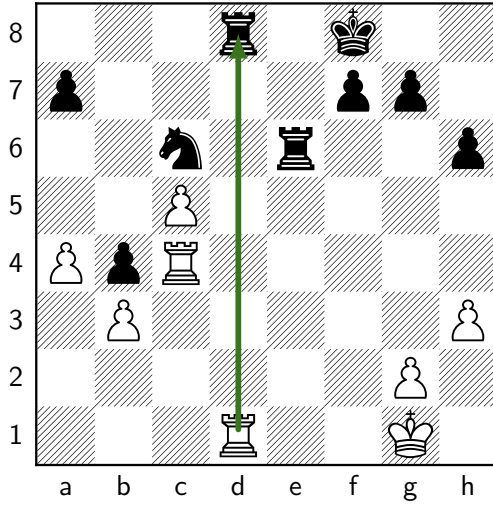
Figure 29: **Move 35, White: Ra7.** (a) Our position; Melanie plays Ra7 (a1a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0012#1277` (ply 59) [structural distance $d=59$, QAP obj=11]: the master played Rd2+; under μ our Ra1 plays the role of that game’s Rd2; cross-type roles: our Ng1 as their Rc8, our Rh1 as their Pf6; transferred move a1a7 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0028#1247` (ply 58) [structural distance $d=59$, QAP obj=4]: the master played R1e7; under μ our Ra1 plays the role of that game’s Re1; cross-type roles: our rb2 as their Pa2, our Ng1 as their pb7; transferred move a1a7 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0003#1459` (ply 82) [structural distance $d=50$, QAP obj=2]: the master played Nb7; under μ our Ra1 plays the role of that game’s Nd6 (a **cross-type** correspondence); cross-type roles: our ph7 as their Rd2, our Ng1 as their Pf2; transferred move a1a7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ra7 (summed QAP objective 16, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



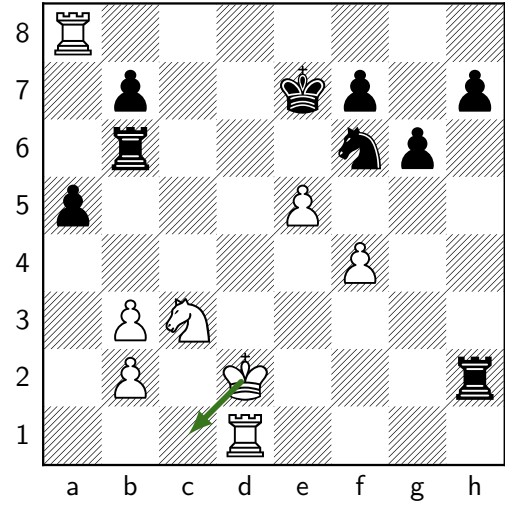
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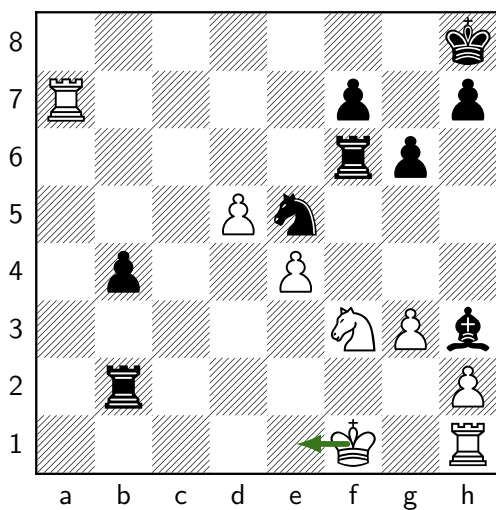


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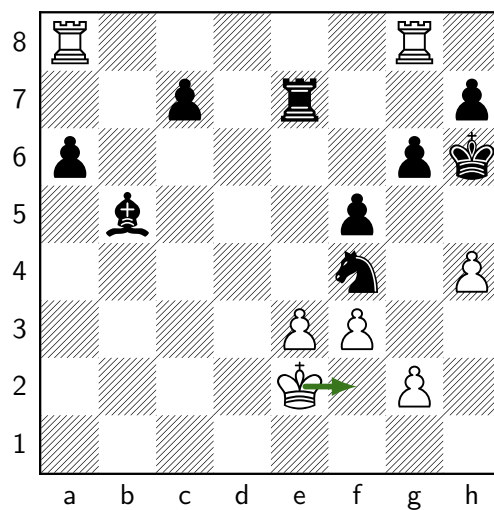


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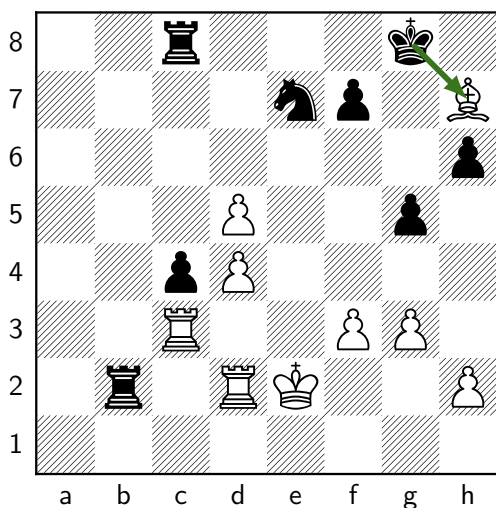
Figure 30: **Move 36, White: Nf3**. (a) Our position; Melanie plays Nf3 (g1f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#3568` (ply 70) [structural distance $d=56$, QAP obj=3]: the master played R1e4; under μ our Ng1 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our rb2 as their Pb3, our Ra7 as their nf3; transferred move g1f3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0021#795` (ply 62) [structural distance $d=57$, QAP obj=2]: the master played Rxd8+; under μ our Ng1 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our bc8 as their Ph3, our rf6 as their nc6; transferred move g1f3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0017#2447` (ply 68) [structural distance $d=60$, QAP obj=55]: the master played Kc1; under μ our Kf1 plays the role of that game’s Kd2; cross-type roles: our ne5 as their Pe5, our bc8 as their nf6; transferred move f1e1. Synthesis: 2 retrieved analogs transferred to Nf3 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



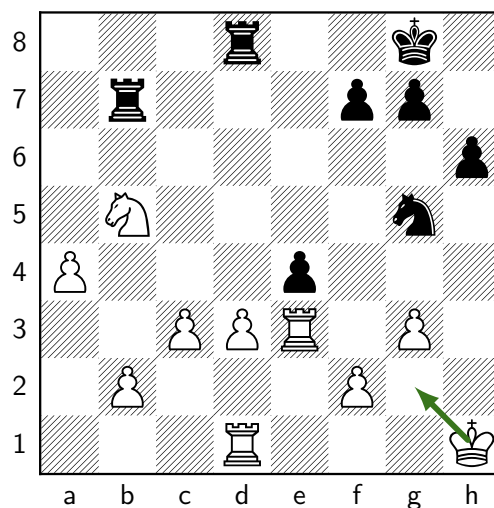
(a)



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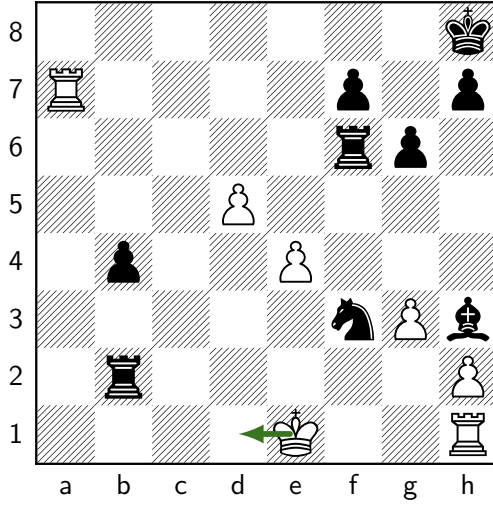


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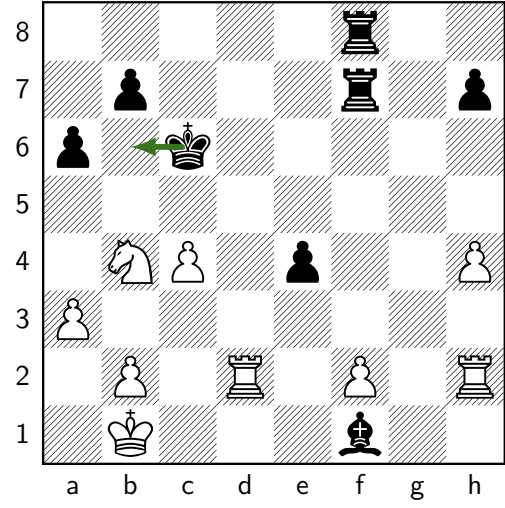


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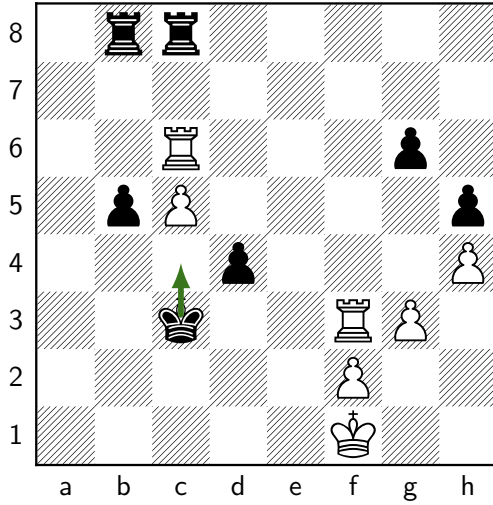
Figure 31: **Move 37, White: Ke1**. (a) Our position; Melanie plays Ke1 (f1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0026#1227` (ply 80) [structural distance $d=67$, QAP obj=113]: the master played Kf2; under μ our Kf1 plays the role of that game’s Ke2; cross-type roles: our Rh1 as their Pg2, our Nf3 as their Pe3; transferred move f1e1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#2177` (ply 73) [structural distance $d=69$, QAP obj=44]: the master played Kxh7; under μ our Kf1 plays the role of that game’s Kg1; cross-type roles: our Pd5 as their Ne7, our ne5 as their pd5; transferred move f1e1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0023#2012` (ply 74) [structural distance $d=65$, QAP obj=16]: the master played Kg2; under μ our Kf1 plays the role of that game’s Kh1; cross-type roles: our bh3 as their Pb2, our ne5 as their Pd3; transferred move f1e1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ke1 (summed QAP objective 174, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



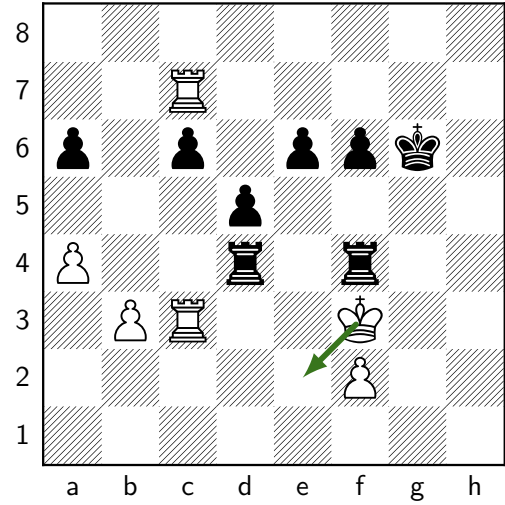
(a)



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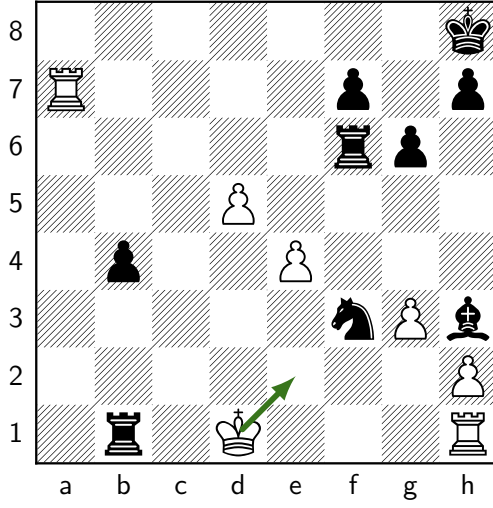


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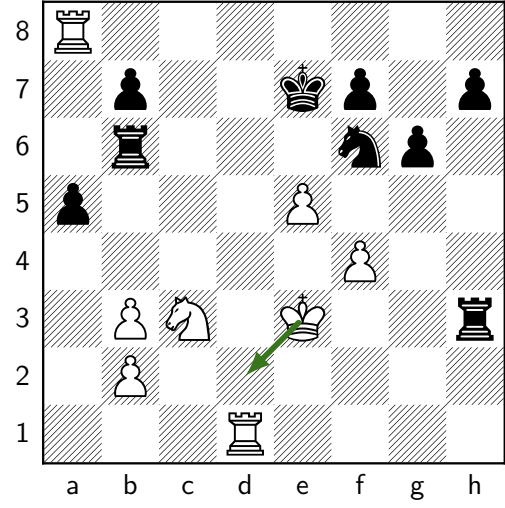


(d)

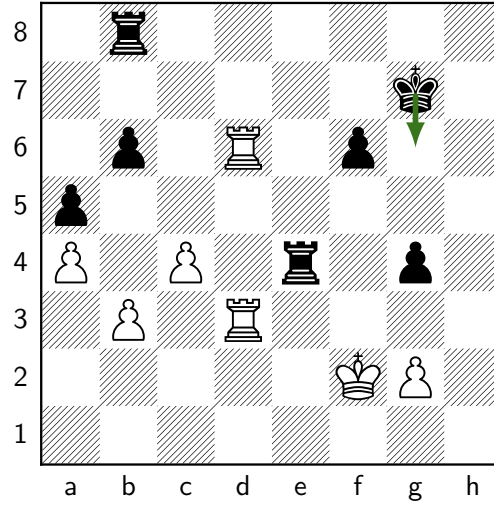
Figure 32: **Move 38, White: Kd1**. (a) Our position; Melanie plays Kd1 (e1d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0025#619` (ply 75) [structural distance $d=51$, QAP obj=68]: the master played Kb6; under μ our *Ke1* plays the role of that game’s *Kc3*; cross-type roles: our *rb2* as their *pc4*, our *bh3* as their *rh2*; transferred move e1d1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0020#3571` (ply 77) [structural distance $d=51$, QAP obj=47]: the master played Kc4; under μ our *Ke1* plays the role of that game’s *Kc6*; cross-type roles: our *bh3* as their *Rc8*, our *Rh1* as their *Pb5*; transferred move e1d1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0025#206` (ply 74) [structural distance $d=49$, QAP obj=38]: the master played Ke2; under μ our *Ke1* plays the role of that game’s *Kf3*; cross-type roles: our *bh3* as their *Pf2*, our *nf3* as their *rf4*; transferred move e1d1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kd1 (summed QAP objective 155, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



(a)

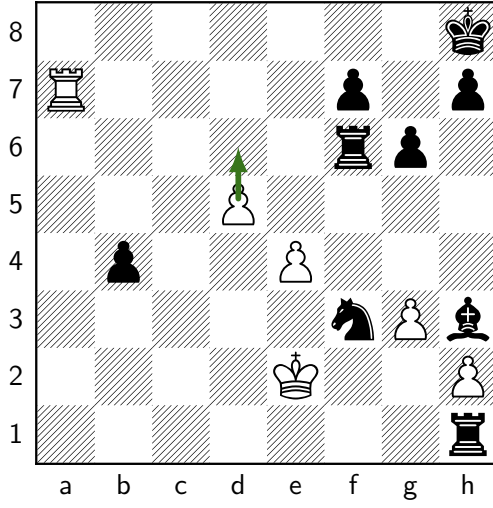


(b)

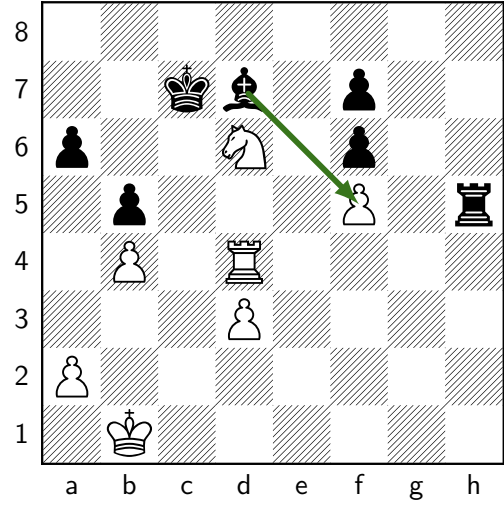


(c)

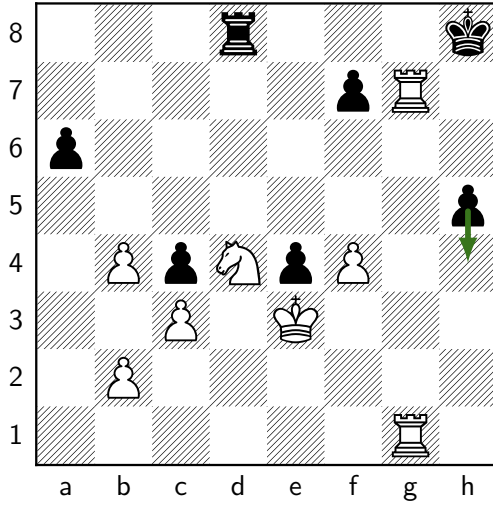
Figure 33: **Move 39, White: Ke2.** (a) Our position; Melanie plays Ke2 (d1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#2447` (ply 66) [structural distance $d=47$, QAP obj=46]: the master played Kd2; under μ our Kd1 plays the role of that game’s Ke3; cross-type roles: our bh3 as their Nc3, our nf3 as their Pf4; transferred move d1e2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0019#873` (ply 89) [structural distance $d=47$, QAP obj=4]: the master played Kg6; under μ our Kd1 plays the role of that game’s Kg2; cross-type roles: our nf3 as their rd6, our bh3 as their pa4; transferred move d1c2. Synthesis: 1 retrieved analog transferred to Ke2 (summed QAP objective 46, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



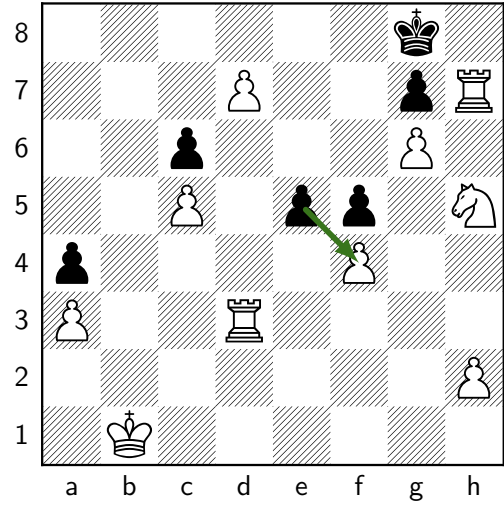
(a)



(b)

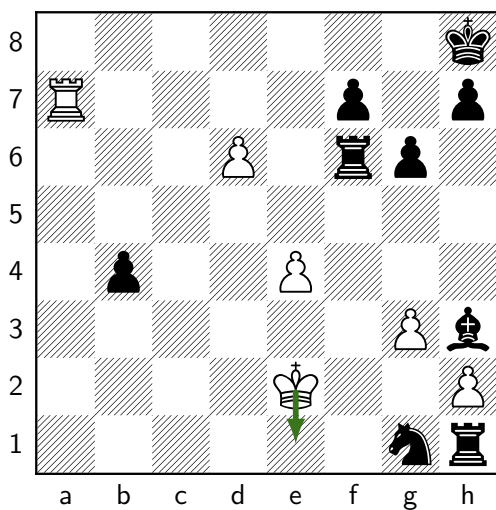


(c)

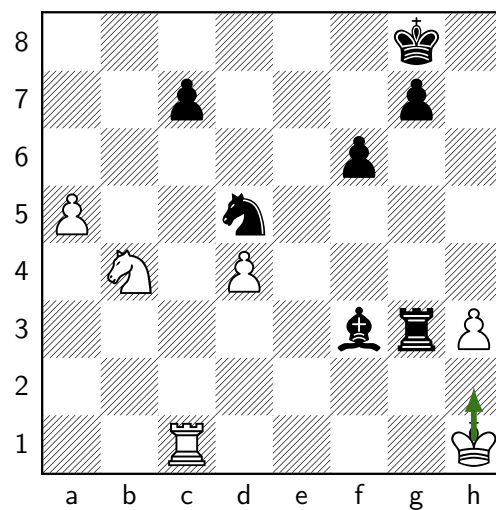


(d)

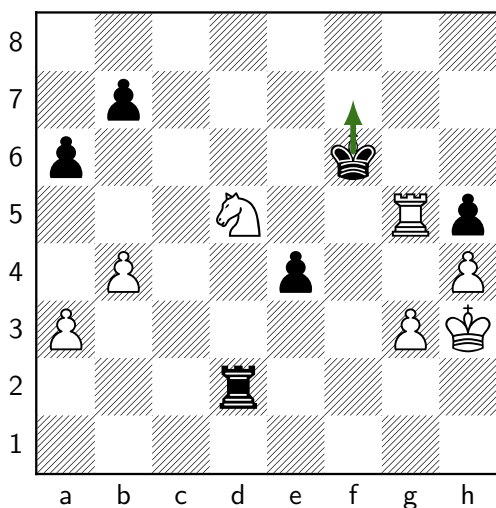
Figure 34: **Move 40, White: d6.** (a) Our position; Melanie plays d6 (d5d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#2847` (ply 77) [structural distance $d=43$, QAP obj=42]: the master played Bxf5; under μ our Pd5 plays the role of that game’s Bd2 (a **cross-type** correspondence); cross-type roles: our bh3 as their pf5; transferred move d5d6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0022#2779` (ply 79) [structural distance $d=41$, QAP obj=24]: the master played h4; under μ our Pg3 plays the role of that game’s Ph4; cross-type roles: our nf3 as their rg7, our bh3 as their Pa6; transferred move g3g4. (d) Retrieved from `lichess_elite_2024-09_p0026#3173` (ply 83) [structural distance $d=41$, QAP obj=18]: the master played exf4; under μ our Pe4 plays the role of that game’s Pe4; cross-type roles: our rh1 as their pd7, our nf3 as their rh7; transferred move e4e5. Synthesis: 1 retrieved analog transferred to d6 (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



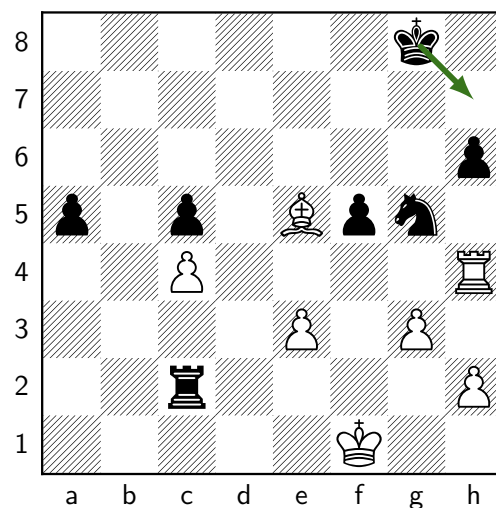
(a)



(b)

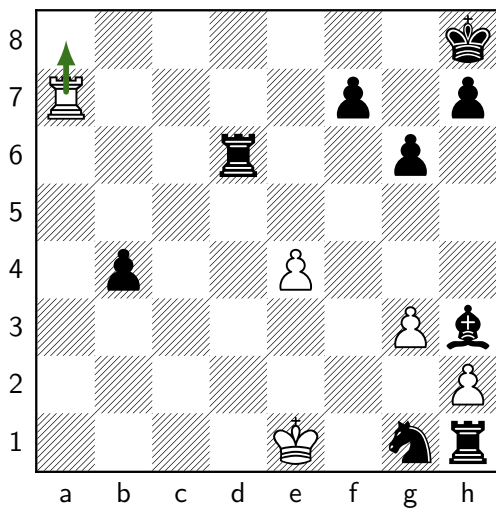


(c)

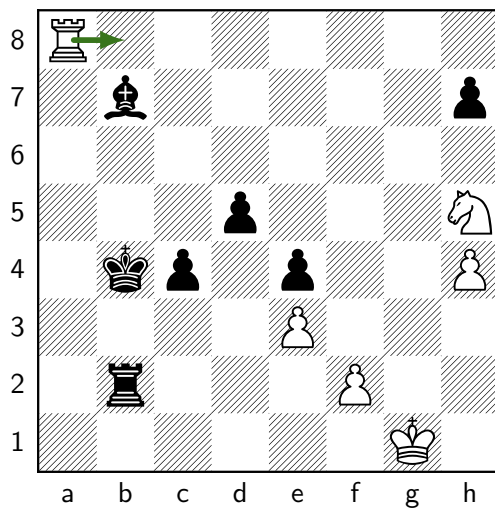


(d)

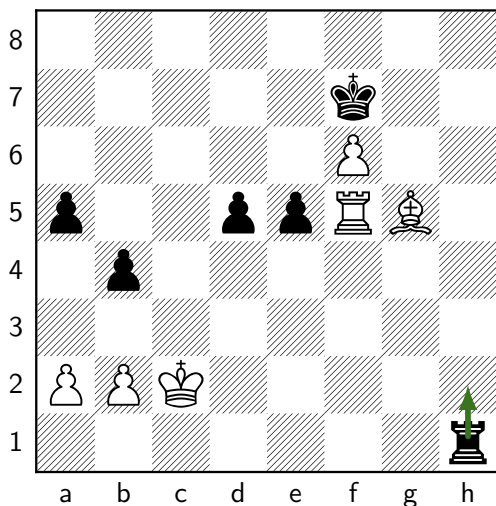
Figure 35: **Move 41, White: Ke1**. (a) Our position; Melanie plays Ke1 (e2e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#1516` (ply 84) [structural distance $d=34$, QAP obj=70]: the master played Kh2; under μ our Ke2 plays the role of that game’s Kh1; cross-type roles: our ng1 as their bf3, our bh3 as their rg3; transferred move e2e1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0023#1039` (ply 97) [structural distance $d=34$, QAP obj=69]: the master played Kf7; under μ our Ke2 plays the role of that game’s Kf3; cross-type roles: our rh1 as their Pe4, our bh3 as their ph4; transferred move e2e1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0016#1533` (ply 73) [structural distance $d=39$, QAP obj=5]: the master played Kh7; under μ our Ke2 plays the role of that game’s Kg1; cross-type roles: our rh1 as their Pa5, our bh3 as their Pc5; transferred move e2e1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ke1 (summed QAP objective 143, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



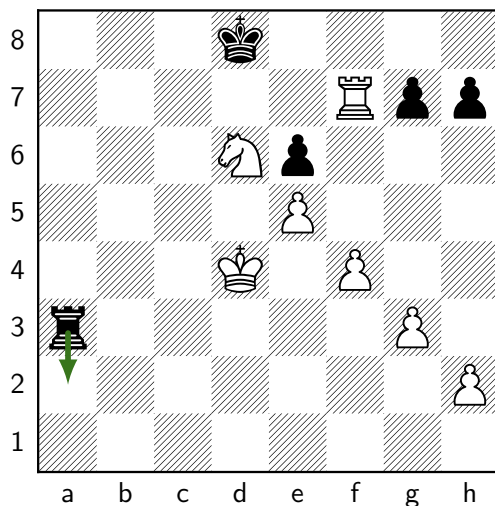
(a)



(b)

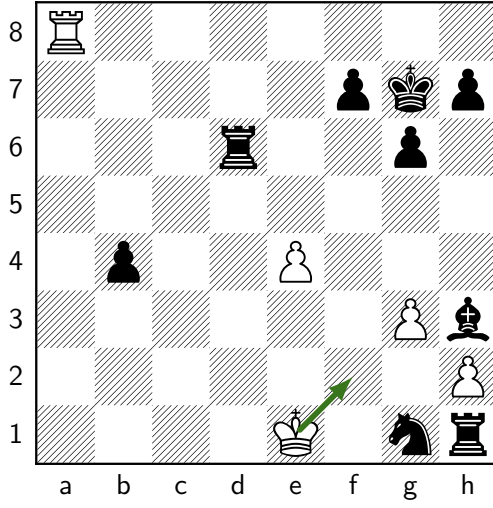


(c)

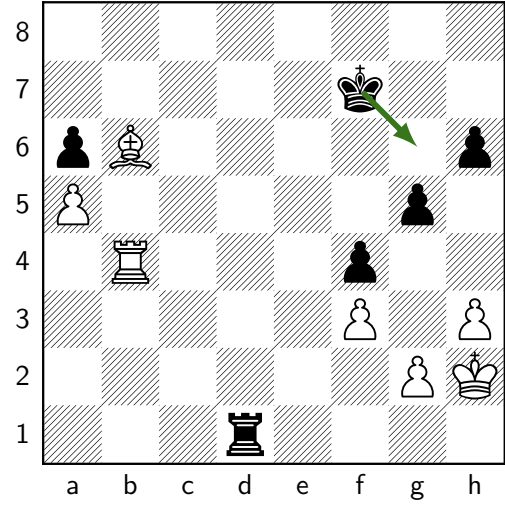


(d)

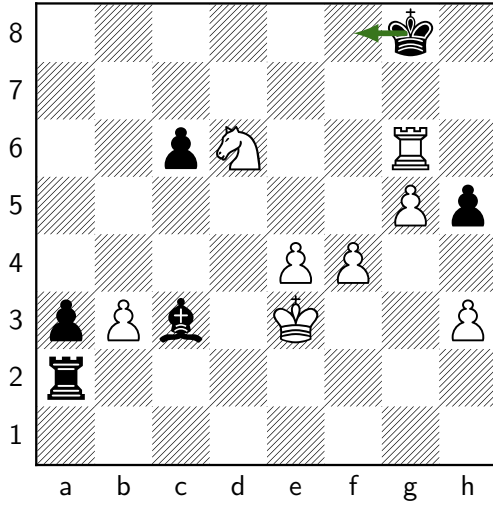
Figure 36: **Move 42, White: Ra8+.** (a) Our position; Melanie plays Ra8+ (a7a8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0012#3568` (ply 86) [structural distance $d=30$, QAP obj=3]: the master played Rb8; under μ our Ra7 plays the role of that game’s Ra8; cross-type roles: our ng1 as their rb2, our bh3 as their Pf2; transferred move a7a8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0012#1421` (ply 91) [structural distance $d=27$, QAP obj=23]: the master played Rh2+; under μ our Pg3 plays the role of that game’s Rh8 (a **cross-type** correspondence); cross-type roles: our bh3 as their pf6, our ng1 as their bg5; transferred move g3g4. (d) Retrieved from `lichess_elite_2024-09_p0019#2197` (ply 89) [structural distance $d=28$, QAP obj=22]: the master played Ra2; under μ our Ra7 plays the role of that game’s Ra6; cross-type roles: our bh3 as their nd6, our ng1 as their pe5; transferred move a7c7. Synthesis: 1 retrieved analog transferred to Ra8+ (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



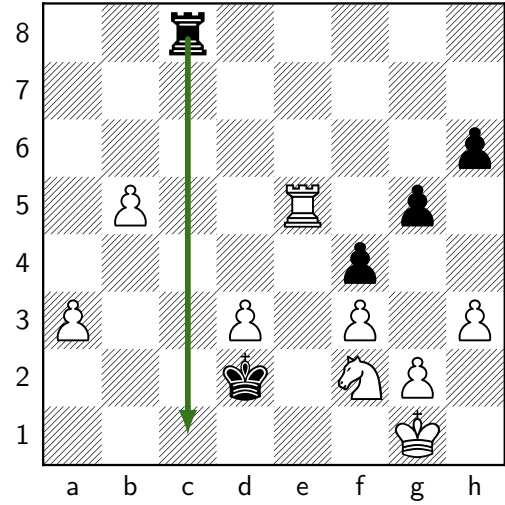
(a)



(b)

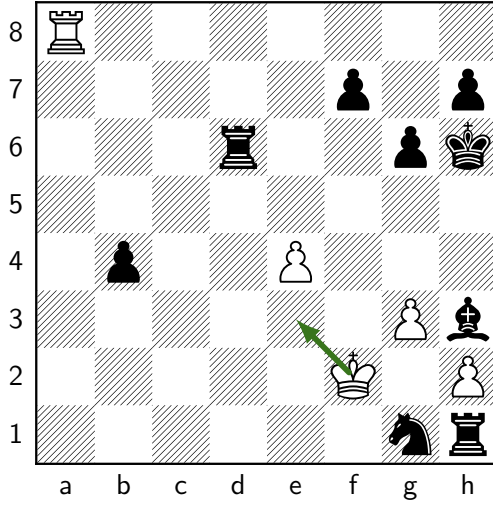


(c)

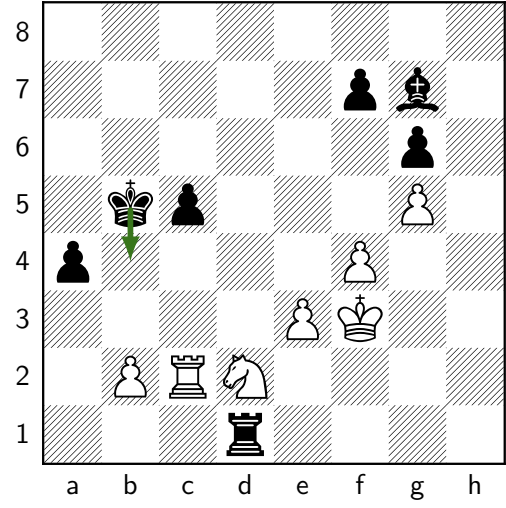


(d)

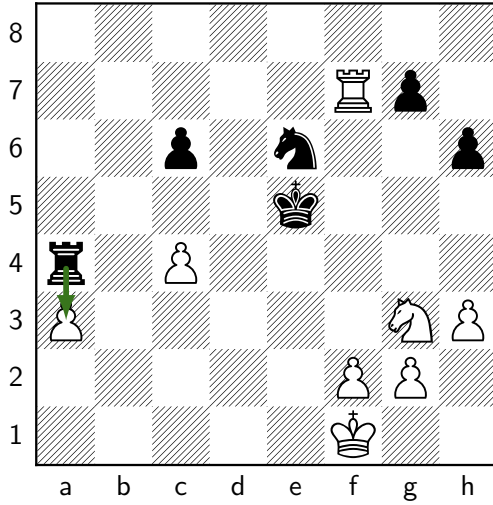
Figure 37: **Move 43, White: Kf2.** (a) Our position; Melanie plays Kf2 (e1f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#1837` (ply 85) [structural distance $d=36$, QAP obj=10]: the master played Kg6; under μ our *Ke1* plays the role of that game’s *Kf2*; cross-type roles: our *ng1* as their *bb6*, our *bh3* as their *pa5*; transferred move e1f2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#657` (ply 93) [structural distance $d=31$, QAP obj=5]: the master played Kf8; under μ our *Ke1* plays the role of that game’s *Kg1*; cross-type roles: our *rd6* as their *nd6*, our *ng1* as their *rg6*; transferred move e1f2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0016#355` (ply 79) [structural distance $d=29$, QAP obj=13]: the master played Rc1+; under μ our *Ra8* plays the role of that game’s *Rc1*; cross-type roles: our *rd6* as their *pb5*, our *ng1* as their *pa3*; transferred move a8b8. Synthesis: 2 retrieved analogs transferred to Kf2 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



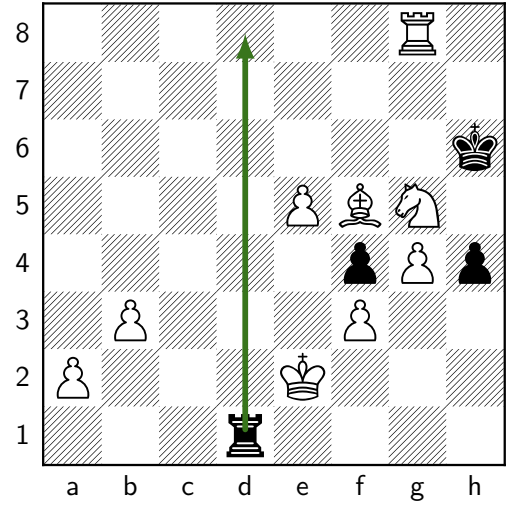
(a)



(b)

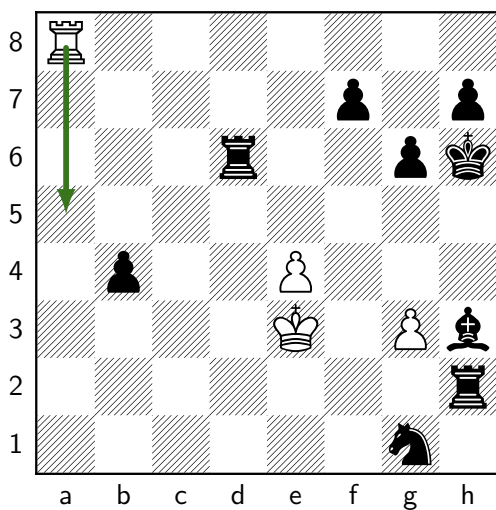


(c)

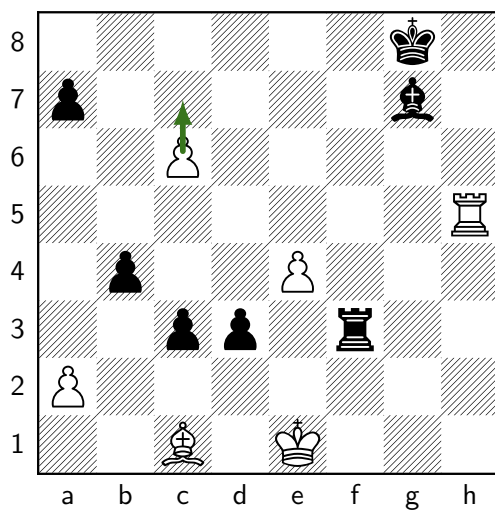


(d)

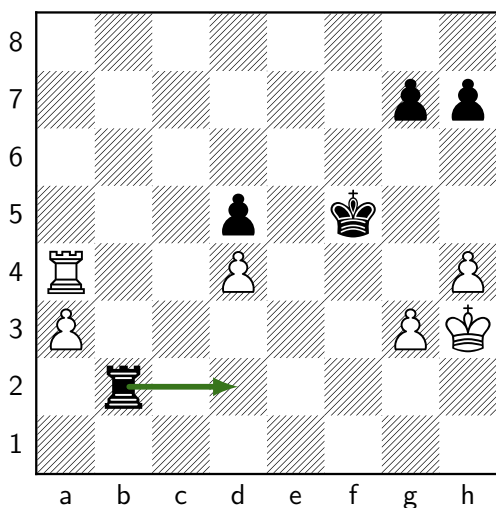
Figure 38: **Move 44, White: Ke3 (f2e3).** (a) Our position; Melanie plays Ke3 (f2e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0026#1032` (ply 87) [structural distance $d=33$, QAP obj=17]: the master played Kb4; under μ our Kf2 plays the role of that game’s Kb4; cross-type roles: our Pe4 as their Bg7, our rd6 as their Pg6; transferred move f2e3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#245` (ply 71) [structural distance $d=34$, QAP obj=9]: the master played Rxa3; under μ our Kf2 plays the role of that game’s Ra5 (a **cross-type** correspondence); cross-type roles: our pb4 as their Ne6, our bh3 as their pc4; transferred move f2e3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0010#753` (ply 103) [structural distance $d=32$, QAP obj=16]: the master played Rd8; under μ our Ra8 plays the role of that game’s Rd8; cross-type roles: our rd6 as their pe5; transferred move a8a1. Synthesis: 2 retrieved analogs transferred to Ke3 (summed QAP objective 26, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



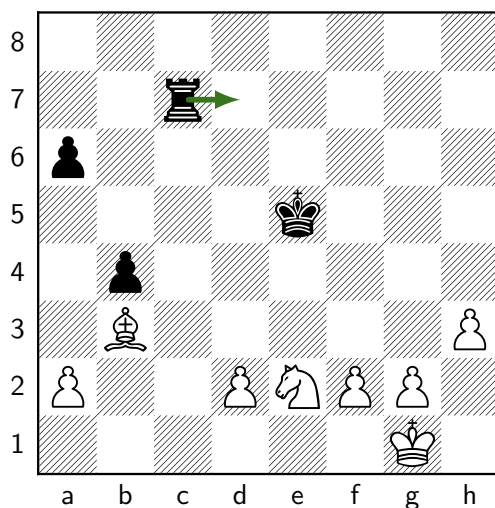
(a)



(b)

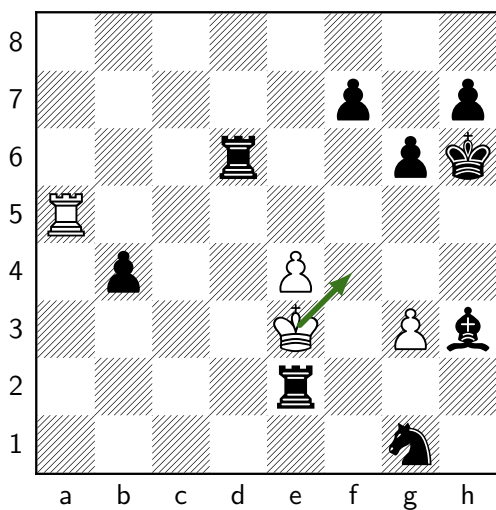


(c)

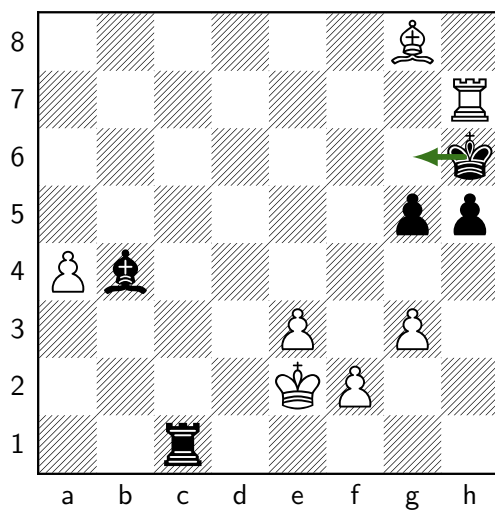


(d)

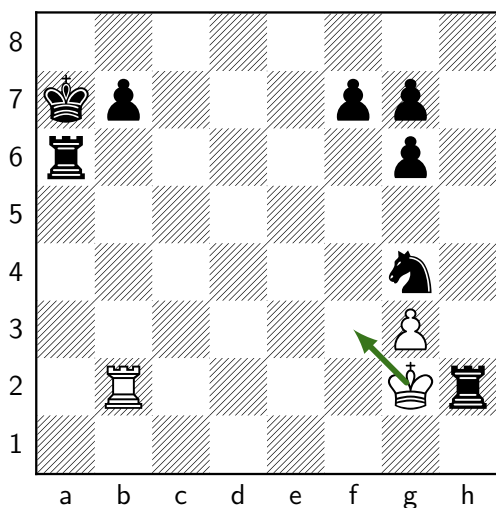
Figure 39: **Move 45, White: Ra5.** (a) Our position; Melanie plays Ra5 (a8a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#3702` (ply 86) [structural distance $d=33$, QAP obj=3]: the master played c7; under μ our $Pg3$ plays the role of that game’s $Pc6$; cross-type roles: our $Pe4$ as their $Bc1$, our $Ra8$ as their $Pa2$; transferred move g3g4. (c) Retrieved from `lichess_elite_2024-09_p0010#3001` (ply 79) [structural distance $d=32$, QAP obj=3]: the master played Rd2; under μ our $Ra8$ plays the role of that game’s $Rb7$; cross-type roles: our $ng1$ as their $Pg7$, our $rh2$ as their $Ph7$; transferred move a8a7. (d) Retrieved from `lichess_elite_2024-09_p0018#1598` (ply 91) [structural distance $d=33$, QAP obj=2]: the master played Rd7; under μ our $Ra8$ plays the role of that game’s $Rc2$; cross-type roles: our $ng1$ as their $Pa6$, our $rd6$ as their $bb3$; transferred move a8a4. Synthesis: no retrieved analog transferred to Ra5 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



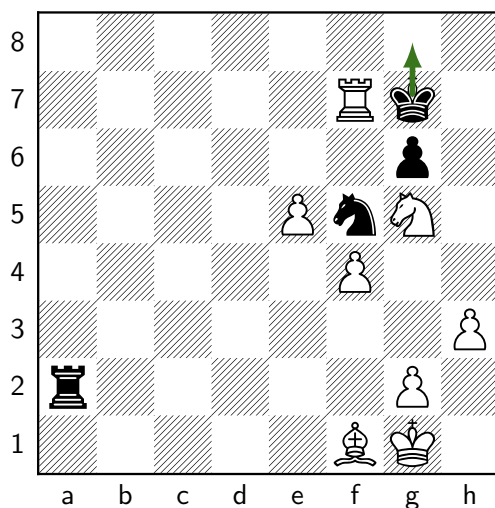
(a)



(b)

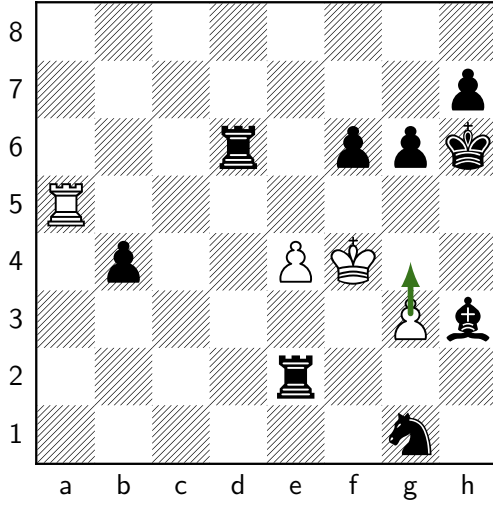


(c)

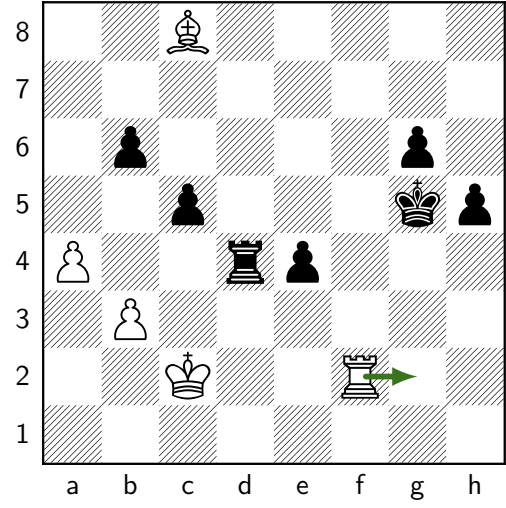


(d)

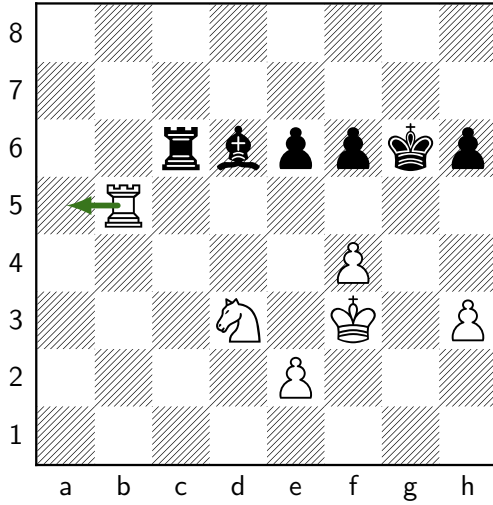
Figure 40: **Move 46, White: Kf4**. (a) Our position; Melanie plays Kf4 (e3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#92` (ply 85) [structural distance $d=35$, QAP obj=72]: the master played Kg6; under μ our Ke3 plays the role of that game’s Kh3; cross-type roles: our ng1 as their bg8, our rd6 as their Ph5; transferred move e3f4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0003#3510` (ply 96) [structural distance $d=38$, QAP obj=70]: the master played Kf3; under μ our Ke3 plays the role of that game’s Kg2; cross-type roles: our ph7 as their ra6, our bh3 as their pg6; transferred move e3f4 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#3144` (ply 99) [structural distance $d=36$, QAP obj=70]: the master played Kg8; under μ our Ke3 plays the role of that game’s Kg2; cross-type roles: our rd6 as their pe5, our pg6 as their Nf5; transferred move e3f4 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kf4 (summed QAP objective 458, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



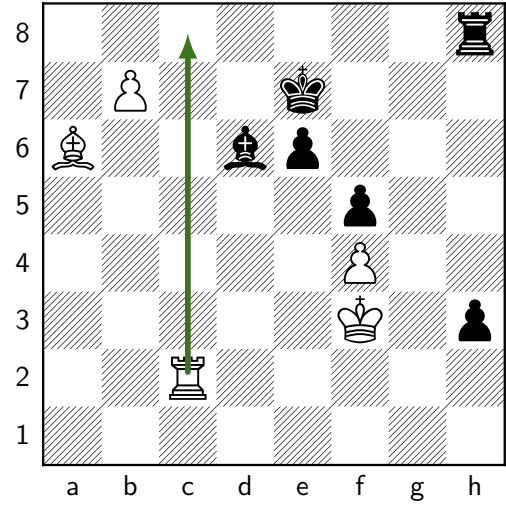
(a)



(b)



(c)



(d)

Figure 41: **Move 47, White: g4**. (a) Our position; Melanie plays g4 (g3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#1003` (ply 100) [structural distance $d=46$, QAP obj=16]: the master played Rg2+; under μ our Pg3 plays the role of that game’s Rf2 (a **cross-type** correspondence); cross-type roles: our ng1 as their pc5, our bh3 as their pb6; transferred move g3g4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#313` (ply 106) [structural distance $d=47$, QAP obj=24]: the master played Ra5; under μ our Ra5 plays the role of that game’s Rb5; cross-type roles: our bh3 as their Nd3, our ng1 as their rc6; transferred move a5b5. (d) Retrieved from `lichess_elite_2024-09_p0026#1933` (ply 80) [structural distance $d=46$, QAP obj=21]: the master played Rc8; under μ our Ra5 plays the role of that game’s Rc2; cross-type roles: our ng1 as their Ba6, our re2 as their bd6; transferred move a5a6. Synthesis: 1 retrieved analog transferred to g4 (summed QAP objective 16, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

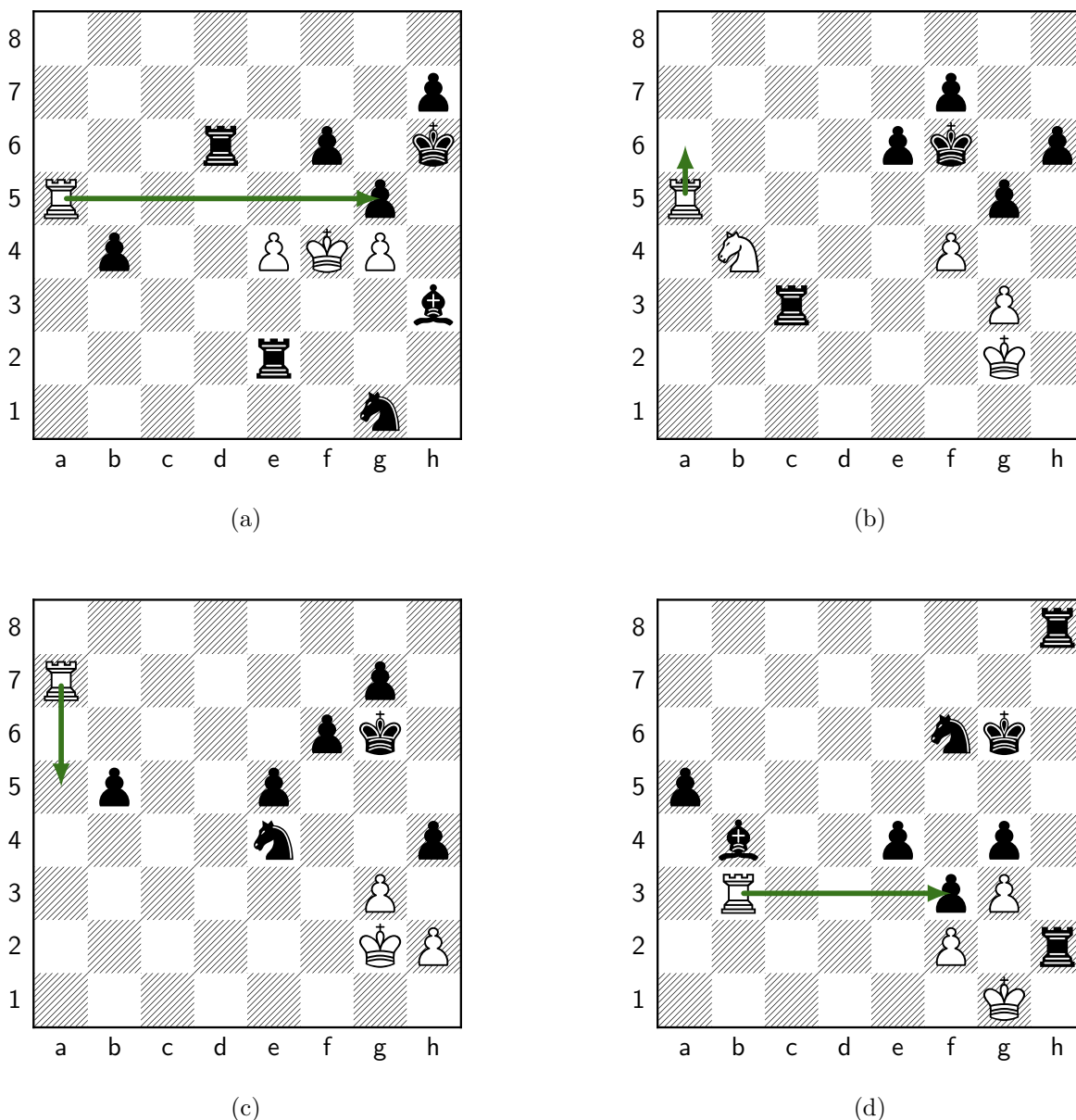
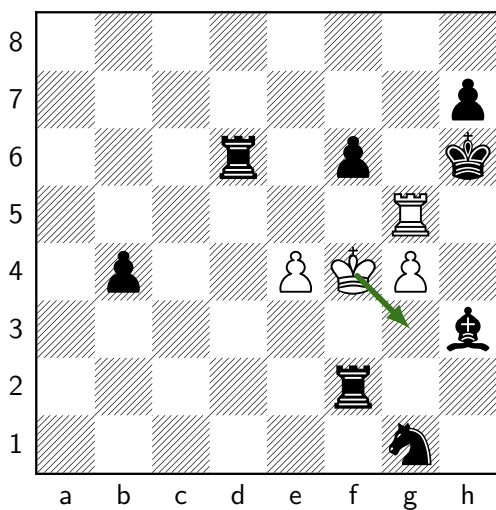
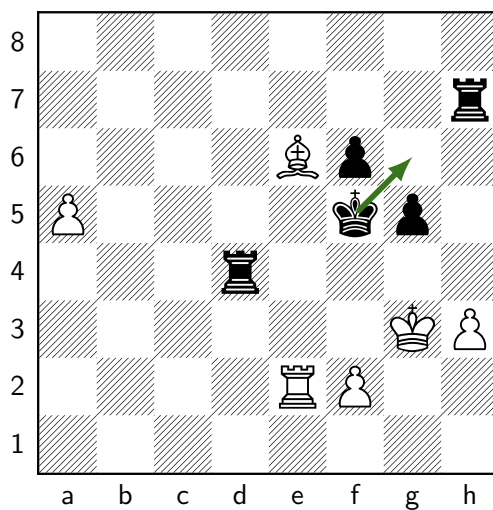


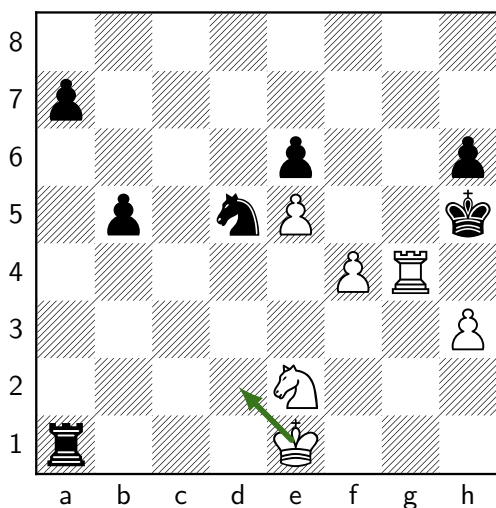
Figure 42: **Move 48, White: $Rxg5$.** (a) Our position; Melanie plays $Rxg5$ (a5g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0016#1251` (ply 88) [structural distance $d=45$, QAP obj=4]: the master played $Ra6$; under μ our $Ra5$ plays the role of that game's $Ra5$; cross-type roles: our $bh3$ as their $Pg3$; transferred move a5g5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#1632` (ply 84) [structural distance $d=43$, QAP obj=4]: the master played $Ra5$; under μ our $Ra5$ plays the role of that game's $Ra7$; cross-type roles: our $bh3$ as their $ne4$, our $ng1$ as their $pe5$; transferred move a5g5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0018#2849` (ply 98) [structural distance $d=38$, QAP obj=4]: the master played $Rxf3$; under μ our $Ra5$ plays the role of that game's $Rb3$; cross-type roles: our $pg5$ as their $rh2$, our $re2$ as their $pf3$; transferred move a5g5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to $Rxg5$ (summed QAP objective 12, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



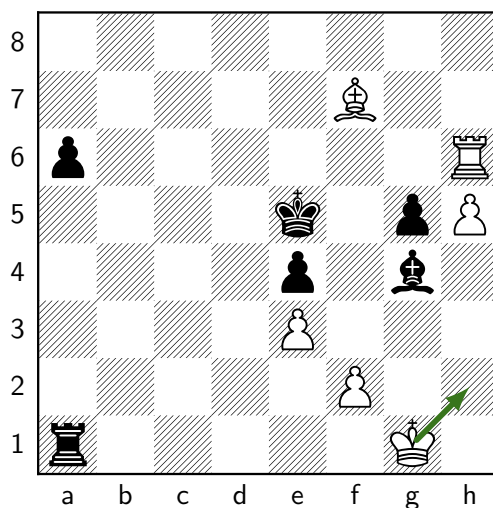
(a)



(b)



(c)



(d)

Figure 43: **Move 49, White: Kg3.** (a) Our position; Melanie plays Kg3 (f4g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#2833` (ply 93) [structural distance $d=38$, QAP obj=38]: the master played Kg6; under μ our Kf4 plays the role of that game’s Kf4; cross-type roles: our Pg4 as their Rh7, our rf2 as their be6; transferred move f4g3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0031#313` (ply 96) [structural distance $d=44$, QAP obj=37]: the master played Kd2; under μ our Kf4 plays the role of that game’s Ke1; cross-type roles: our pf6 as their Ne2, our ng1 as their Ph3; transferred move f4g3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0017#3763` (ply 84) [structural distance $d=44$, QAP obj=25]: the master played Kh2; under μ our Kf4 plays the role of that game’s Kg1; cross-type roles: our rd6 as their pe4, our ng1 as their pg5; transferred move f4g3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg3 (summed QAP objective 102, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

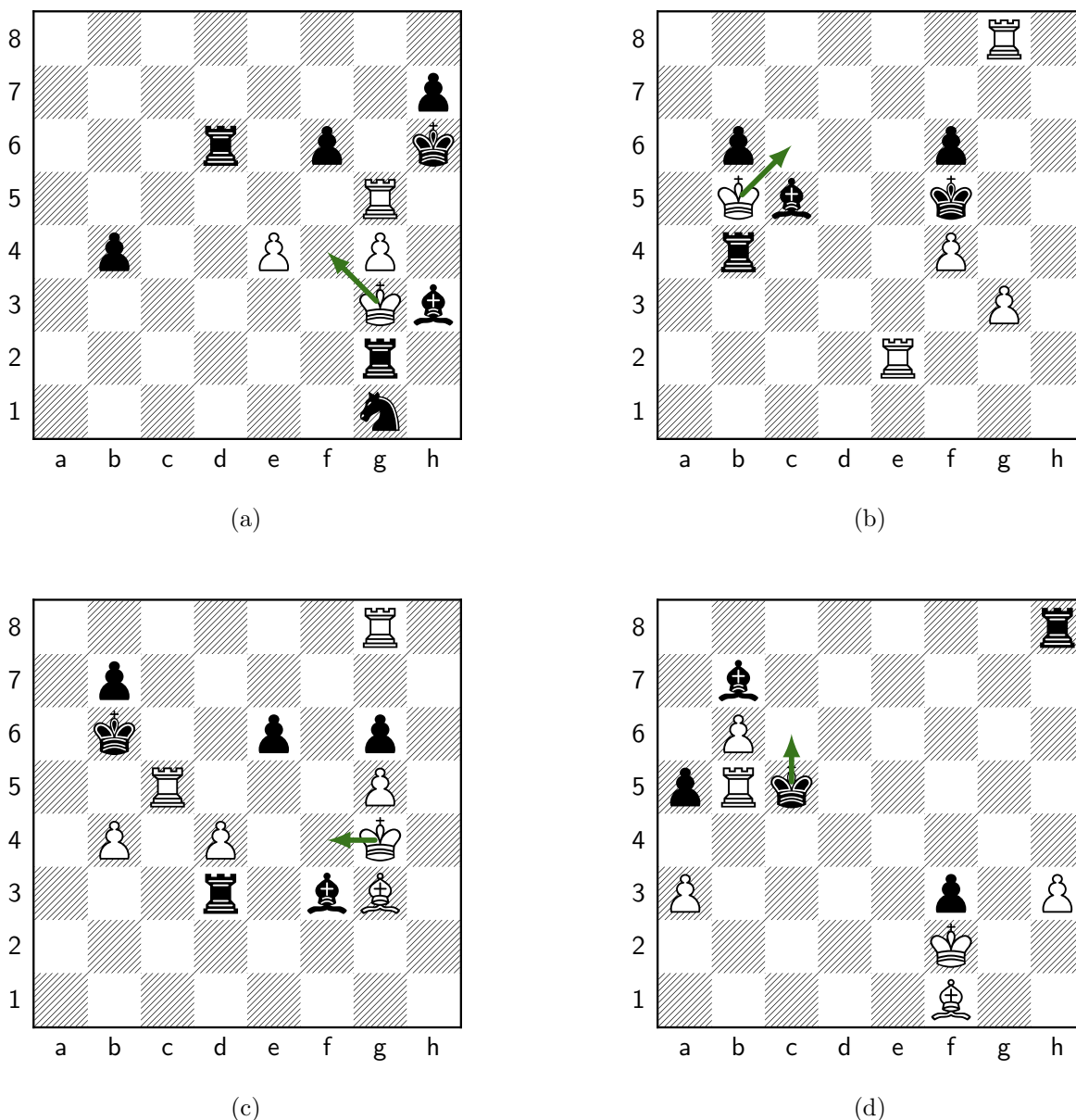
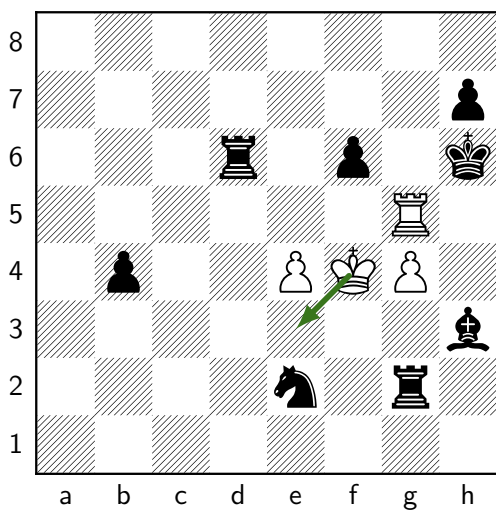
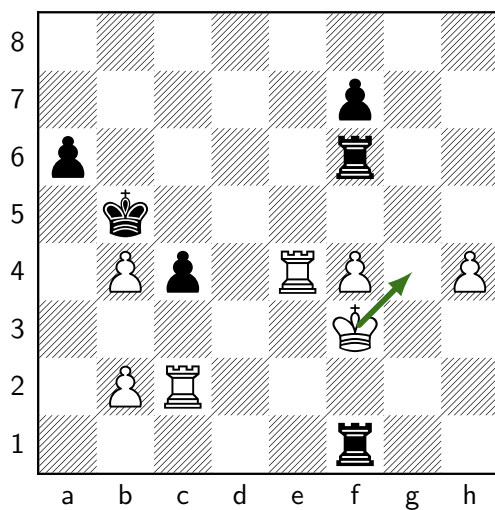


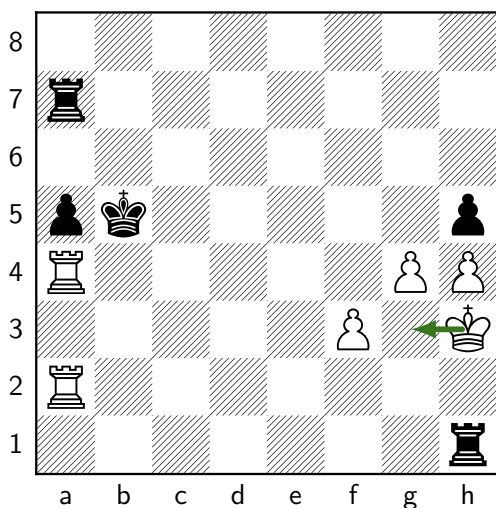
Figure 44: **Move 50, White: Kf4.** (a) Our position; Melanie plays Kf4 (g3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0030#1930 (ply 102) [structural distance $d=44$, QAP obj=73]: the master played Kc6; under μ our Kg3 plays the role of that game's Kb5; cross-type roles: our ng1 as their pb6, our pb4 as their Rg8; transferred move g3f4 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0025#3949 (ply 96) [structural distance $d=49$, QAP obj=69]: the master played Kf4; under μ our Kg3 plays the role of that game's Kg4; cross-type roles: our bh3 as their rd3, our rg2 as their bf3; transferred move g3f4 (**endorses this move**). (d) Retrieved from lichess_elite_2025-11_p0000#671 (ply 103) [structural distance $d=48$, QAP obj=69]: the master played Kc6; under μ our Kg3 plays the role of that game's Kc4; cross-type roles: our pf6 as their Bb7, our ng1 as their pa3; transferred move g3f4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kf4 (summed QAP objective 280, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



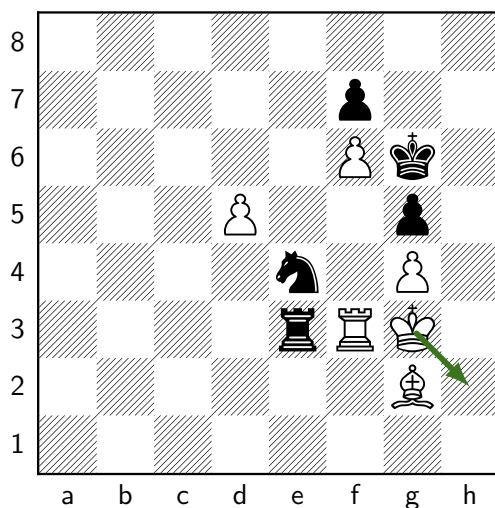
(a)



(b)

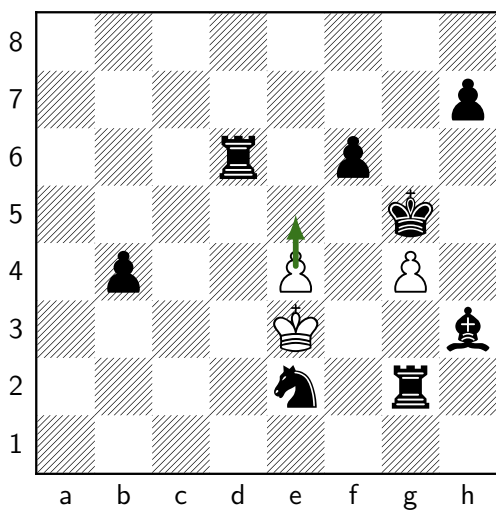


(c)

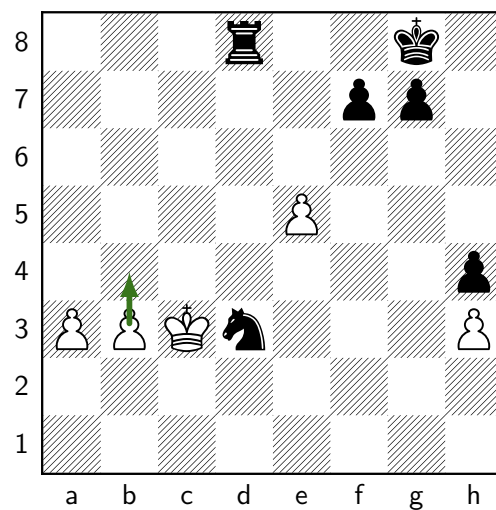


(d)

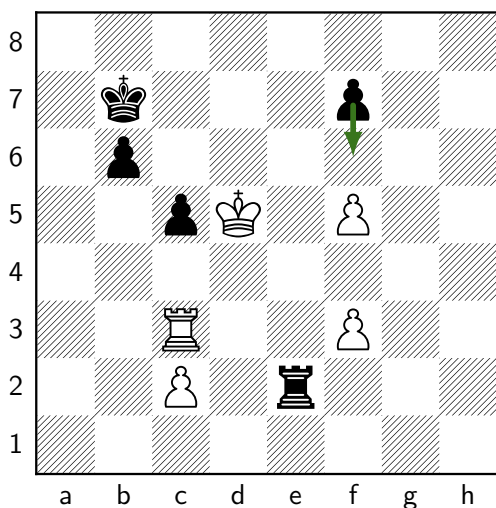
Figure 45: **Move 51, White: Ke3**. (a) Our position; Melanie plays Ke3 (f4e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#2291` (ply 96) [structural distance $d=53$, QAP obj=88]: the master played Kg4; under μ our Kf4 plays the role of that game’s Kf3; cross-type roles: our ne2 as their rf1, our pb4 as their Rc2; transferred move f4e3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#2881` (ply 110) [structural distance $d=53$, QAP obj=78]: the master played Kg3; under μ our Kf4 plays the role of that game’s Kh3; cross-type roles: our ne2 as their rh1, our ph7 as their Ra2; transferred move f4e3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0010#825` (ply 96) [structural distance $d=52$, QAP obj=72]: the master played Kh2; under μ our Kf4 plays the role of that game’s Kg3; cross-type roles: our Rg5 as their Bg2, our Pg4 as their Rf3; transferred move f4e3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ke3 (summed QAP objective 341, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



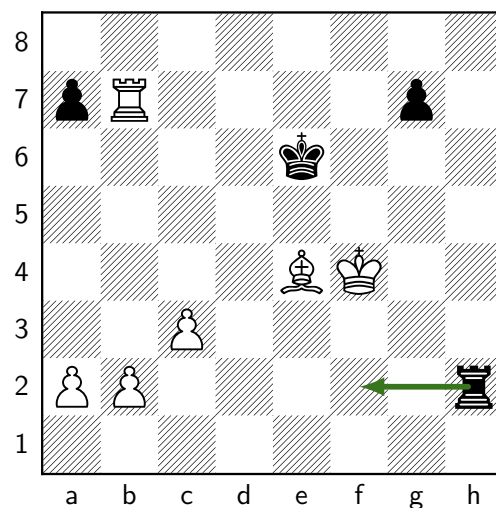
(a)



(b)

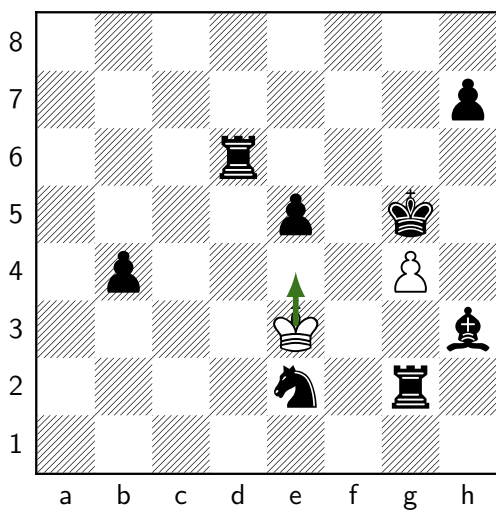


(c)

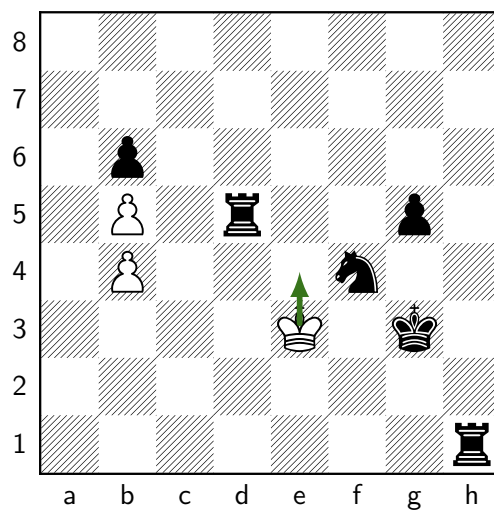


(d)

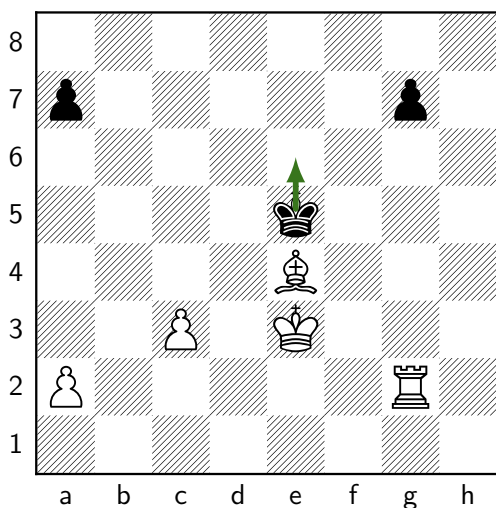
Figure 46: **Move 52, White: e5**. (a) Our position; Melanie plays e5 (e4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#1803` (ply 100) [structural distance $d=32$, QAP obj=20]: the master played b4; under μ our Pe_4 plays the role of that game’s Pb_3 ; cross-type roles: our bh_3 as their Pa_3 , our rd_6 as their Pe_5 ; transferred move e4e5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0020#1966` (ply 93) [structural distance $d=30$, QAP obj=12]: the master played f6; under μ our Pe_4 plays the role of that game’s Pf_2 ; cross-type roles: our rd_6 as their pf_5 , our bh_3 as their rc_3 ; transferred move e4e5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0013#320` (ply 89) [structural distance $d=35$, QAP obj=7]: the master played Rf2+; under μ our Pe_4 plays the role of that game’s Rh_7 (a **cross-type** correspondence); cross-type roles: our bh_3 as their rb_7 , our ne_2 as their Pg_7 ; transferred move e4e5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to e5 (summed QAP objective 39, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



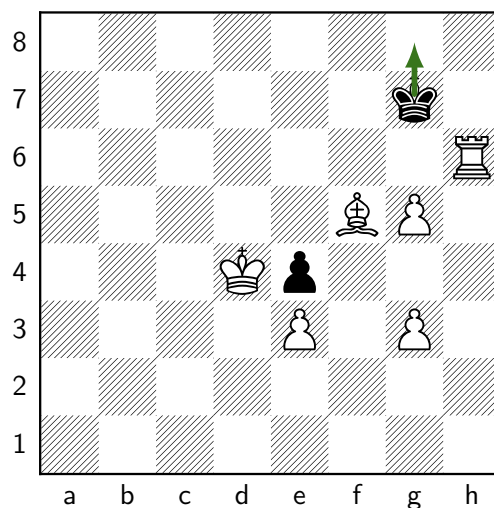
(a)



(b)

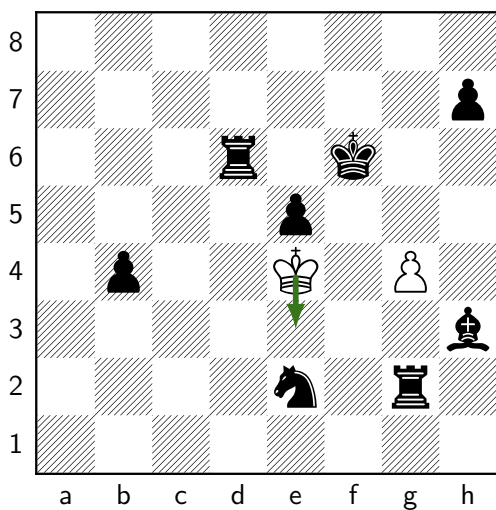


(c)

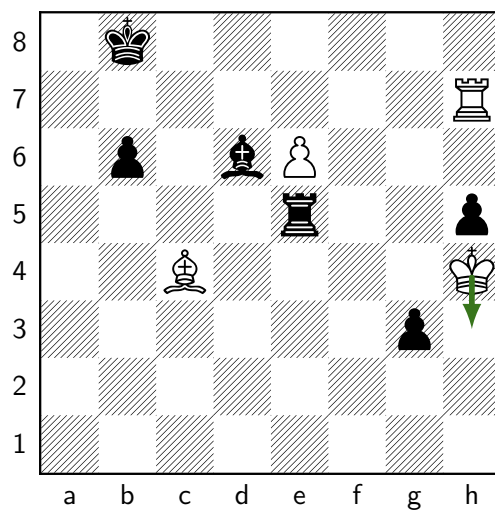


(d)

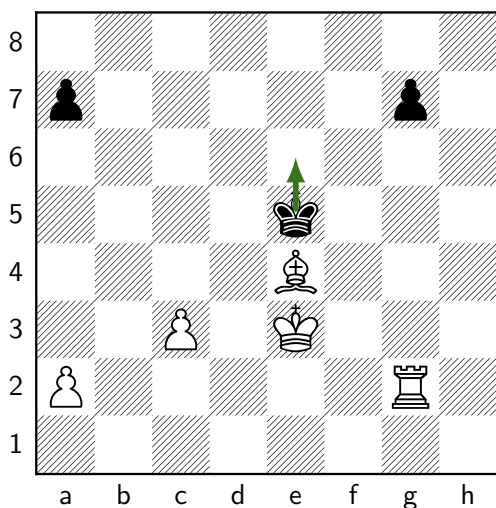
Figure 47: **Move 53, White: Ke4**. (a) Our position; Melanie plays Ke4 (e3e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0023#950` (ply 110) [structural distance $d=31$, QAP obj=20]: the master played Ke4; under μ our Ke3 plays the role of that game’s Ke3; cross-type roles: our pe5 as their rd5, our rg2 as their pg5; transferred move e3e4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#320` (ply 95) [structural distance $d=30$, QAP obj=19]: the master played Ke6; under μ our Ke3 plays the role of that game’s Ke4; cross-type roles: our bh3 as their Pa7, our ne2 as their be4; transferred move e3e4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0028#785` (ply 119) [structural distance $d=29$, QAP obj=17]: the master played Kg8; under μ our Ke3 plays the role of that game’s Kg2; cross-type roles: our ne2 as their rh6, our rg2 as their pe3; transferred move e3e4 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Ke4 (summed QAP objective 77, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



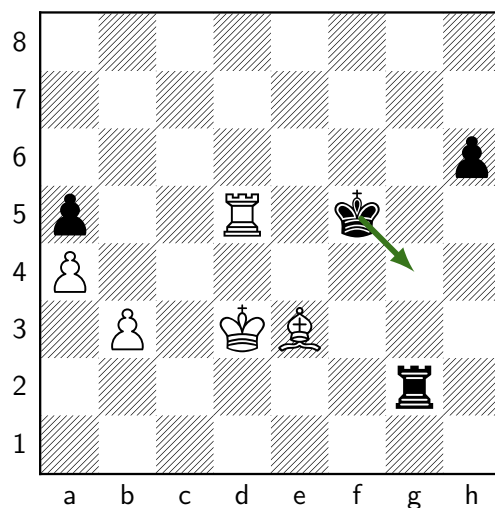
(a)



(b)

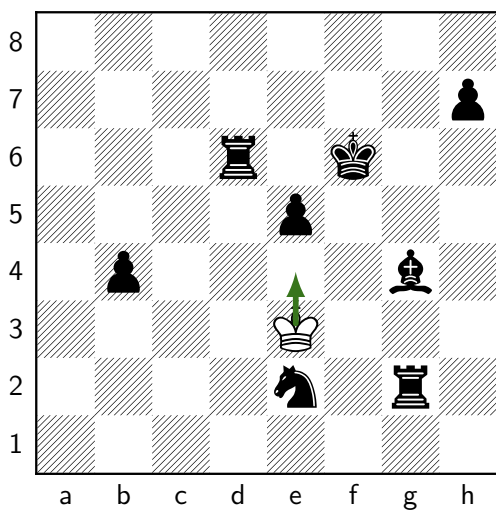


(c)

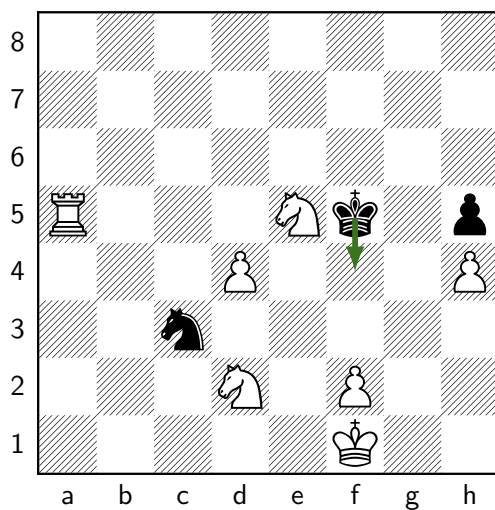


(d)

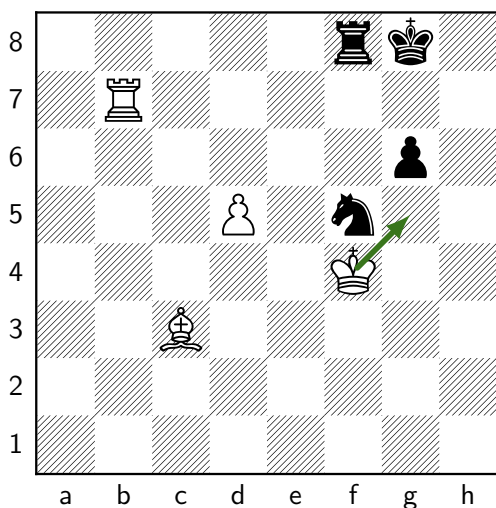
Figure 48: **Move 54, White: Ke3.** (a) Our position; Melanie plays Ke3 (e4e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#571` (ply 104) [structural distance $d=29$, QAP obj=56]: the master played Kh3; under μ our Ke4 plays the role of that game’s Kh4; cross-type roles: our ne2 as their Bc4, our bh3 as their re5; transferred move e4e3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#320` (ply 95) [structural distance $d=25$, QAP obj=8]: the master played Ke6; under μ our Ke4 plays the role of that game’s Ke4; cross-type roles: our pe5 as their be4, our ne2 as their pc3; transferred move e4e3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0013#717` (ply 113) [structural distance $d=30$, QAP obj=3]: the master played Kg4; under μ our Ke4 plays the role of that game’s Kf4; cross-type roles: our bh3 as their pa4, our rg2 as their pb3; transferred move e4e3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Ke3 (summed QAP objective 72, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



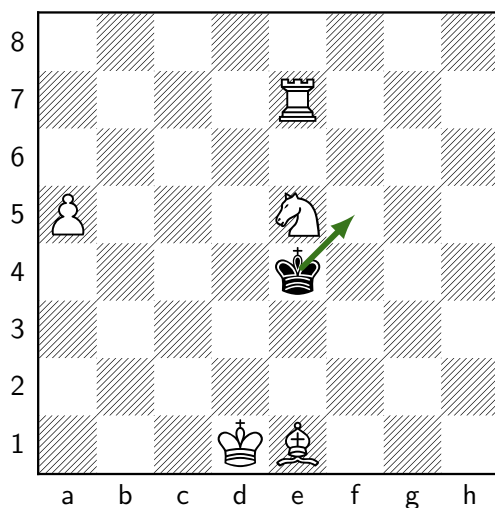
(a)



(b)



(c)



(d)

Figure 49: **Move 55, White: Ke4**. (a) Our position; Melanie plays Ke4 (e3e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#140` (ply 95) [structural distance $d=30$, QAP obj=22]: the master played Kf4; under μ our Ke3 plays the role of that game’s Kf4; cross-type roles: our bg4 as their pd4, our rd6 as their nd2; transferred move e3e4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#2651` (ply 118) [structural distance $d=28$, QAP obj=20]: the master played Kg5; under μ our Ke3 plays the role of that game’s Kf4; cross-type roles: our pb4 as their Bc3, our bg4 as their Pd5; transferred move e3e4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0028#1123` (ply 95) [structural distance $d=31$, QAP obj=20]: the master played Kf5; under μ our Ke3 plays the role of that game’s Ke5; cross-type roles: our bg4 as their pa5, our pb4 as their be1; transferred move e3e4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Ke4 (summed QAP objective 93, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

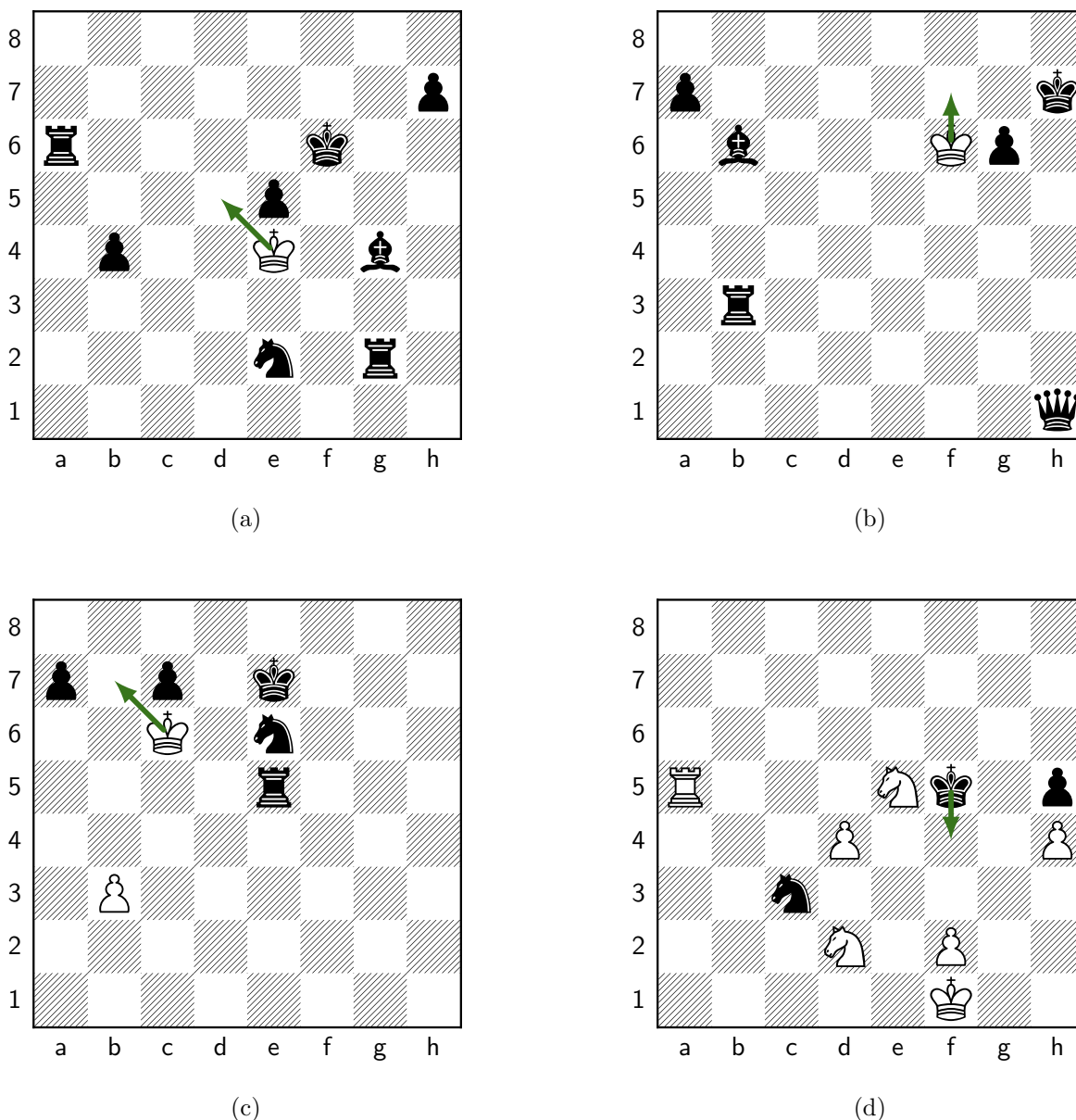


Figure 50: **Move 56, White: Kd5.** (a) Our position; Melanie plays Kd5 (e4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0016#1722 (ply 102) [structural distance $d=31$, QAP obj=21]: the master played Kf7; under μ our Ke4 plays the role of that game's Kf6; cross-type roles: our ra6 as their qh1, our ne2 as their bb6; transferred move e4d5 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0021#1941 (ply 112) [structural distance $d=27$, QAP obj=20]: the master played Kb7; under μ our Ke4 plays the role of that game's Kc6; cross-type roles: our rg2 as their Pb3, our bg4 as their re5; transferred move e4d5 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0005#140 (ply 95) [structural distance $d=34$, QAP obj=7]: the master played Kf4; under μ our Ke4 plays the role of that game's Kf4; cross-type roles: our bg4 as their ra5, our pe5 as their ne5; transferred move e4d5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kd5 (summed QAP objective 55, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

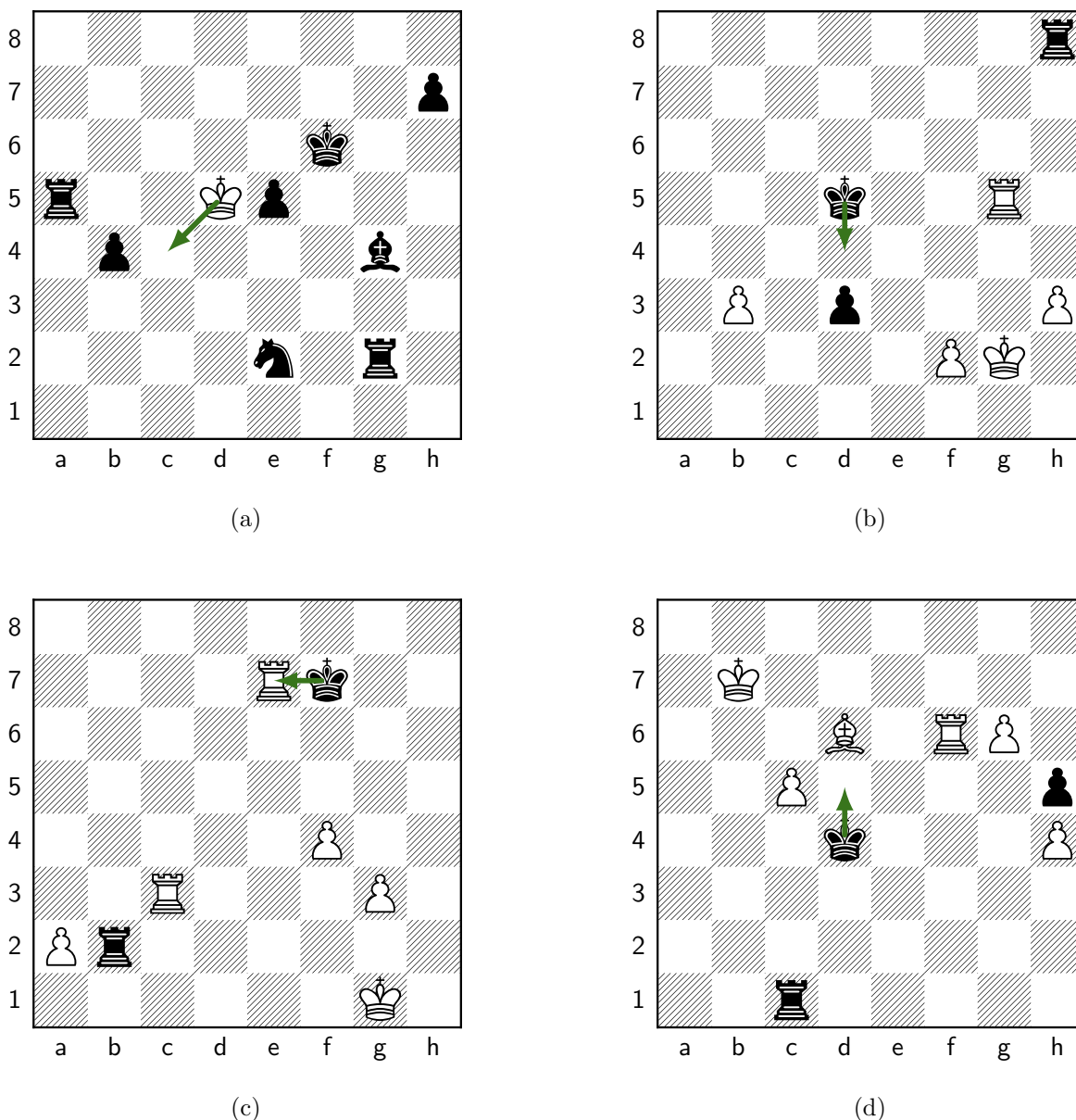


Figure 51: **Move 57, White: Kc4.** (a) Our position; Melanie plays Kc4 (d5c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#2328` (ply 115) [structural distance $d=30$, QAP obj=67]: the master played Kd4; under μ our Kd5 plays the role of that game’s Kd4; cross-type roles: our pb4 as their Rh8, our ne2 as their Pd3; transferred move d5c4 (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#3680` (ply 103) [structural distance $d=29$, QAP obj=23]: the master played Kxe7; under μ our Kd5 plays the role of that game’s Kf2; cross-type roles: our ne2 as their pf4, our bg4 as their pg3; transferred move d5c4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0028#527` (ply 111) [structural distance $d=27$, QAP obj=3]: the master played Kd5; under μ our Kd5 plays the role of that game’s Kd5; cross-type roles: our ne2 as their bd6, our bg4 as their pg6; transferred move d5c4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kc4 (summed QAP objective 94, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

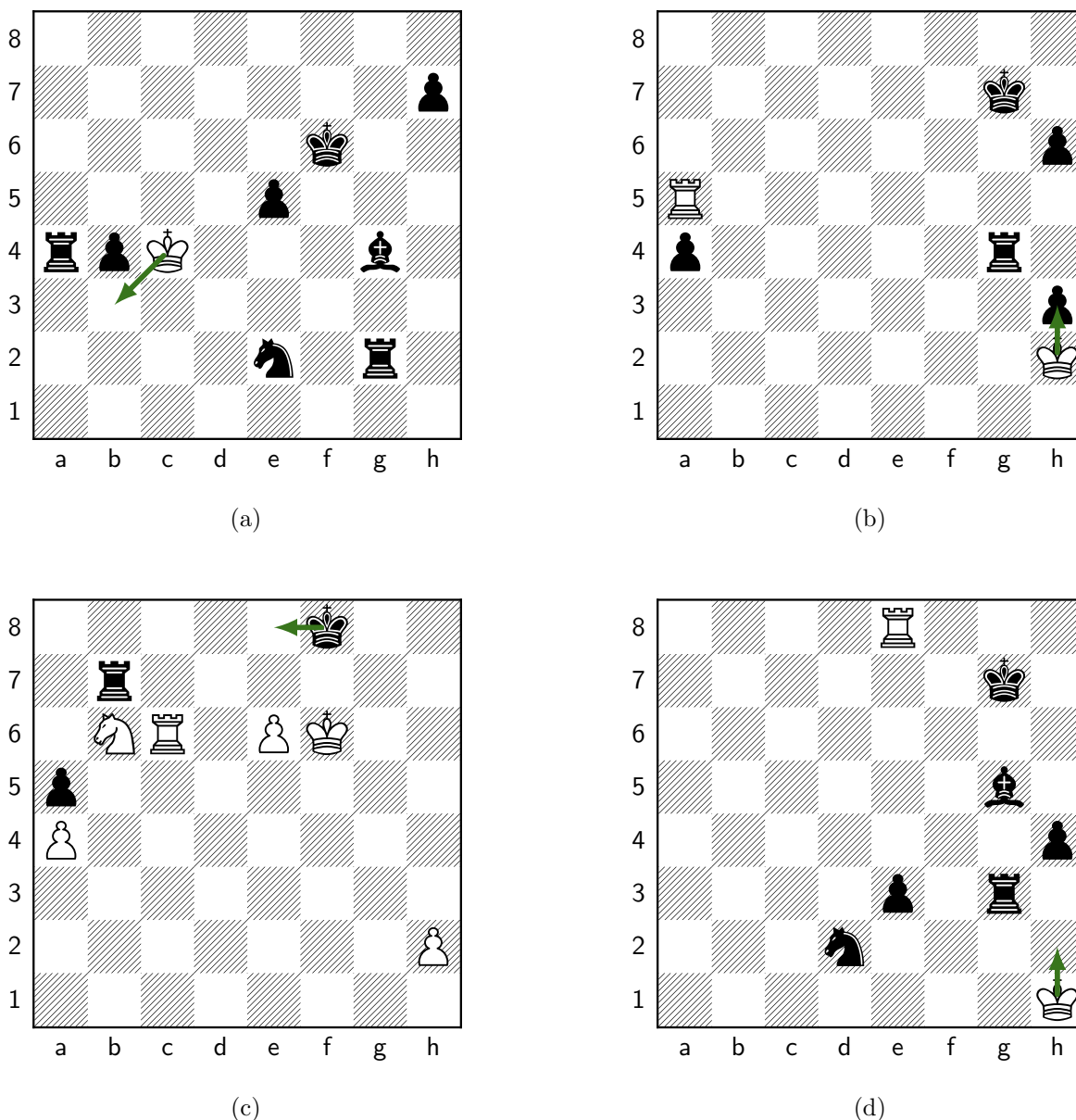
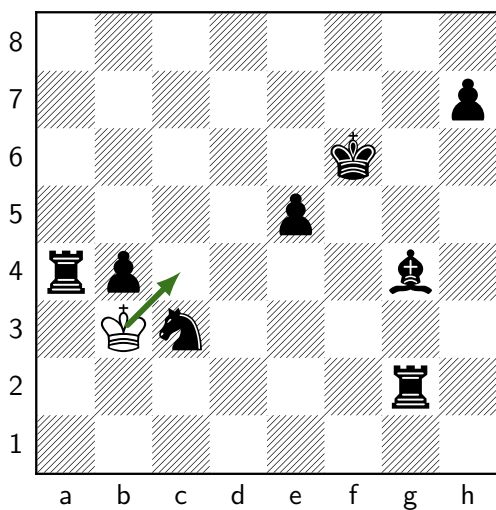
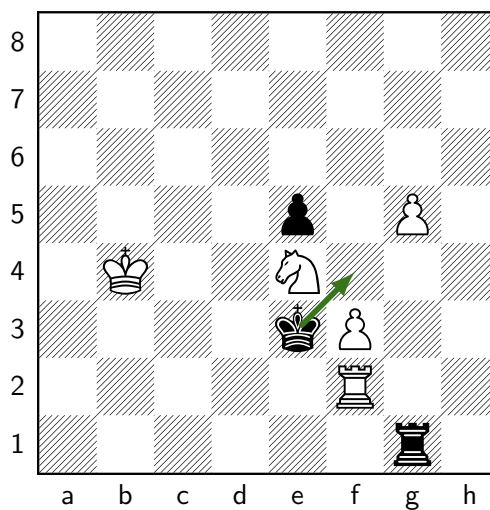


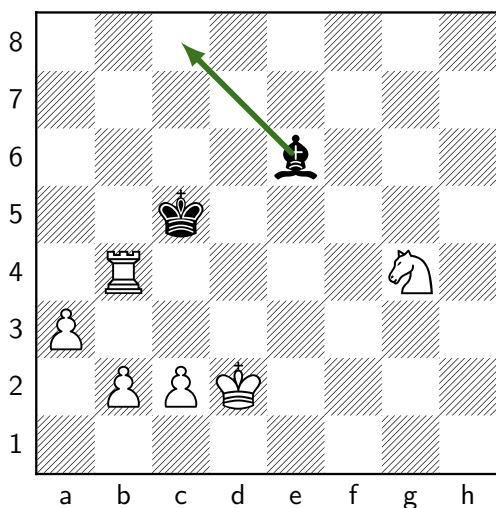
Figure 52: **Move 58, White: Kb3.** (a) Our position; Melanie plays Kb3 (**c4b3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#1550` (ply 114) [structural distance $d=30$, QAP obj=28]: the master played Kxh3; under μ our Kc4 plays the role of that game’s Kh2; cross-type roles: our Kc4 as their Ra5, our ne2 as their ph6; transferred move c4b3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#2922` (ply 111) [structural distance $d=28$, QAP obj=4]: the master played Ke8; under μ our Kc4 plays the role of that game’s Kf1; cross-type roles: our Kc4 as their Rb7, our pb4 as their nb6; transferred move c4b3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0019#238` (ply 122) [structural distance $d=29$, QAP obj=2]: the master played Kh2; under μ our Kc4 plays the role of that game’s Kh1; cross-type roles: our ra4 as their nd2, our ne2 as their ph4; transferred move c4b3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kb3 (summed QAP objective 33, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



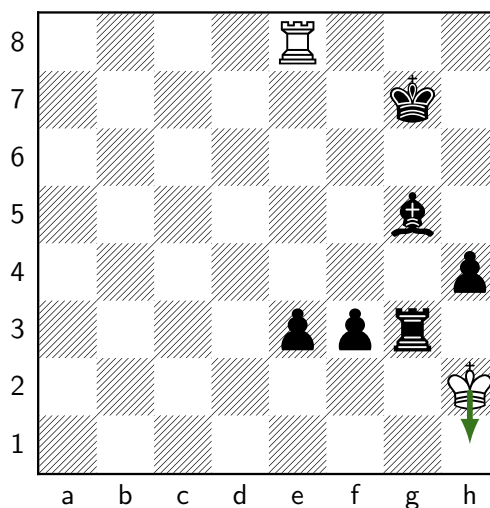
(a)



(b)

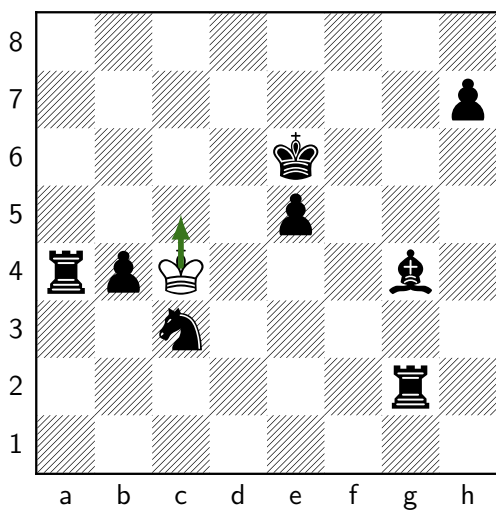


(c)

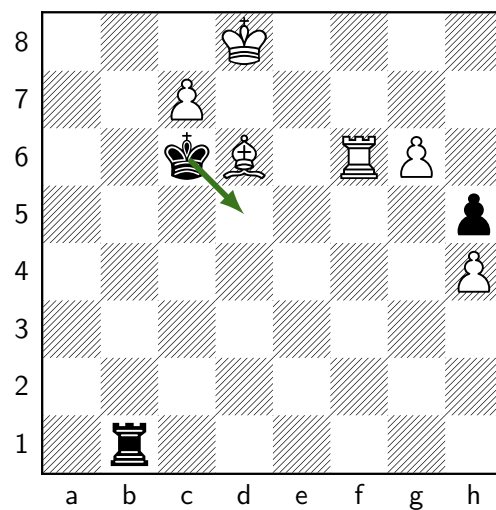


(d)

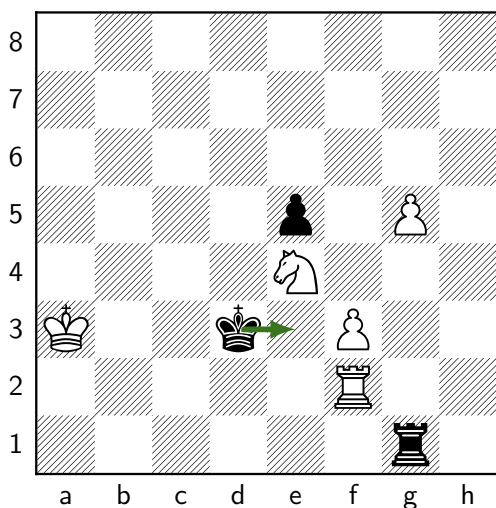
Figure 53: **Move 59, White: Kc4**. (a) Our position; Melanie plays Kc4 (b3c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#2031` (ply 107) [structural distance $d=20$, QAP obj=60]: the master played Kf4; under μ our Kb3 plays the role of that game’s Ke6; cross-type roles: our bg4 as their Pe5, our rg2 as their pg5; transferred move b3c4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0025#700` (ply 109) [structural distance $d=23$, QAP obj=21]: the master played Bc8; under μ our Kb3 plays the role of that game’s Be3 (a **cross-type** correspondence); cross-type roles: our rg2 as their pb2; transferred move b3c4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0019#238` (ply 116) [structural distance $d=30$, QAP obj=20]: the master played Kh1; under μ our Kb3 plays the role of that game’s Kh2; cross-type roles: our rg2 as their pe3, our nc3 as their ph4; transferred move b3c4 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kc4 (summed QAP objective 235, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



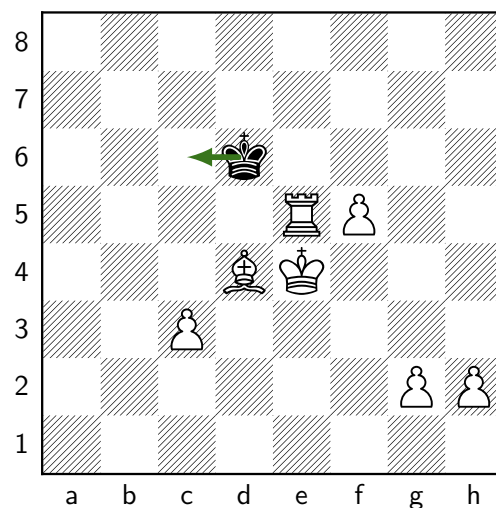
(a)



(b)



(c)



(d)

Figure 54: **Move 60, White: Kc5**. (a) Our position; Melanie plays Kc5 (c4c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#527` (ply 119) [structural distance $d=17$, QAP obj=36]: the master played Kd5; under μ our $Kc4$ plays the role of that game’s $Kc3$; cross-type roles: our $nc3$ as their $bd6$, our $bg4$ as their $Ph5$; transferred move c4c5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0019#2031` (ply 105) [structural distance $d=23$, QAP obj=21]: the master played Ke3; under μ our $Kc4$ plays the role of that game’s $Kd6$; cross-type roles: our $bg4$ as their $Pe5$; transferred move c4c5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0010#2131` (ply 111) [structural distance $d=25$, QAP obj=13]: the master played Kc6; under μ our $Kc4$ plays the role of that game’s $Kd3$; cross-type roles: our $nc3$ as their $re5$, our $ra4$ as their $bd4$; transferred move c4c5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kc5 (summed QAP objective 91, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

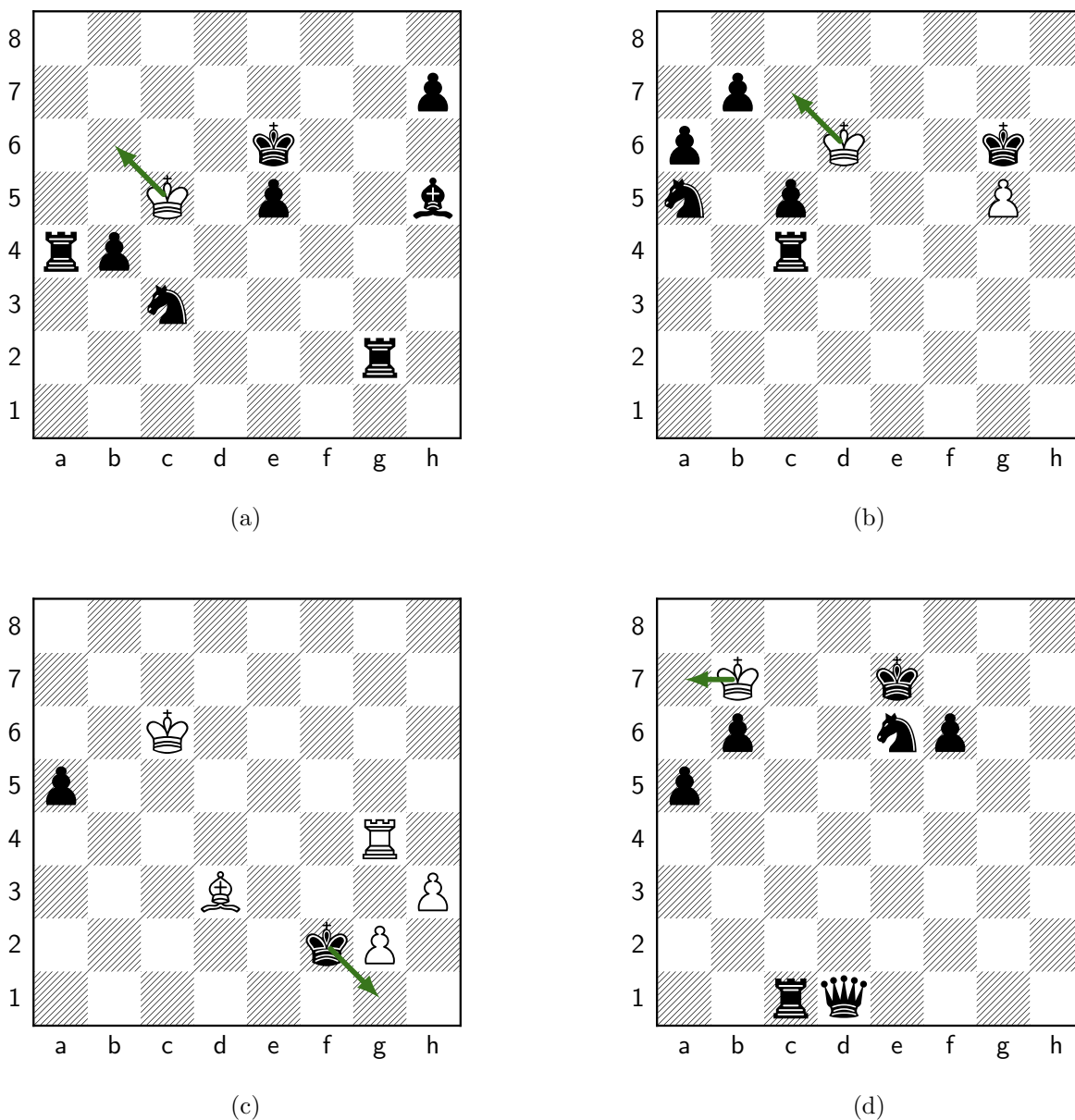
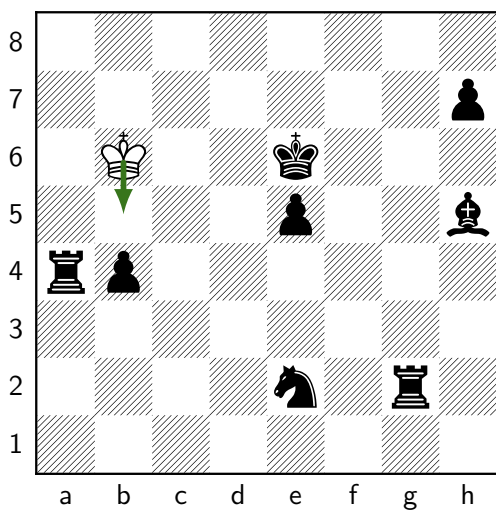
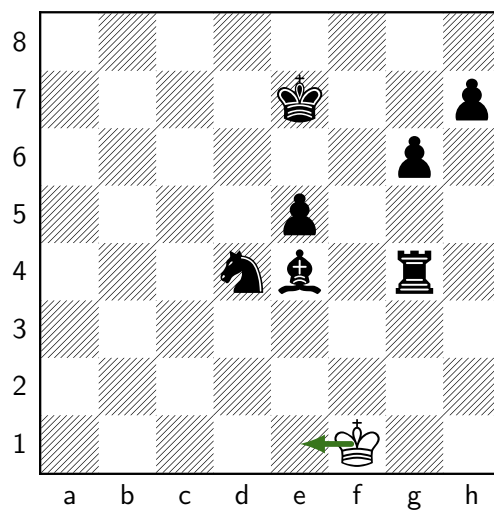


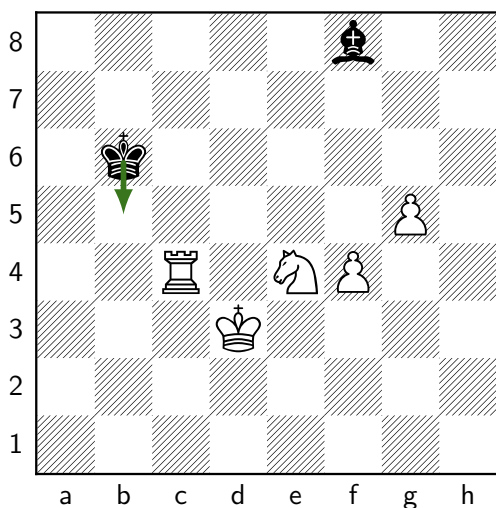
Figure 55: **Move 61, White: Kb6.** (a) Our position; Melanie plays Kb6 (c5b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0016#57 (ply 124) [structural distance $d=20$, QAP obj=22]: the master played Kc7; under μ our Kc5 plays the role of that game's Kd6; cross-type roles: our rg2 as their Pg5, our bh5 as their pa6; transferred move c5b6 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0020#1043 (ply 125) [structural distance $d=21$, QAP obj=21]: the master played Kg1; under μ our Kc5 plays the role of that game's Kf7; cross-type roles: our rg2 as their bd3, our nc3 as their ph3; transferred move c5b6 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0018#1948 (ply 128) [structural distance $d=17$, QAP obj=20]: the master played Ka7; under μ our Kc5 plays the role of that game's Kb7; cross-type roles: our bh5 as their qd1, our nc3 as their pa5; transferred move c5b6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kb6 (summed QAP objective 78, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



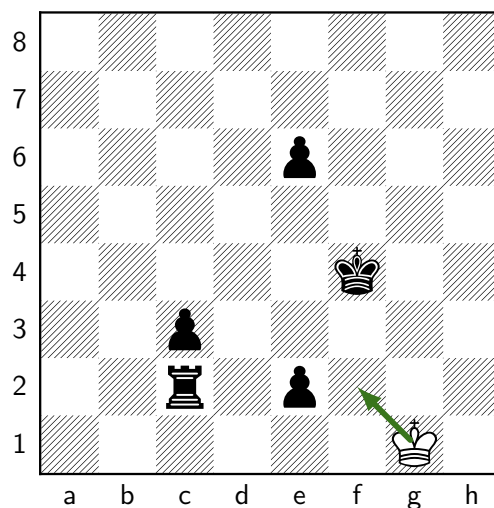
(a)



(b)



(c)



(d)

Figure 56: **Move 62, White: Kb5.** (a) Our position; Melanie plays Kb5 (b6b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#98` (ply 106) [structural distance $d=18$, QAP obj=2]: the master played Kc1; under μ our Kb6 plays the role of that game’s Kf1; cross-type roles: our ra4 as their nd4, our ne2 as their be4; transferred move b6b5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0012#2185` (ply 119) [structural distance $d=24$, QAP obj=1]: the master played Kb5; under μ our Kb6 plays the role of that game’s Kb3; cross-type roles: our rg2 as their Bf8, our ne2 as their pg5; transferred move b6b5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0013#1157` (ply 136) [structural distance $d=27$, QAP obj=1]: the master played Kf2; under μ our Kb6 plays the role of that game’s Kg1; cross-type roles: our bh5 as their rc2, our ra4 as their pe2; transferred move b6b5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kb5 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

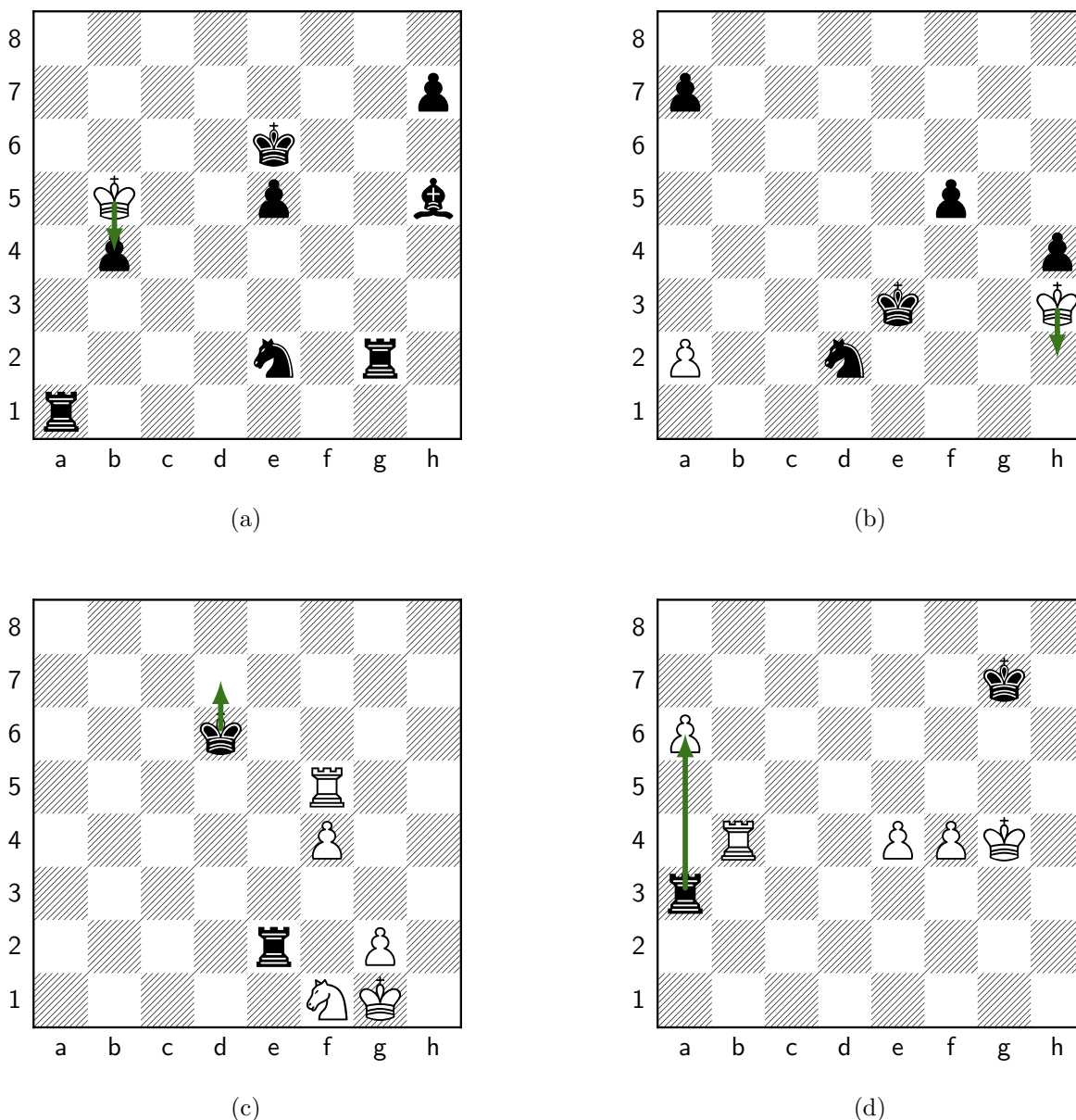
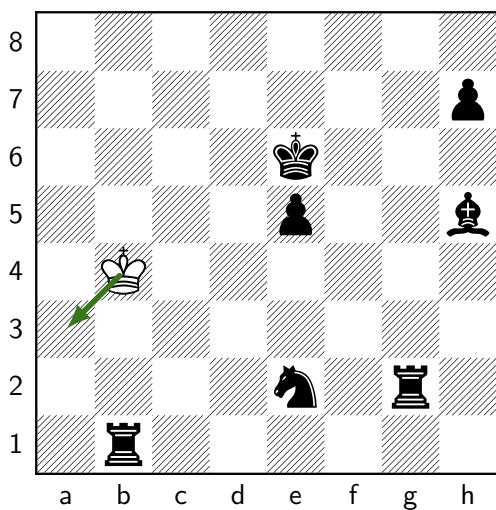
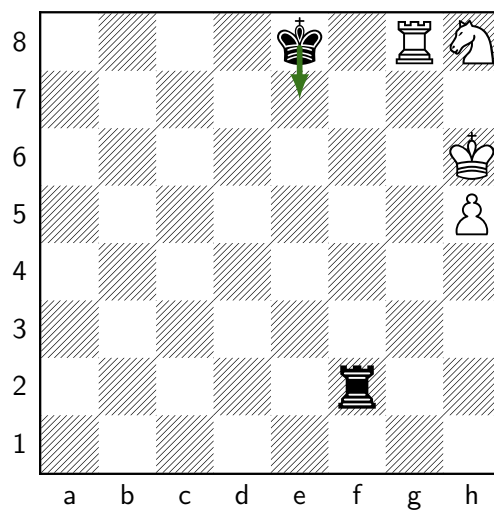


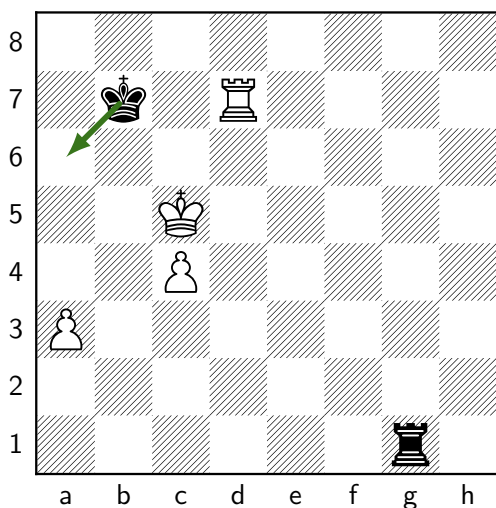
Figure 57: **Move 63, White: Kxb4.** (a) Our position; Melanie plays Kxb4 (b5b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0026#1678` (ply 126) [structural distance $d=24$, QAP obj=19]: the master played Kh2; under μ our Kb5 plays the role of that game’s Kh3; cross-type roles: our rg2 as their Pa2, our pe5 as their nd2; transferred move b5b4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0017#1093` (ply 119) [structural distance $d=23$, QAP obj=5]: the master played Kd7; under μ our Kb5 plays the role of that game’s Kd3; cross-type roles: our ne2 as their pf4, our Kb5 as their Re2; transferred move b5b4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0022#428` (ply 137) [structural distance $d=25$, QAP obj=4]: the master played Rxa6; under μ our Kb5 plays the role of that game’s Ra6 (a **cross-type** correspondence); cross-type roles: our ne2 as their pe4; transferred move b5b4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kxb4 (summed QAP objective 46, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



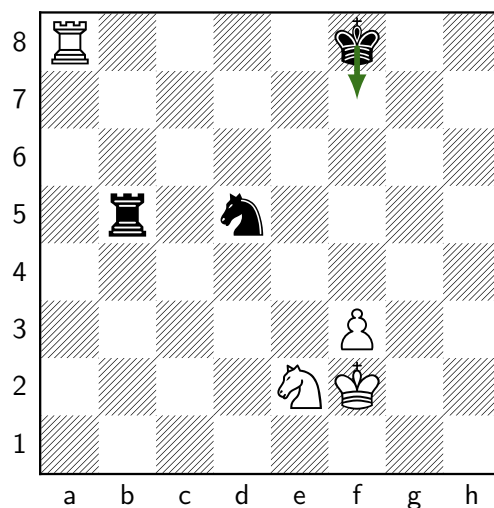
(a)



(b)

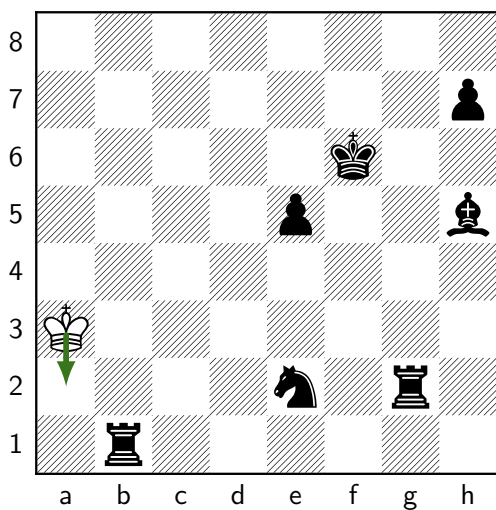


(c)

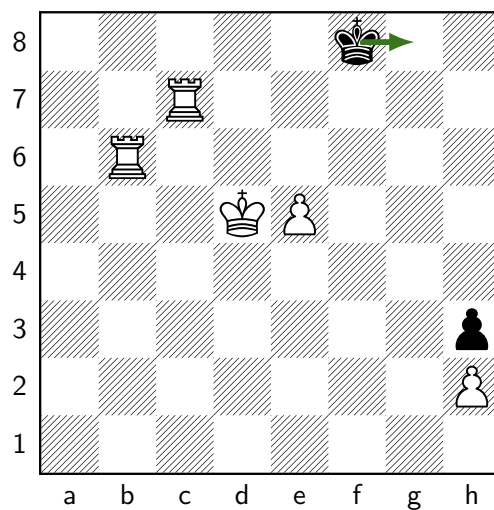


(d)

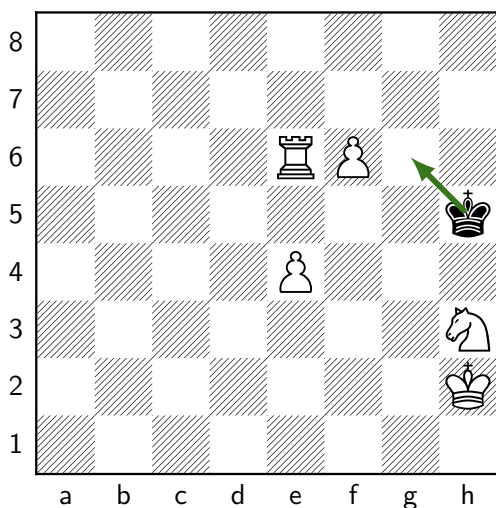
Figure 58: **Move 64, White: Ka3**. (a) Our position; Melanie plays Ka3 (b4a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#3074` (ply 133) [structural distance $d=16$, QAP obj=46]: the master played Ke7; under μ our Kb4 plays the role of that game’s Ke1; transferred move b4a3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#2166` (ply 119) [structural distance $d=23$, QAP obj=45]: the master played Ka6; under μ our Kb4 plays the role of that game’s Kb2; cross-type roles: our ne2 as their pa3; transferred move b4a3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0010#750` (ply 141) [structural distance $d=21$, QAP obj=42]: the master played Kf7; under μ our Kb4 plays the role of that game’s Kf1; cross-type roles: our bh5 as their ne2; transferred move b4a3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Ka3 (summed QAP objective 182, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



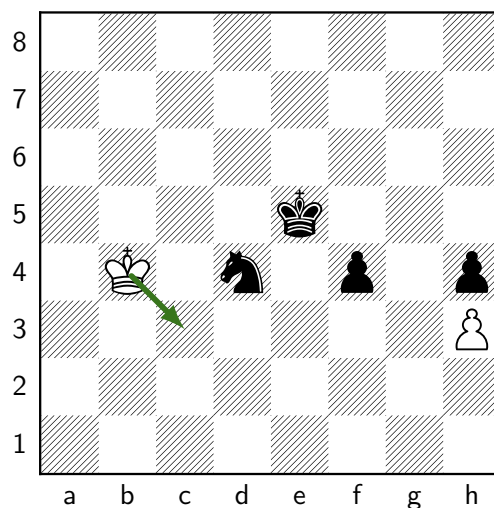
(a)



(b)

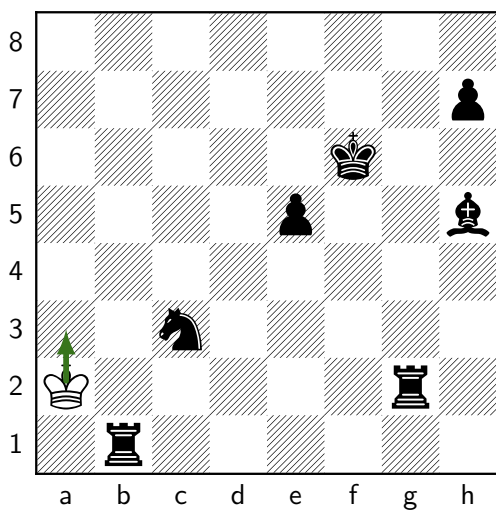


(c)

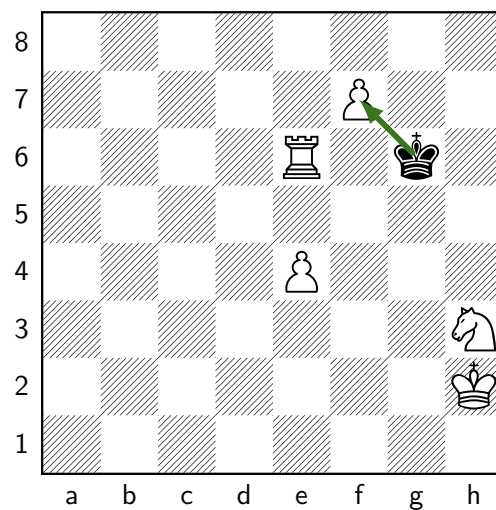


(d)

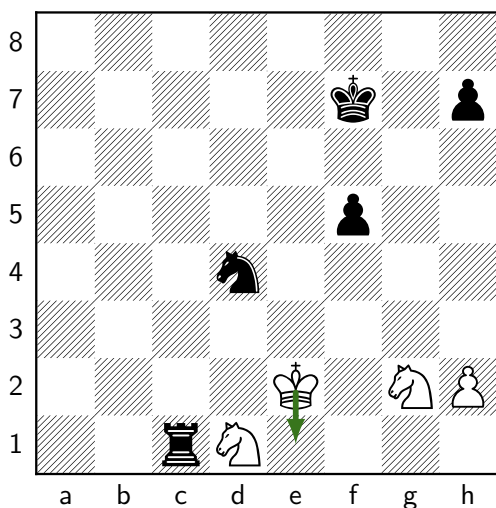
Figure 59: **Move 65, White: Ka2.** (a) Our position; Melanie plays Ka2 (a3a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#1179` (ply 139) [structural distance $d=19$, QAP obj=2]: the master played Kg8; under μ our Ka3 plays the role of that game’s Kf1; cross-type roles: our bh5 as their Ph3, our ne2 as their ph2; transferred move a3a2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#1580` (ply 117) [structural distance $d=17$, QAP obj=1]: the master played Kg6; under μ our Ka3 plays the role of that game’s Kh4; cross-type roles: our ne2 as their pf6, our rb1 as their pe4; transferred move a3a2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0019#3773` (ply 138) [structural distance $d=22$, QAP obj=1]: the master played Kc3; under μ our Ka3 plays the role of that game’s Kb4; cross-type roles: our bh5 as their Ph3, our rb1 as their nd4; transferred move a3a2 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Ka2 (summed QAP objective 7, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



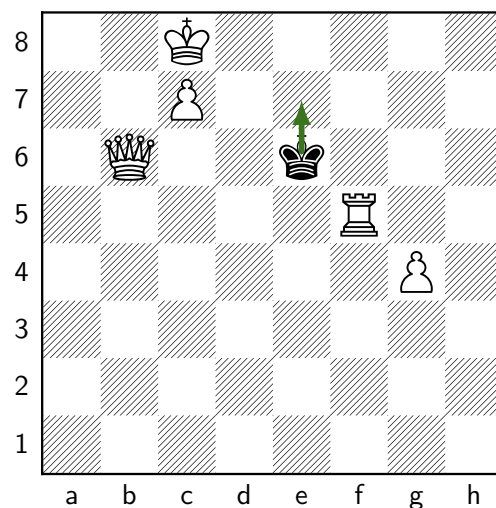
(a)



(b)



(c)



(d)

Figure 60: **Move 66, White: Ka3.** (a) Our position; Melanie plays Ka3 (a2a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#1580` (ply 119) [structural distance $d=23$, QAP obj=73]: the master played Kxf7; under μ our Ka2 plays the role of that game’s Kg3; cross-type roles: our rb1 as their pf7, our nc3 as their re6; transferred move a2a3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0028#1909` (ply 130) [structural distance $d=23$, QAP obj=66]: the master played Ke1; under μ our Ka2 plays the role of that game’s Ke2; cross-type roles: our rg2 as their Nd1, our bh5 as their pf5; transferred move a2a3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0020#1311` (ply 121) [structural distance $d=25$, QAP obj=60]: the master played Ke7; under μ our Ka2 plays the role of that game’s Ke3; cross-type roles: our nc3 as their qb6, our rg2 as their pg4; transferred move a2a3 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to Ka3 (summed QAP objective 539, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.